Chapter 5

MULTISTAGE, POWER, AND DIFFERENTIAL AMPLIFIERS

CHAPTER OUTLINE

5-1 Coupled Amplifiers
5-2 Power Amplifiers
5-3 Differential Amplifiers
5-4 Frequency Response of Amplifiers
5-5 Negative Feedback in Amplifiers

5-1 Coupled Amplifiers

5-1-1 Capacitively Coupled Amplifiers

5-1-2 Transformer-Coupled Amplifiers

5-1-3 Direct-Coupled Amplifiers

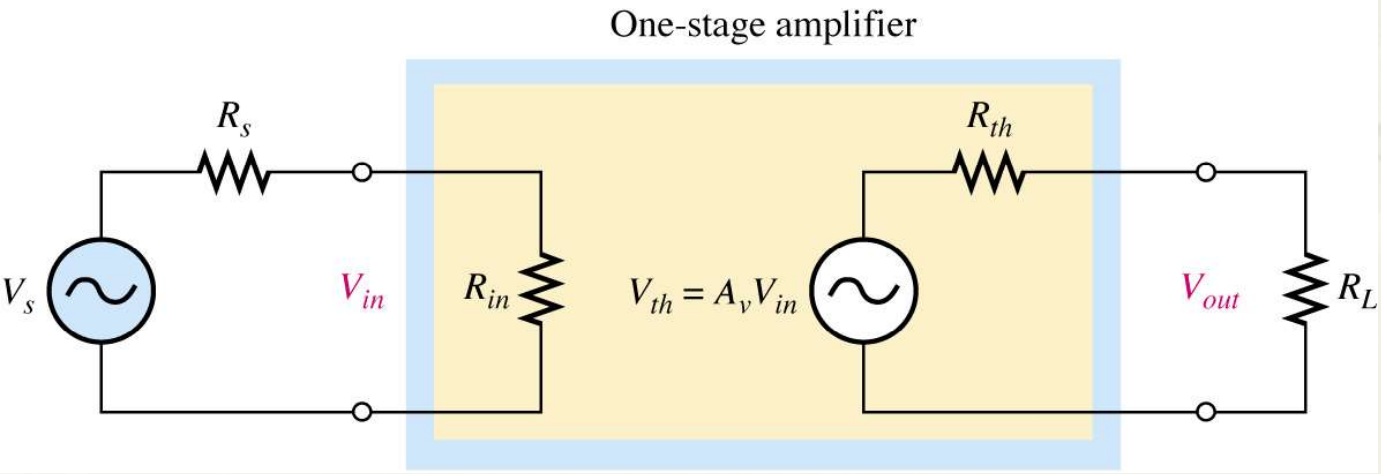
5-1-1 Capacitively Coupled Amplifiers

- Amplifier Model
- Cascaded Stages
- Transistor Amplifiers
- Loading Effects
- Unwanted Oscillation and Noise

Two or more transistors can be connected together to form an amplifier called a multistage amplifier.

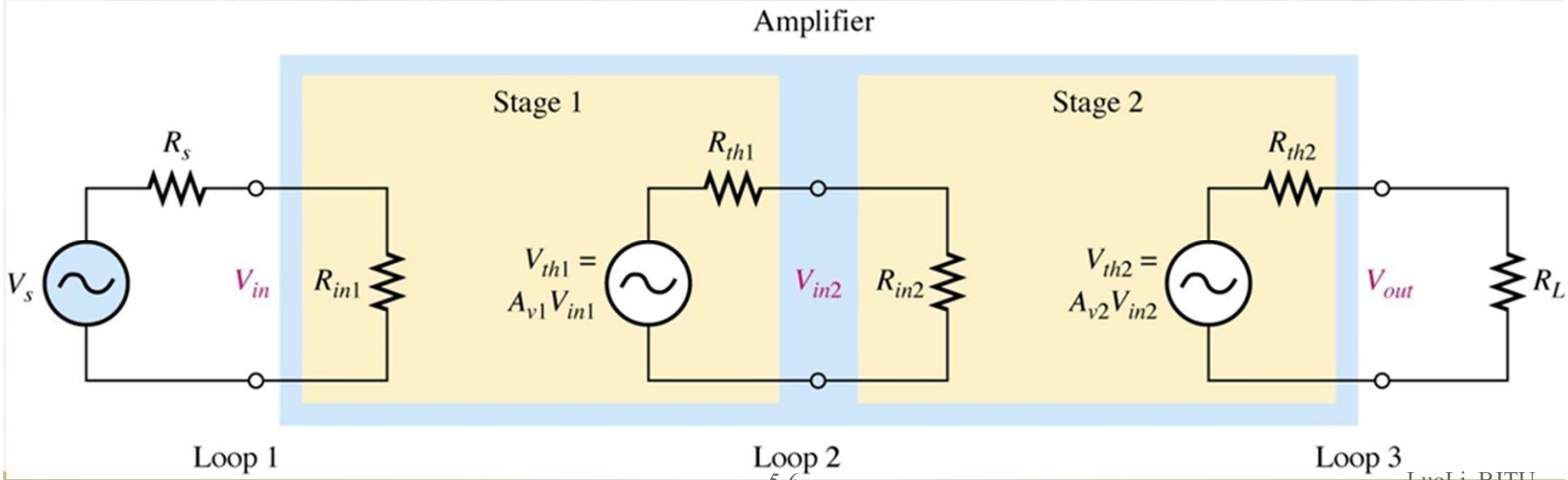
Amplifier Model

- **Amplifier**
  - An amplifier is a device that increase the magnitude of a signal for use by a load.
  - **Input:** Presents an input resistance, R_{in}, forms a voltage divider with the source resistance.
  - **Output:** A dependent voltage source with a series resistance, the output is dependent on the unloaded gain (A_v) and the input voltage.

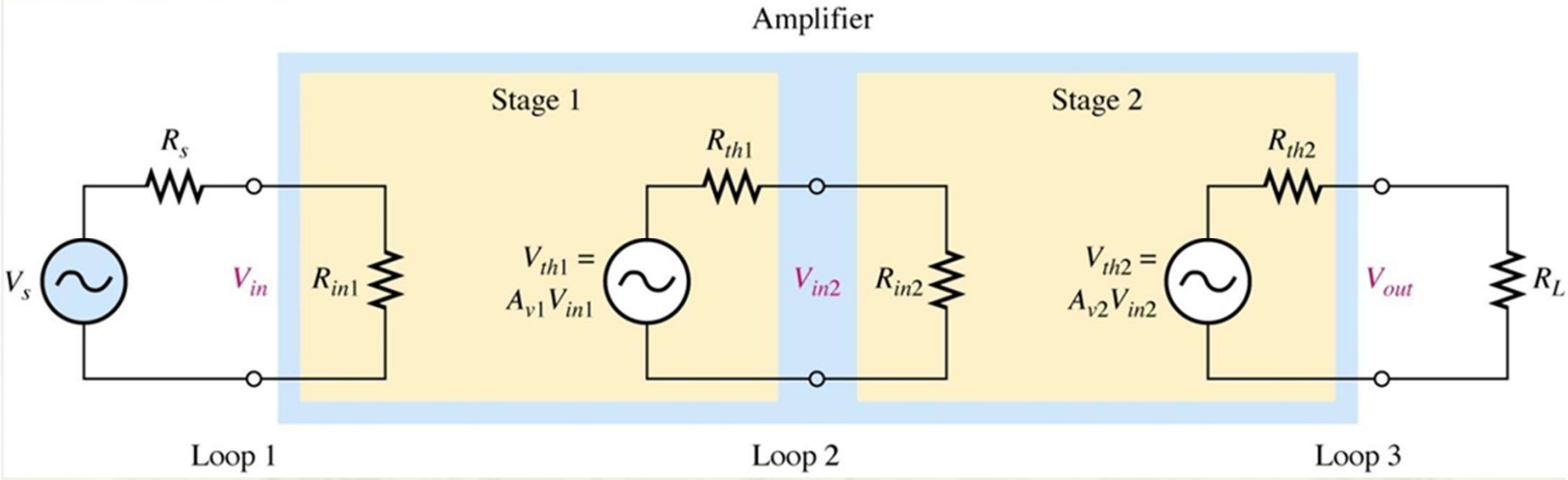


Cascaded Stages

- **Stage**
  - Each functional part that amplifies a signal is considered to be a stage.
  - Every stage is considered for source and load effects.



Cascaded Stages



$$V_{in1} = V_s\left(\frac{R_{in1}}{R_{in1} + R_s}\right),$$

$$V_{th1} = A_{v1}V_{in1},$$

$$V_{in2} = V_{th1}\left(\frac{R_{in2}}{R_{in2} + R_{th1}}\right),$$

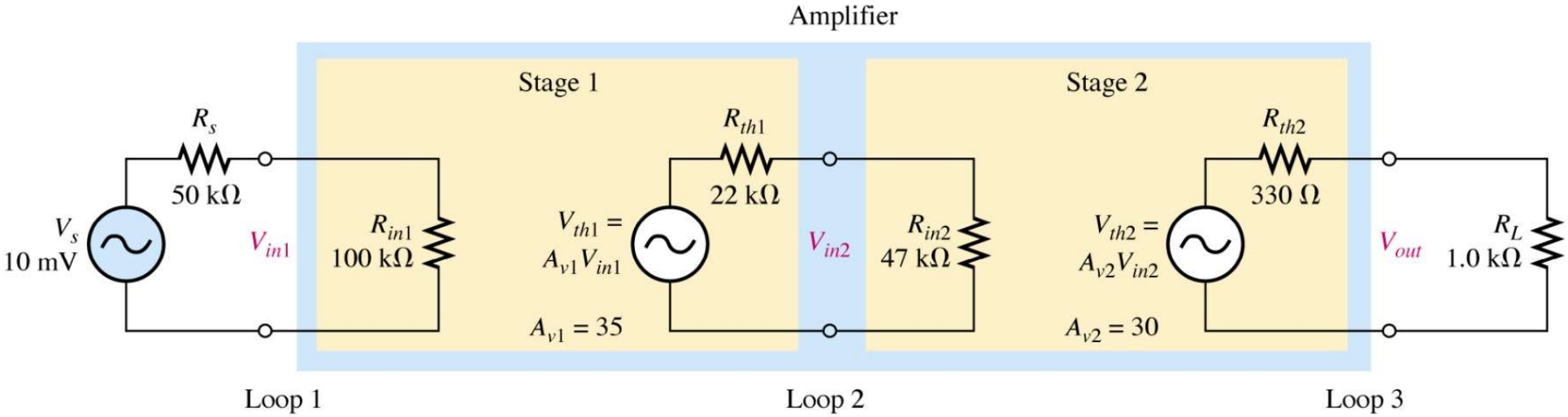
$$V_{th2} = A_{v2}V_{in2},$$

$$V_{R_L} = V_{th2}\left(\frac{R_L}{R_L + R_{th2}}\right)$$

Example 5-1

Problem

Assume a transducer with a Thevenin (unloaded) source, V_s, of 10 mV and a Thevenin source resistance, R_s, of 50 kΩ is connected to a two-stage cascaded amplifier, as shown in Figure 5–3. Compute the voltage across a 1.0 kΩ load.



Example 5-1

Solution

Compute the input voltage to stage 1 from the voltage-divider rule applied to loop 1.

$$V_{in1} = V_s\left(\frac{R_{in1}}{R_{in1} + R_s}\right) = 10\text{ mV}\left(\frac{100\text{ k}\Omega}{100\text{ k}\Omega + 50\text{ k}\Omega}\right) = 6.67\text{ mV}$$

The Thevenin voltage for stage 1 is

$$V_{th1} = A_{v1}V_{in1} = (35)(6.67\text{ mV}) = 233\text{ mV}$$

Compute the input voltage to stage 2 again from the voltage-divider rule, this time applied to loop 2.

$$V_{in2} = V_{th1}\left(\frac{R_{in2}}{R_{in2} + R_{th1}}\right) = 233\text{ mV}\left(\frac{47\text{ k}\Omega}{47\text{ k}\Omega + 22\text{ k}\Omega}\right) = 159\text{ mV}$$

Example 5-1

Solution

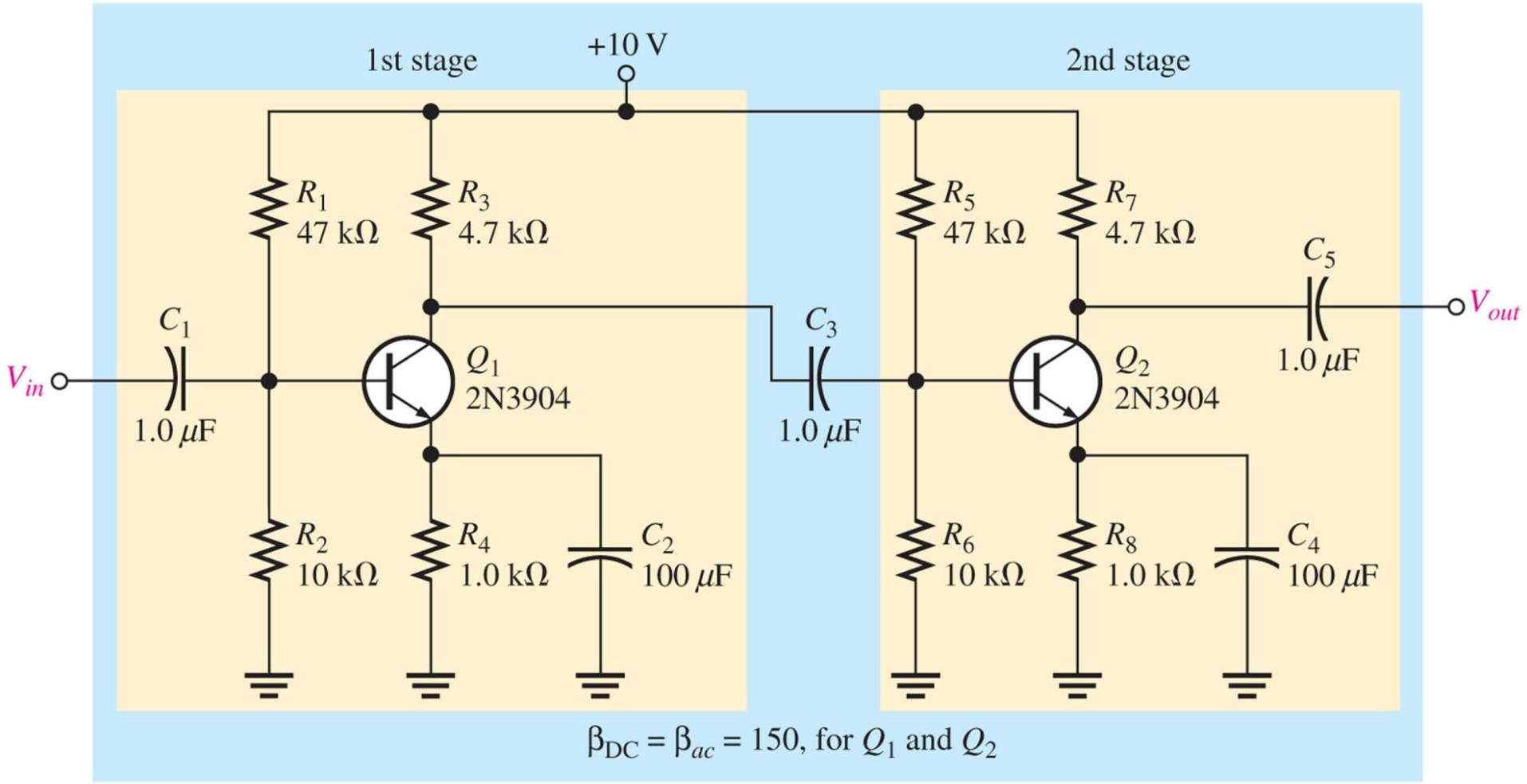
The Thevenin voltage for stage 2 is

$$V_{th2} = A_{v2}V_{in2} = (30)(159\text{ mV}) = 4.77\text{ V}$$

Apply the voltage-divider rule one more time to loop 3. The voltage across the 1.0 kΩ load is

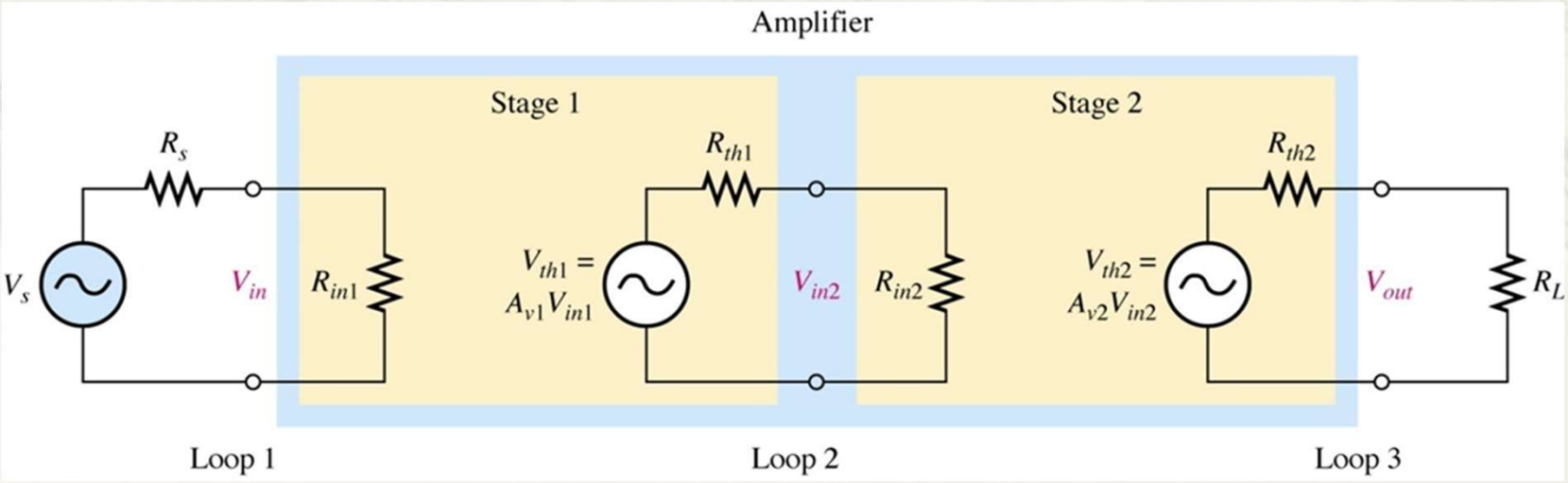
$$V_{R_L} = V_{th2}\left(\frac{R_L}{R_L + R_{th2}}\right) = 4.77\text{ V}\left(\frac{1.0\text{ k}\Omega}{1.0\text{ k}\Omega + 330\ \Omega}\right) = \mathbf{3.59\text{ V}}$$

Transistor Amplifiers



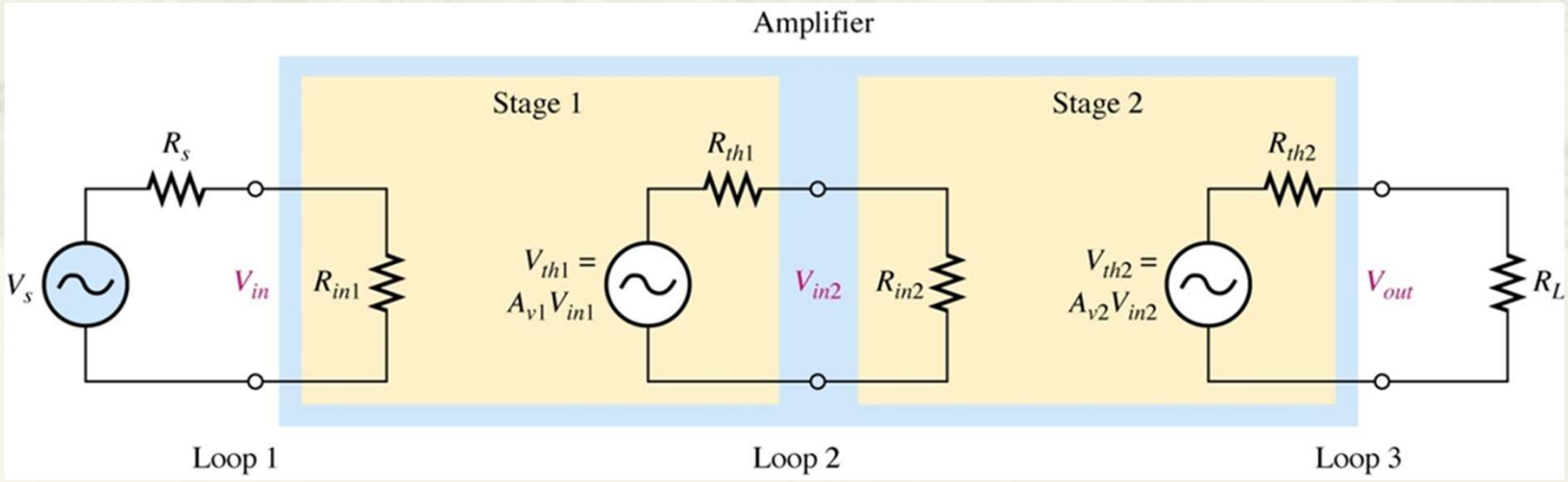
Transistor Amplifiers

- **Multi-transistor amplifier**
  - Enhance the performance of the amplifier
  - The first stage
    - ✓ High input resistance to avoid loading the source
    - ✓ Low noise for the small signal voltage



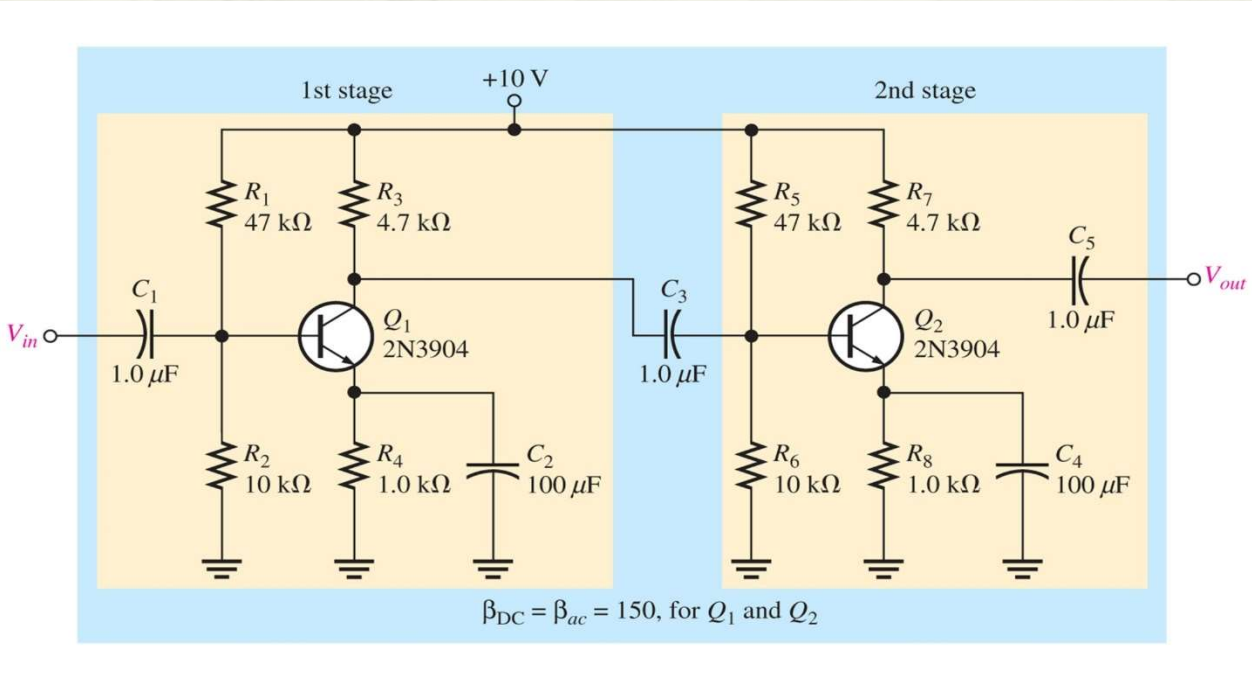
Transistor Amplifiers

- **Multi-transistor amplifier**
  - Succeeding stages
    - ✓ Increase the amplitude of the signal without adding distortion



Transistor Amplifiers

- **Multi-transistor amplifier**
  - Capacitive coupling
    - ✓ Prevents the dc bias of one stage from affecting the dc bias of another stage

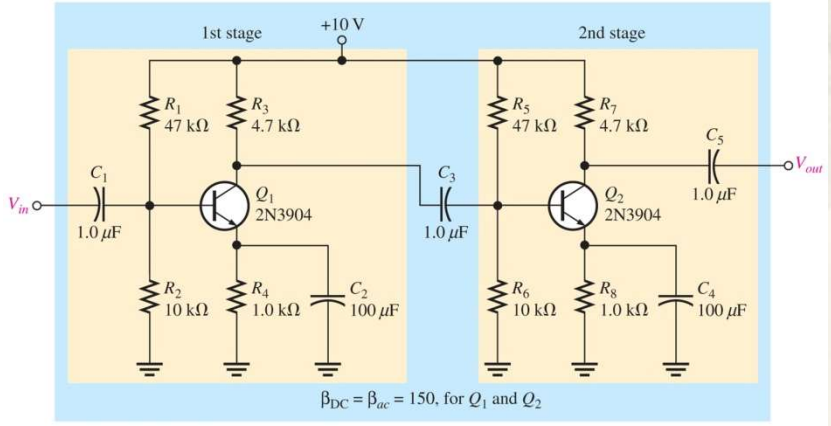


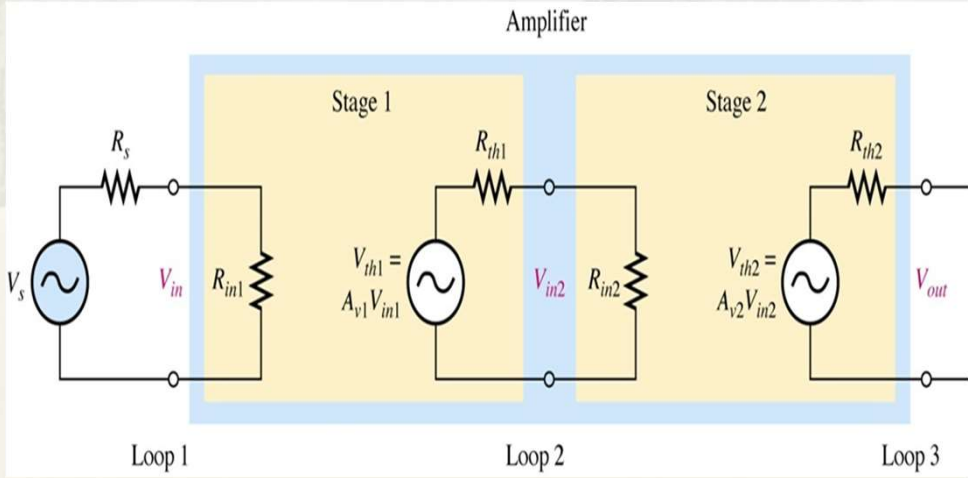
Transistor Amplifiers

- **Multi-transistor amplifier**
  - Enhance the performance of the amplifier
  - The first stage
    - ✓ High input resistance to avoid loading the source
    - ✓ Low noise for the small signal voltage
  - Succeeding stages
    - ✓ Increase the amplitude of the signal without adding distortion
  - Capacitive coupling
    - ✓ Prevents the dc bias of one stage from affecting the dc bias of another stage

Loading Effects

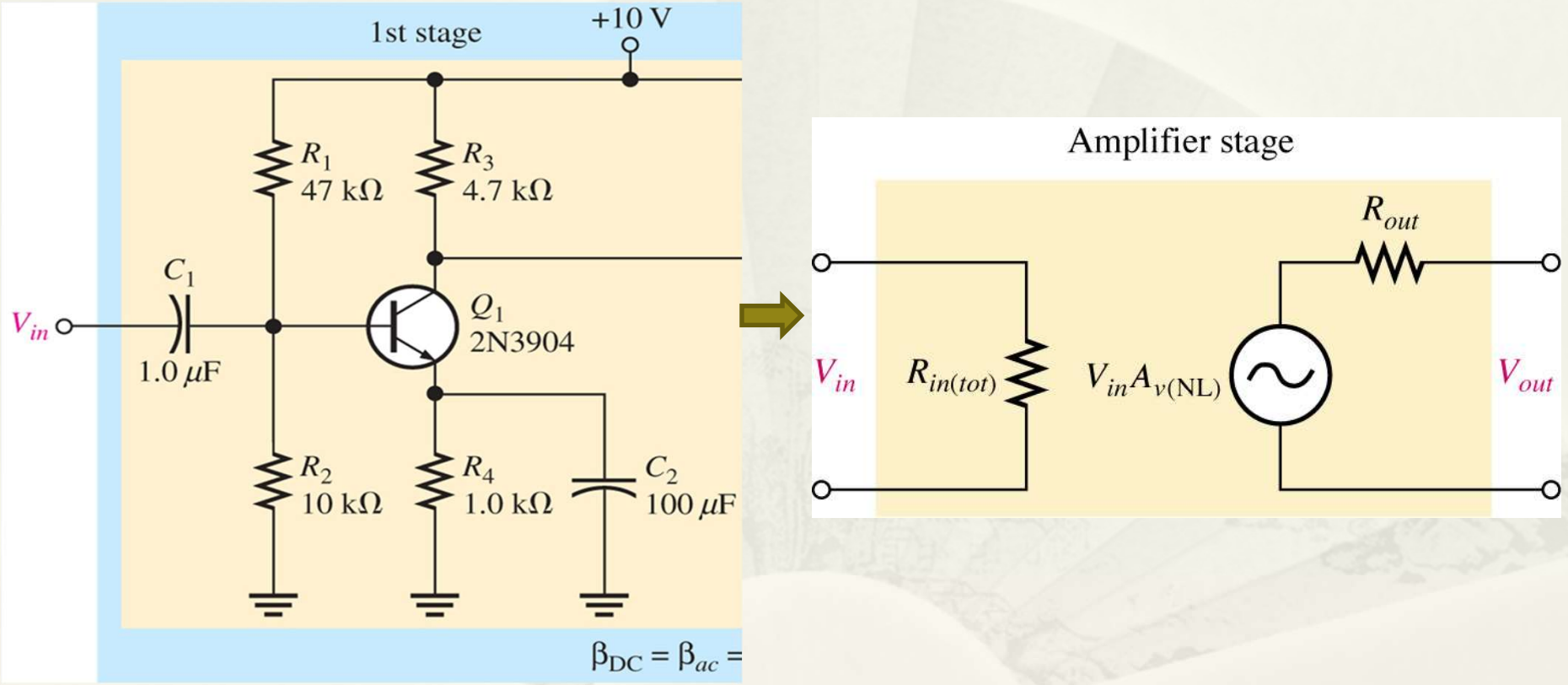
- **Compute the overall gain of the amplifier**
  - Three parameters need to be known
    - ✓ The unloaded voltage gain ($A_{v(\mathrm{NL})}$)
    - ✓ The total input resistance ($R_{in(tot)}$)
    - ✓ The output resistance (R_{out})
  - $V_{out(\mathrm{NL})} = V_{in} \cdot A_{v(\mathrm{NL})}$





Loading Effects

- **Compute the overall gain of the amplifier**

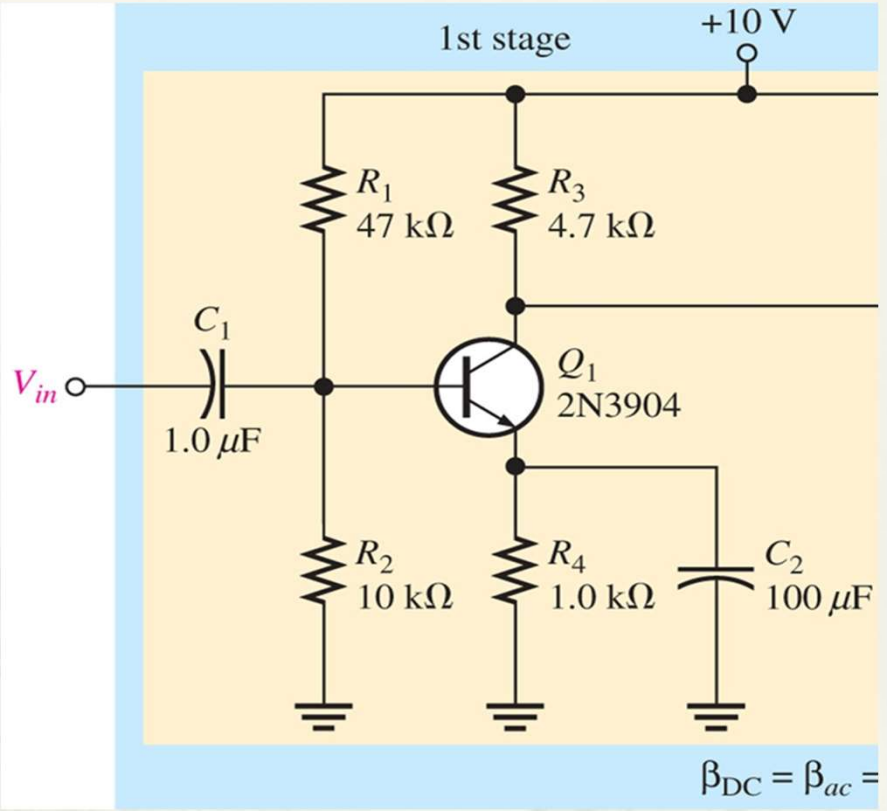


Loading Effects

- **DC operation point**
  - $V_B \cong \frac{R_2}{R_1 + R_2} V_{CC}$

    $= \frac{10\text{k}\Omega}{47\text{k}\Omega + 10\text{k}\Omega} 10\text{V} = 1.7\text{V}$
  - $V_E = V_B - V_{BE}$

    $= 1.7\text{V} - 0.7\text{V} = 1\text{V}$
  - $I_E = \frac{V_E}{R_E} = \frac{1.0\text{V}}{1.0\text{K}\Omega} = 1.0\text{mA}$
  - $r'_e \cong \frac{25\text{mV}}{I_E} = \frac{25\text{mV}}{1.0\text{mA}} = 25\Omega$



The two stages are identical.

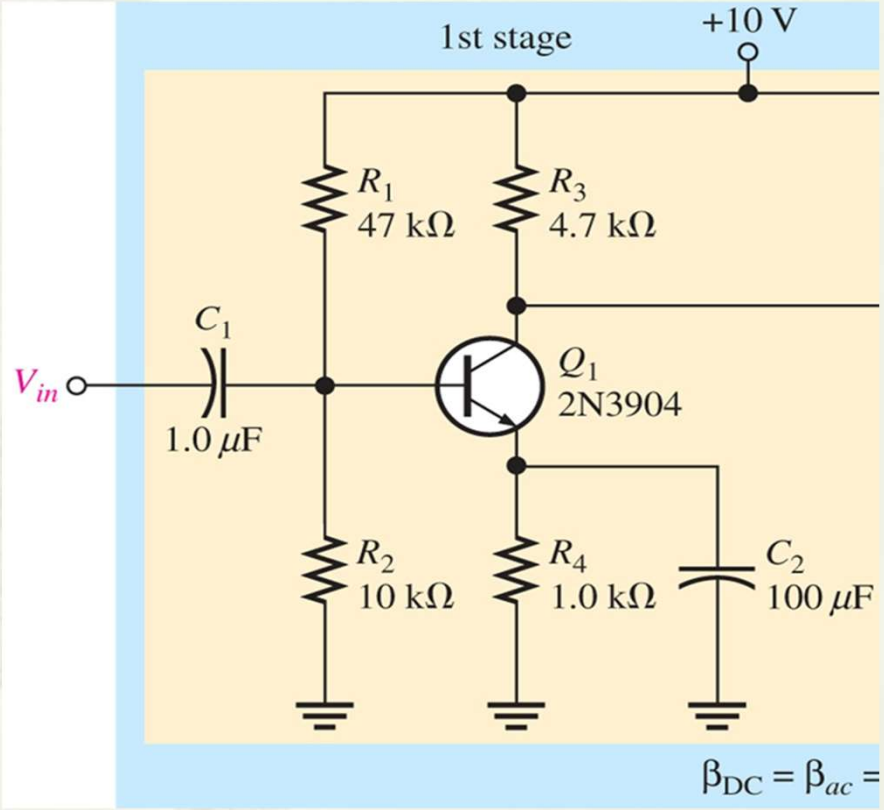
Loading Effects

- **AC parameters**
  - $A_{v(\mathrm{NL})} = -\dfrac{R_c}{R_e} = -\dfrac{R_\mathrm{C}}{r'_e}$

    $= -\dfrac{4.7\mathrm{K}\Omega}{25\Omega} = -188$
  - $R_{in(tot)} = R_1 \| R_2 \| (\beta_{ac} r'_e)$

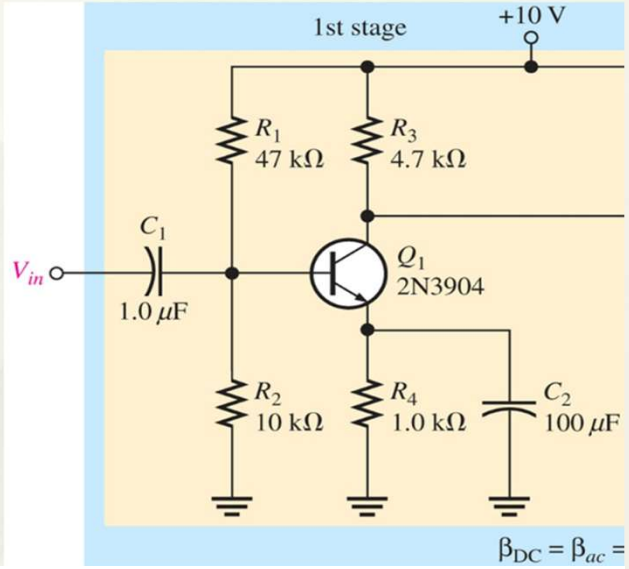
    $\cong 47\mathrm{K}\Omega \| 10\mathrm{K}\Omega \| [150(25\Omega)]$

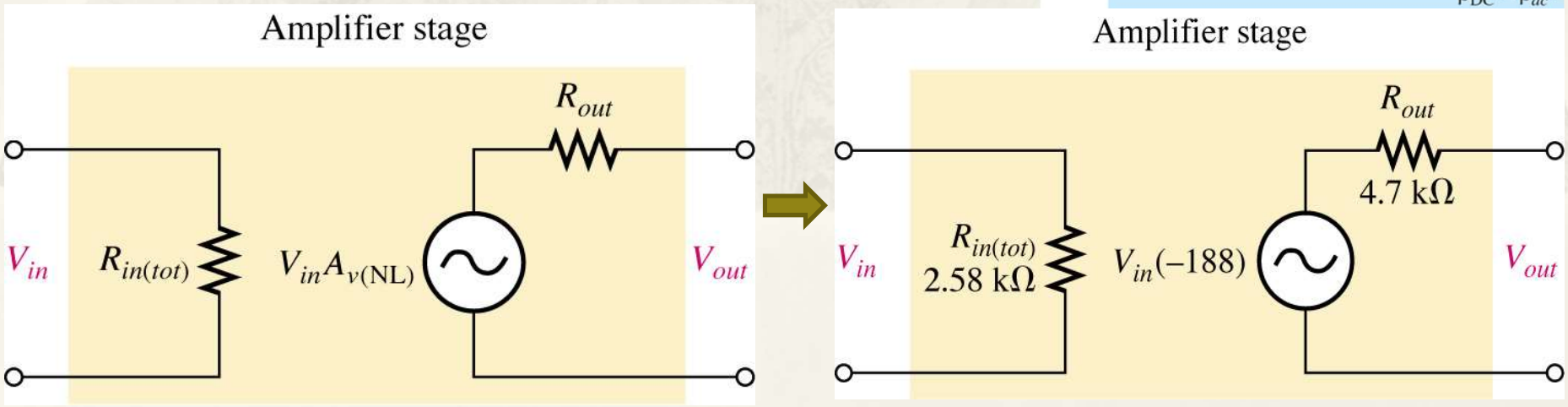
    $\cong 2.58\mathrm{K}\Omega$
  - $R_{out} = R_\mathrm{C} = 4.7\mathrm{K}\Omega$



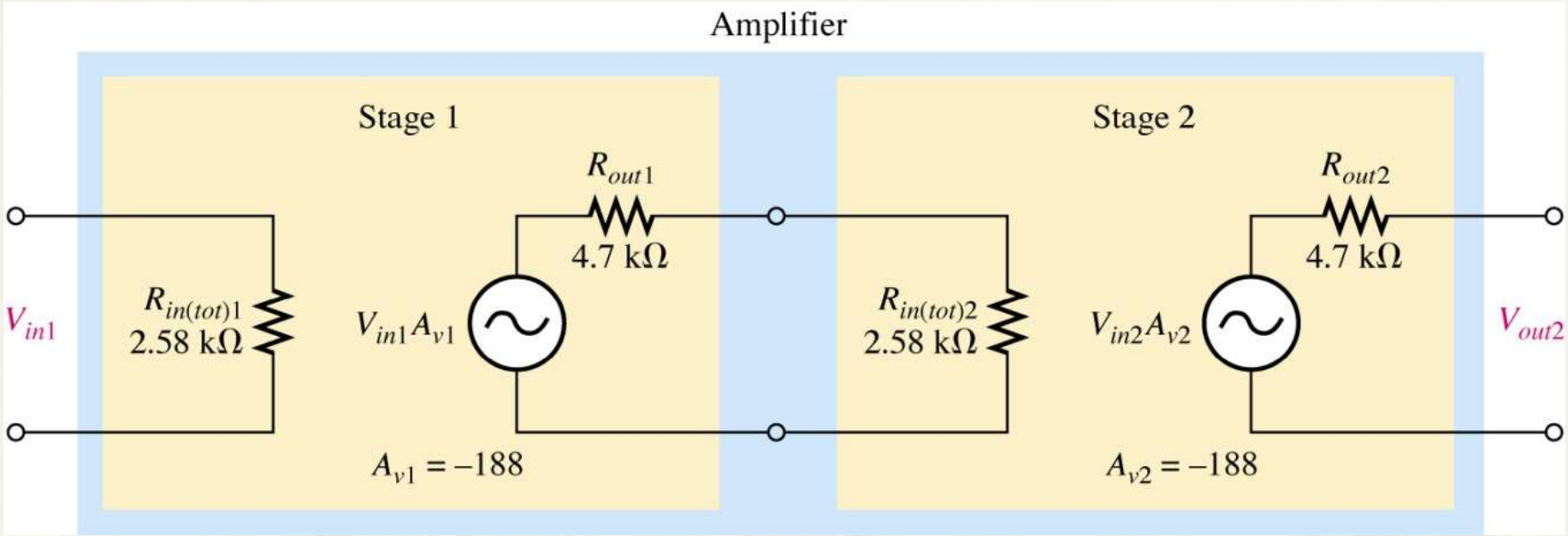
Loading Effects

- **AC parameters**
  - $A_{v(\mathrm{NL})} = -188$
  - $R_{in(tot)} \cong 2.58\mathrm{K}\Omega$
  - $R_{out} = 4.7\mathrm{K}\Omega$





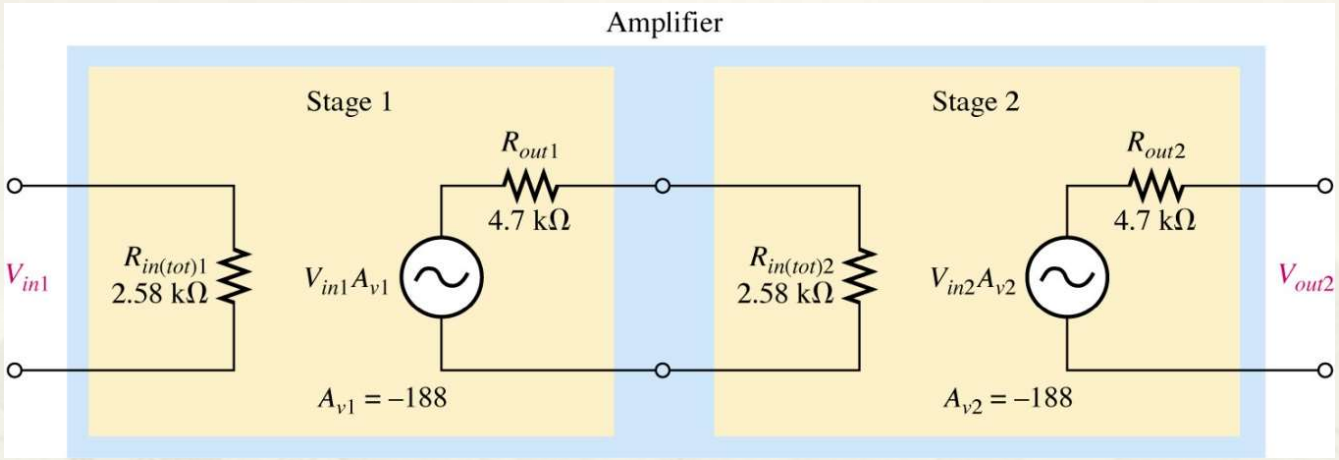
Loading Effects



- **The overall gain is the product of three terms**
  - The unloaded voltage gain of the first stage (A_{V1})
  - The gain of the voltage divider consisting of the input resistance of the second stage with the output resistance of the first stage $(A_{V(divider)})$
  - The unloaded gain of the second stage (A_{V2})

Loading Effects

- **The overall gain**



- $A_{v(divider)} = \dfrac{R_{in(tot)}}{R_{out1} + R_{in(tot)2}} = \dfrac{2.58\text{k}\Omega}{4.7\text{k}\Omega + 2.58\text{k}\Omega} = 0.35$
- $A_{v(tot)} = A_{v1}A_{v(divider)}A_{v2} = (-188)(0.35)(-188) \cong 12{,}400$

The answer is approximate because the gain is very dependent on the value of r'_e and the specific transistor used.

Unwanted Oscillation and Noise

- When large signals are present in the same circuit as small signals, the large signal can have an adverse effect on the small signal due to unwanted feedback paths.
- The ratio of signal to noise (SNR) determines if the noise is sufficient to disrupt the signal.
- FETs have the advantage for high-impedance sources; but when the source impedance is lower, (<1MΩ), BJT can provide excellent low-noise performance.

Unwanted Oscillation and Noise

- **Suggestions for avoiding noise problems**
  1. Keep wiring short to avoid “antennas” in circuit (particularly low-level input lines) and make signal return loops as small as possible.
  2. Use capacitors between power supply and ground at each stage and make sure the power supply is properly filtered.
  3. Reduce noise source, if possible, and separate or shield the noise source and the circuit. Use shield wiring, twisted pair, or shielded twisted pair wiring for low-level signals.

Unwanted Oscillation and Noise

- **Suggestions for avoiding noise problems**

4. Ground circuits at a single point, and isolate grounds that have high currents from those with low currents by running separate ground lines back to the single point. Ground current from a high current ground can generate noise in another part of a circuit because of *IR* drops in the conductive paths.
5. Keep the bandwidth of amplifiers no larger than necessary to amplify only the desired signal, not extra noise.

5-1-2 Transformer-Coupled Amplifiers

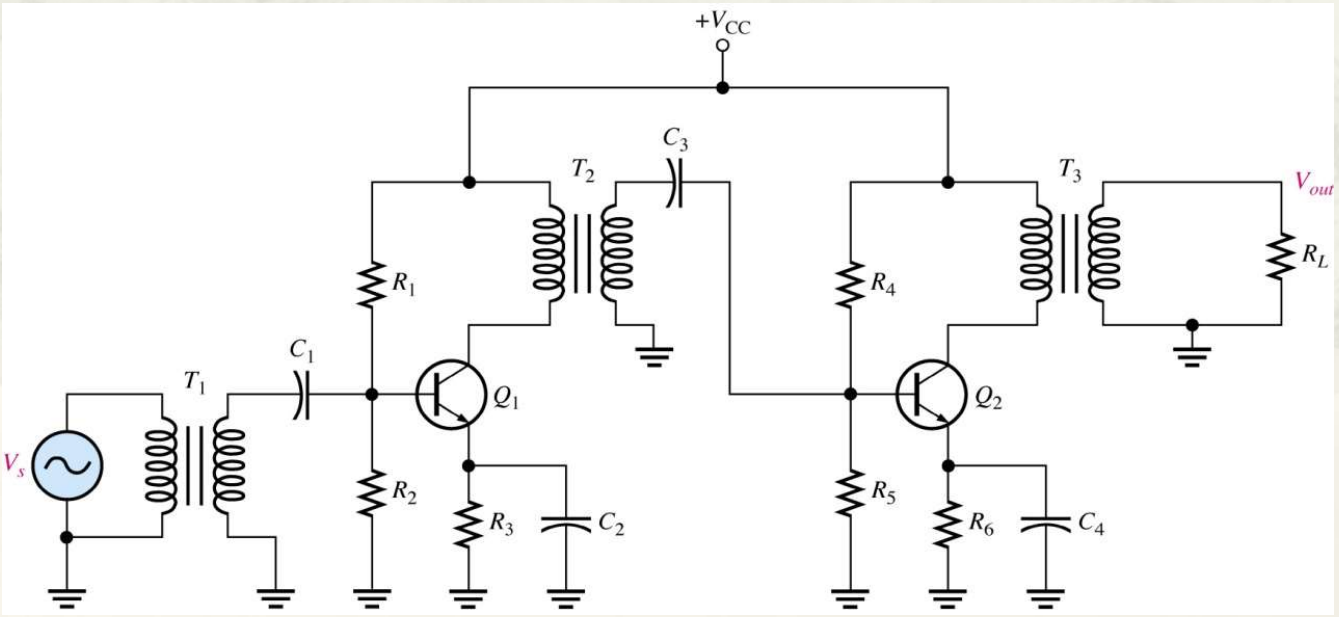
- Low-Frequency Applications
- High-Frequency Applications
- Tuned Amplifiers

Low-Frequency Applications

- **Block dc and pass ac**
- **Descript with impedance**
  - Combination of reactance and resistance
  - Form ac opposition
- **Match the impedance of one part to another**
  - $R'_{\mathrm{L}} = \left(\frac{N_{pri}}{N_{sec}}\right)^2 R_{\mathrm{L}}$
    - ✓ R'_{L}: the reflected resistance on the primary side
    - ✓ N_{pri}/N_{sec}: the ratio of primary turns to secondary turns
    - ✓ R_{L}: the load resistance on the secondary side

Low-Frequency Applications

- **Usage of transformer coupling**
  - **At the input:** matching the impedance of a source
  - **At the output:** matching the impedance of the load, for maximum power
  - **Between stages:** coupling the ac signal from one part of a circuit to another



Low-Frequency Applications

- **Major drawbacks of transformer coupling**
  - Transformers are more expensive and much bulkier than capacitors.
  - Transformers tend to have poorer response at high frequencies due to the reactance of the coils.

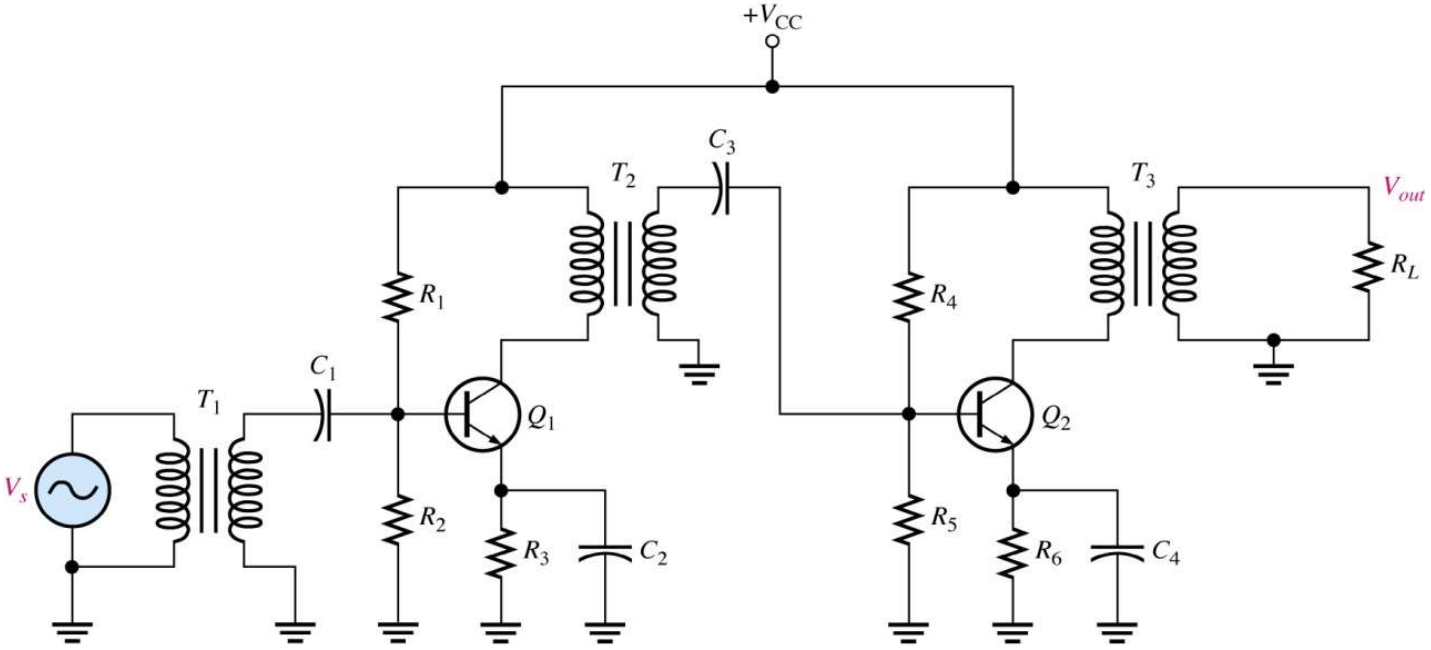
Example 5-2

Problem

Assume the component values for the second stage of Figure 5–8 are as follows: $R_4 = 5.1\,\text{k}\Omega$, $R_5 = 2.7\,\text{k}\Omega$, $R_6 = R_E = 680\,\Omega$, and $R_L = 50\,\Omega$; transformer T_3 is a 5:1 step-down transformer and $V_{CC} = 12$ V.

(a) Find V_{CE}.

(b) Calculate the gain of the second stage.



Example 5-2

Solution

(a) Find the base voltage by applying the voltage-divider rule to the bias resistors.

$$V_B = \left(\frac{R_5}{R_4 + R_5}\right)V_{CC} = \left(\frac{2.7\,\text{k}\Omega}{5.1\,\text{k}\Omega + 2.7\,\text{k}\Omega}\right)12\,\text{V} = 4.2\,\text{V}$$

Next, calculate the emitter voltage and current.

$$V_E = V_B - V_{BE} = 4.2\,\text{V} - 0.7\,\text{V} = 3.5\,\text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{3.5\,\text{V}}{680\,\Omega} = 5.15\,\text{mA}$$

$$V_{CE} \cong V_{CC} - V_E = 12\,\text{V} - 3.5\,\text{V} = \mathbf{8.5\,V}$$

Example 5-2

Solution

(b) Calculate the reflected resistance of the load resistor on the primary side and the ac emitter resistance.

$$R'_L = \left(\frac{N_{pri}}{N_{sec}}\right)^2 R_L = \left(\frac{5}{1}\right)^2 50\ \Omega = 1.25\ \text{k}\Omega$$

$$r'_e = \frac{25\ \text{mV}}{I_E} = \frac{25\ \text{mV}}{5.15\ \text{mA}} = 4.85\ \Omega$$

Calculate the gain by dividing the ac resistance in the collector circuit by the ac resistance of the emitter circuit.

$$A_v = \frac{R'_L}{r'_e} = \frac{1.25\ \text{k}\Omega}{4.85\ \Omega} = \mathbf{257}$$

Low-Frequency Applications

- **The load line**
  - The ac load line of transformer-couple is not as steep as the dc load line.
  - The ac saturation current is lower than the dc saturation current.
  - The ac cutoff voltage is larger than the dc cutoff voltage.

High-Frequency Applications

- **Radio Frequency (RF)**
  - Frequencies greater than 100kHz, used for carrying radio signals
  - Transformers are much smaller, less expensive
- **Parallel resonant circuit**
  - Connect transformer with a parallel capacitor
  - Impedance maximum at the resonant frequency
  - The gain of amplifier can be very high (as high as 1000) at frequencies near the resonant frequency (typically 10kHz bandwidth) while offering little opposition to dc.

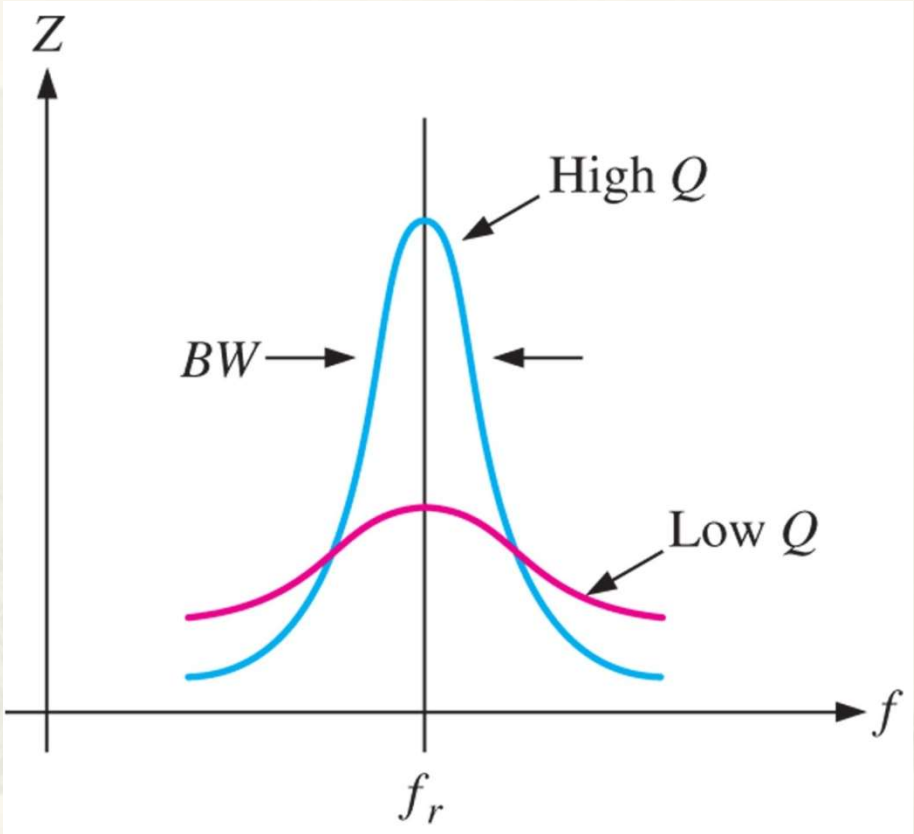
Tuned Amplifiers

- **Tuned amplifier**
  - Be designed to amplify a specific band of frequencies while eliminating any signals outside the band.
  - Use a parallel *LC* resonant circuit for a load
  - The center frequency: $f_r = \frac{1}{2\pi\sqrt{LC}}$
  - The bandwidth: determined by the Q (quality factor) of the resonant circuit

Tuned Amplifiers

Quality factor (Q)

- The quality factor (Q) is a dimensionless number that is the ratio of the maximum energy stored in a cycle to the energy lost in a cycle.
- $Q = \frac{X_L}{R} = \frac{f_r}{BW}$
- Range from 50 to 250 for ferrite-core inductors and even higher for air-core inductors



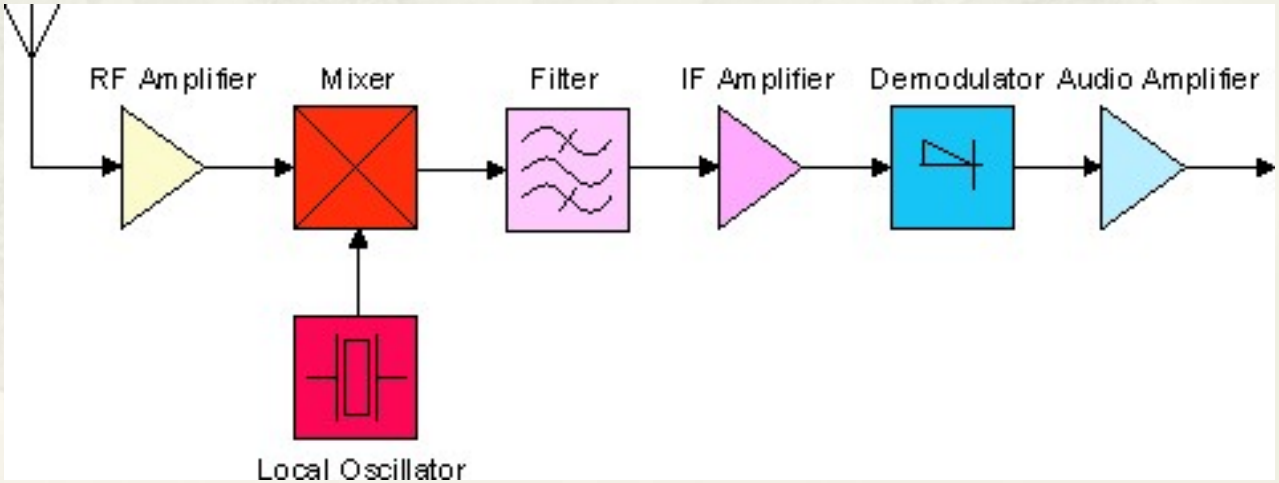
Impedance of a parallel resonant circuit as a function of frequency

Tuned Amplifiers



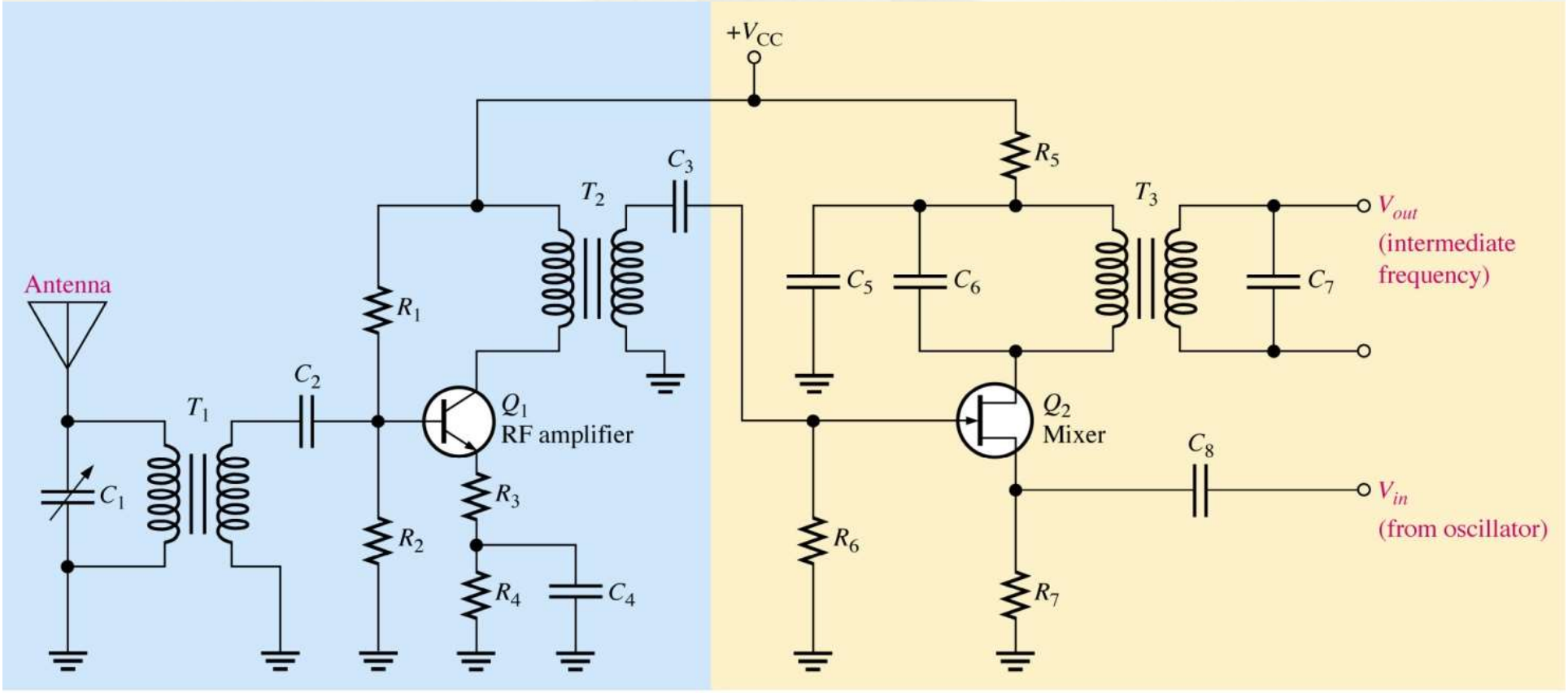
Edwin Howard Armstrong
(1890 – 1954)

- **Superheterodyne receiver (superhet)**
  - Local oscillator (RF tracker)
  - Mixer (nonlinear circuit)
  - Intermediate frequency (IF)



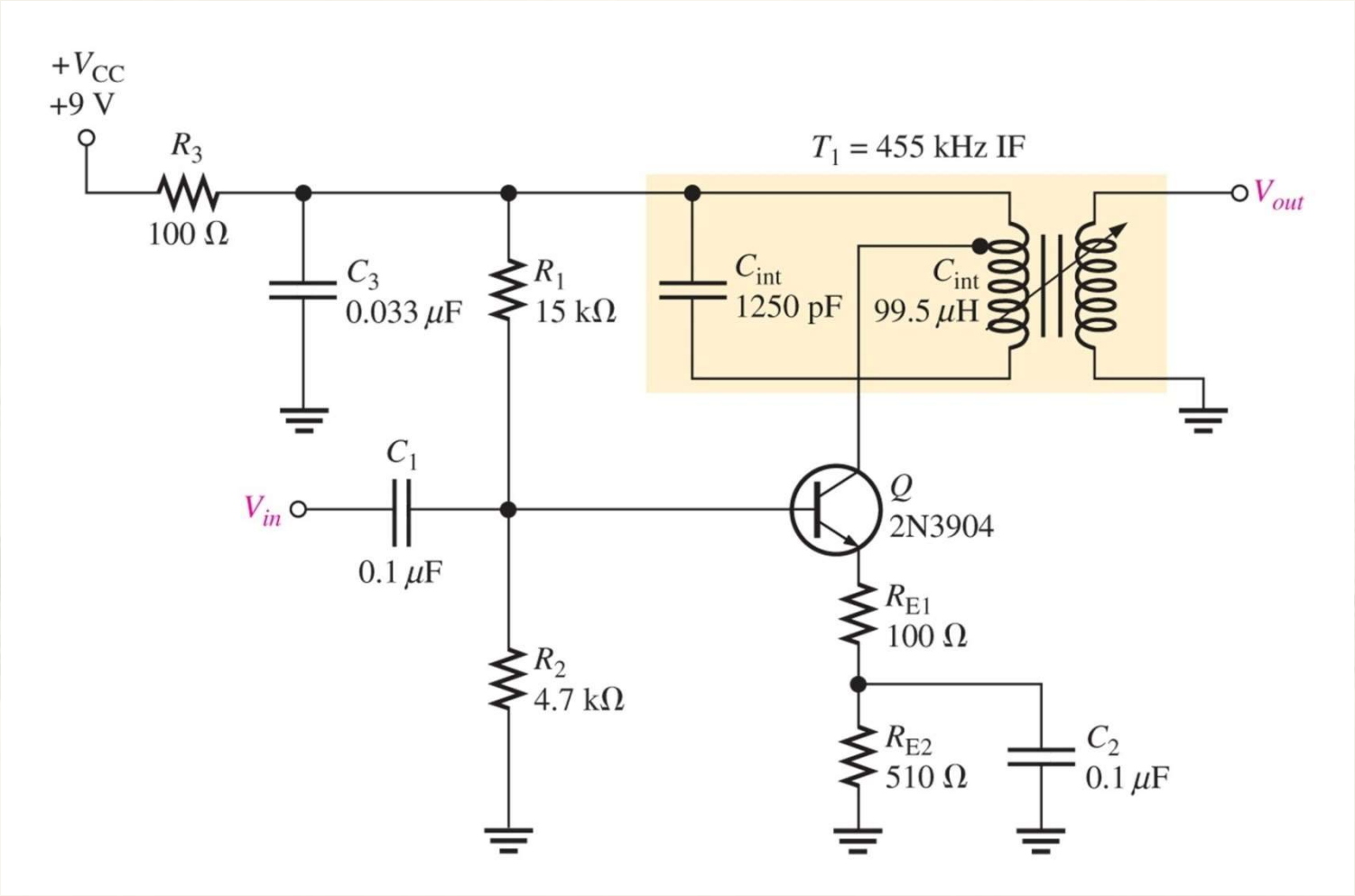
Tuned Amplifiers

- A tuned amplifier consisting of an RF stage and a mixer

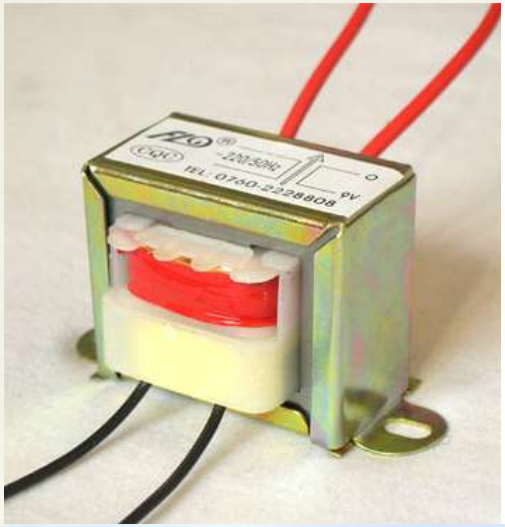


Tuned Amplifiers

- **An IF Amplifier**

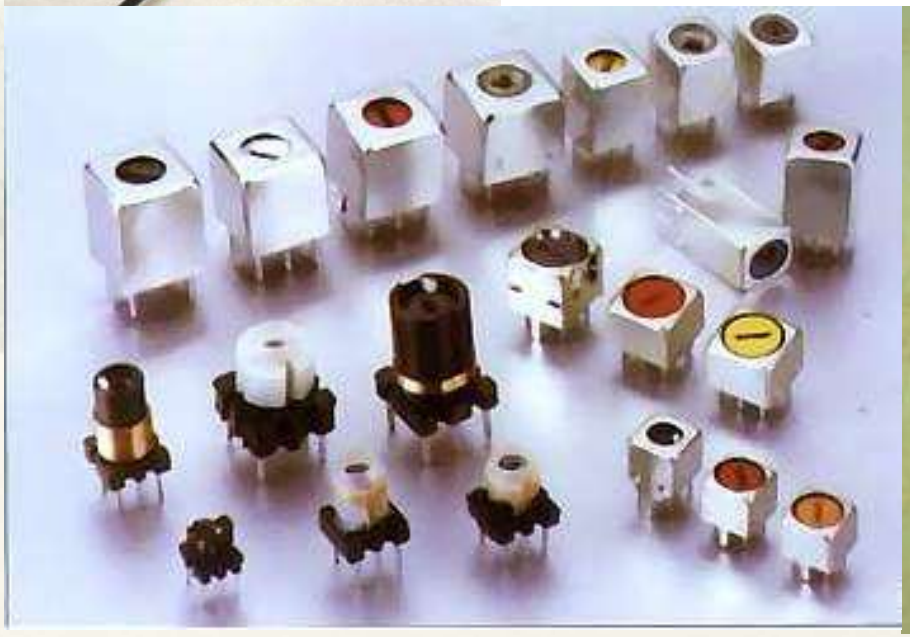


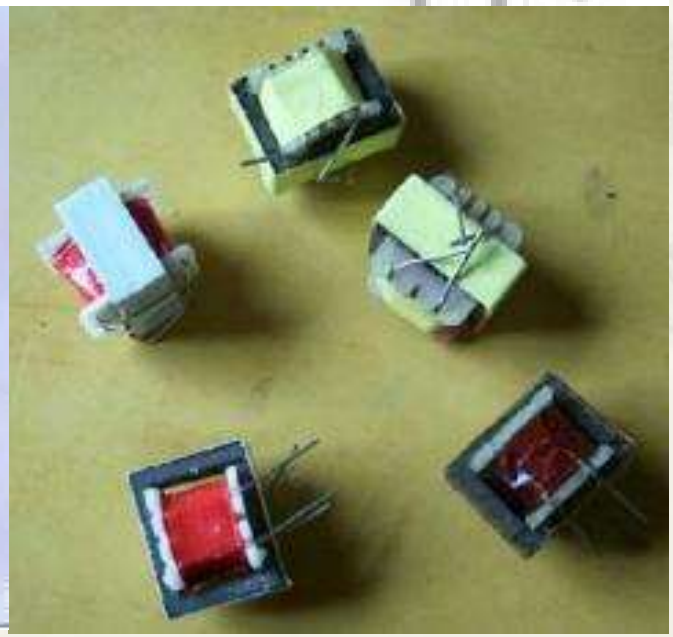
Transformers





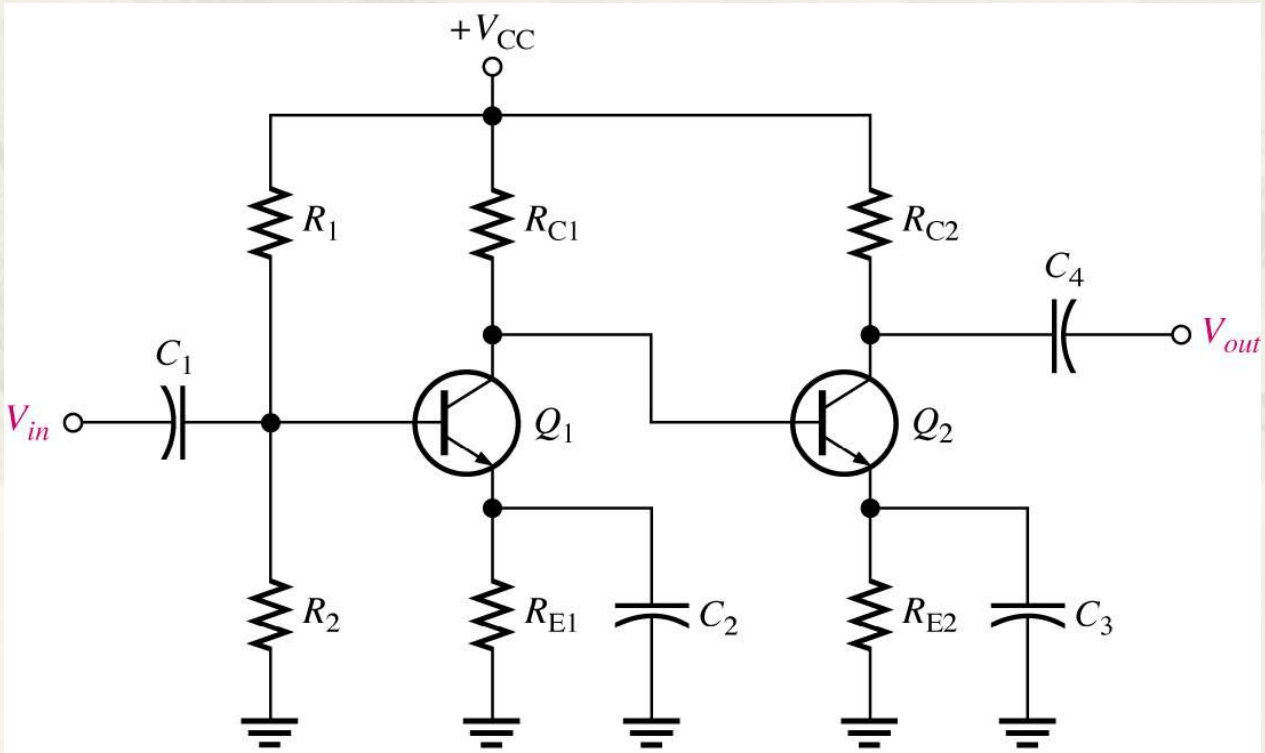






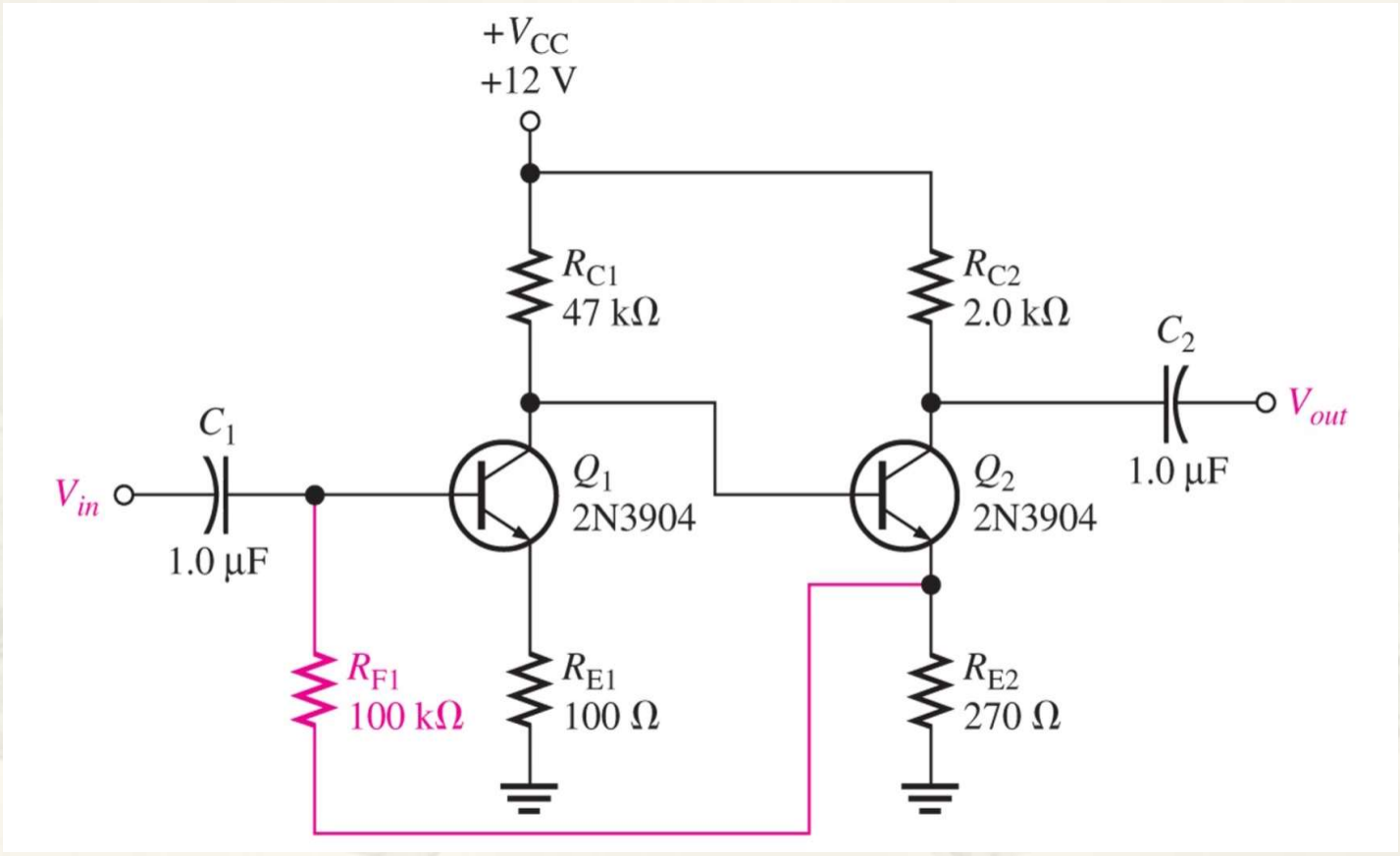
5-1-3 Direct-Coupled Amplifiers

- **No coupling capacitors or transformers between stages.**
  - Advantages: fewer components
  - Drawbacks: bias for Q_2 is not stable



5-1-3 Direct-Coupled Amplifiers

- **The direct-coupled amplifier with feedback**



Negative feedback for bias and gain stability

5-1-3 Direct-Coupled Amplifiers

- **The differential amplifier**

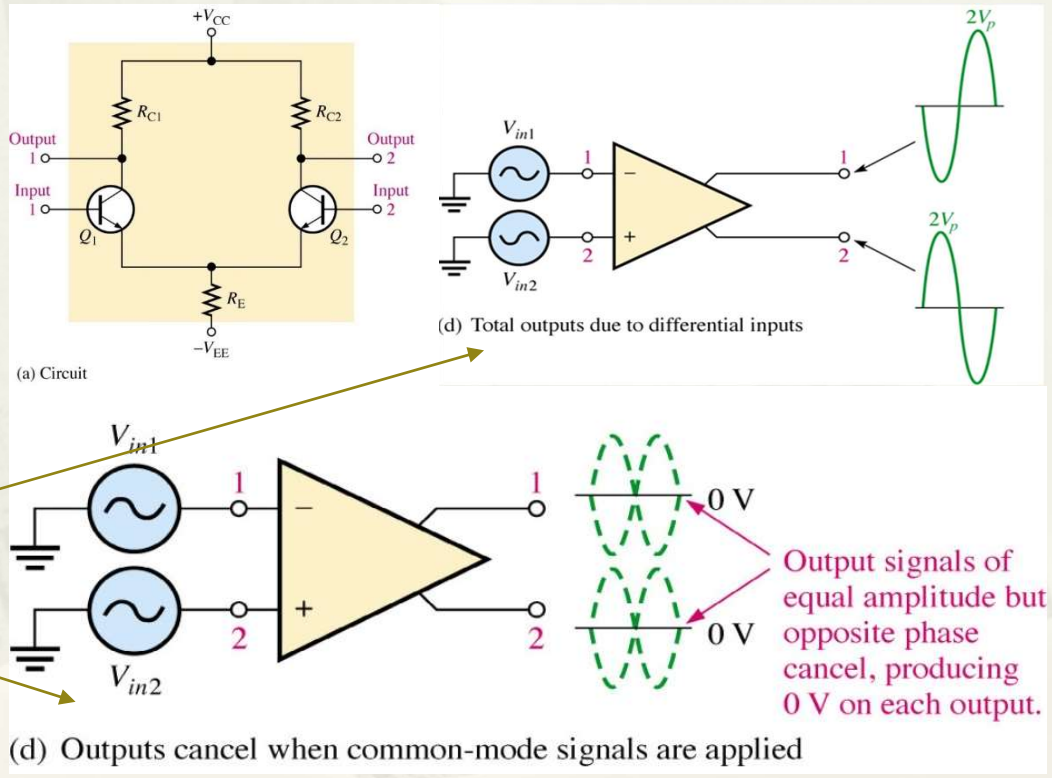
Ideally

1. overcome zero drift

2. zero input/output

3. amplify differential signals

4. reject common-mode signals



5-2 Power Amplifiers

5-2-1 Class A Power Amplifiers

5-2-2 Class B Power Amplifiers

5-2-1 Class A Power Amplifiers

- Heat Dissipation
- Centered Q-point
- Power Gain
- DC Quiescent Power
- Output Power
- A Two-Stage Class A Amplifier
- Efficiency

5-2-1 Class A Power Amplifiers

- **Amplifiers for different types of signals**
  - Small-signal type
  - Large-signal type
- **Class A amplifier**
  - Bias: linear region
  - Output: an amplified replica of the input signal – non-distortion
- **Power amplifier**
  - Objective: delivering power to a load.
  - Large-signal amplifier
  - a rule of thumb: an amplifier may be considered to be a power amplifier if it is necessary to consider the problem of heat dissipation in components (>1/4W).

Heat Dissipation

- **Heat dissipation**
  - Power transistors and other power devices must dissipate excessive internally generated heat.
  - For bipolar power transistors, the collector terminal is the critical junction
- **Heat sink**
- **Cooling fan**



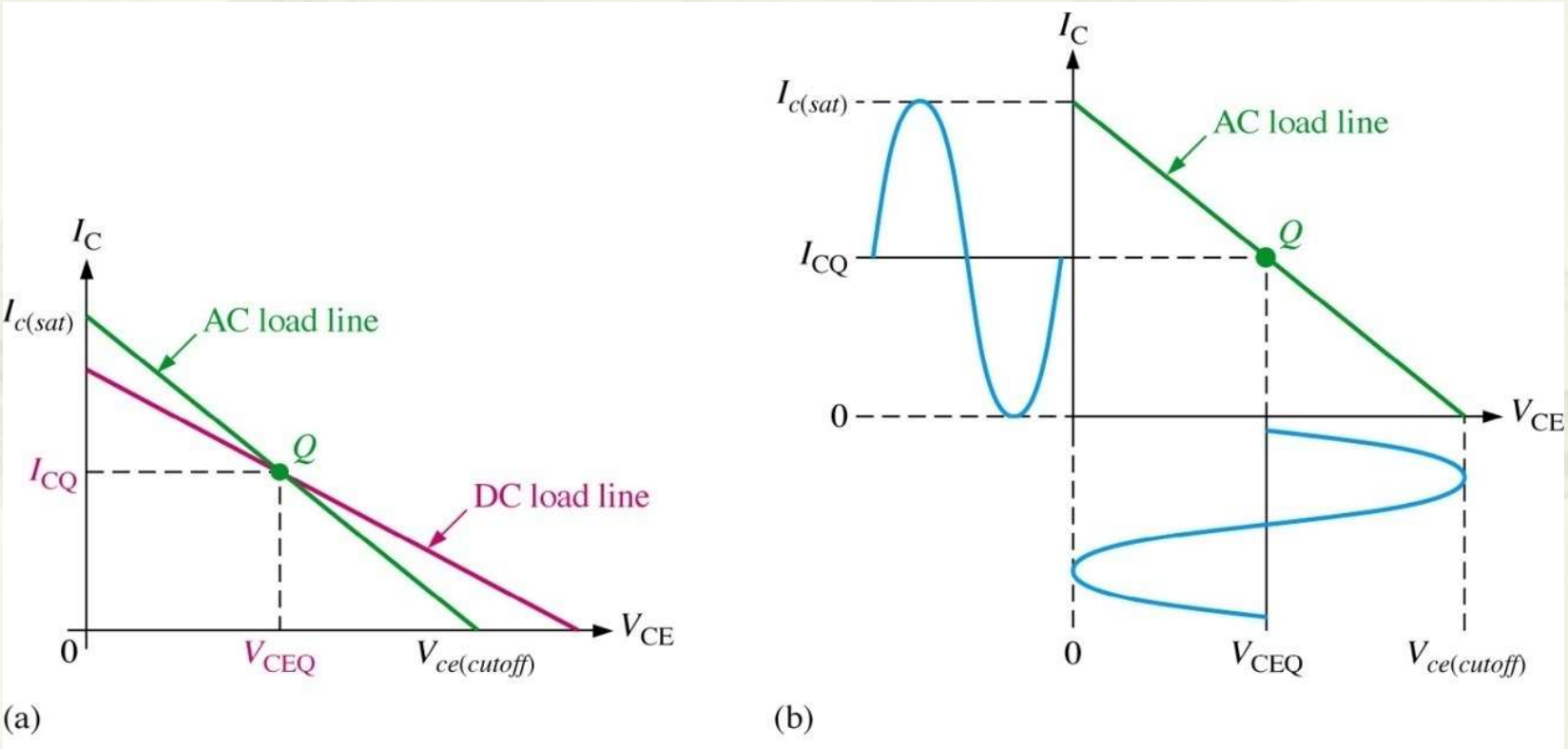
the case is always connected to the collector terminal





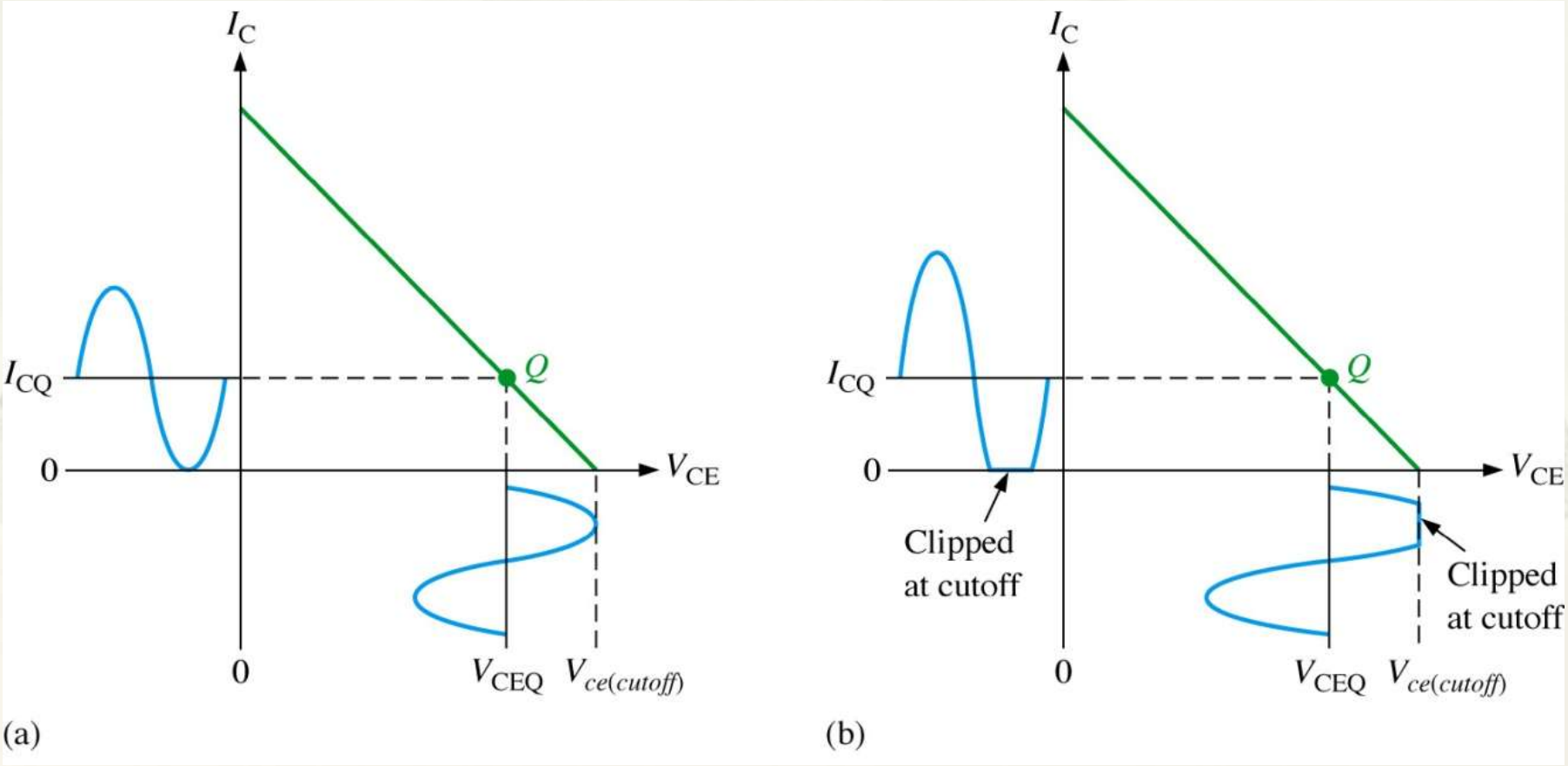
Centered Q-point

- **Centered Q-point**
  - Maximum class A output occurs when the Q-point is at the center of the ac load line.



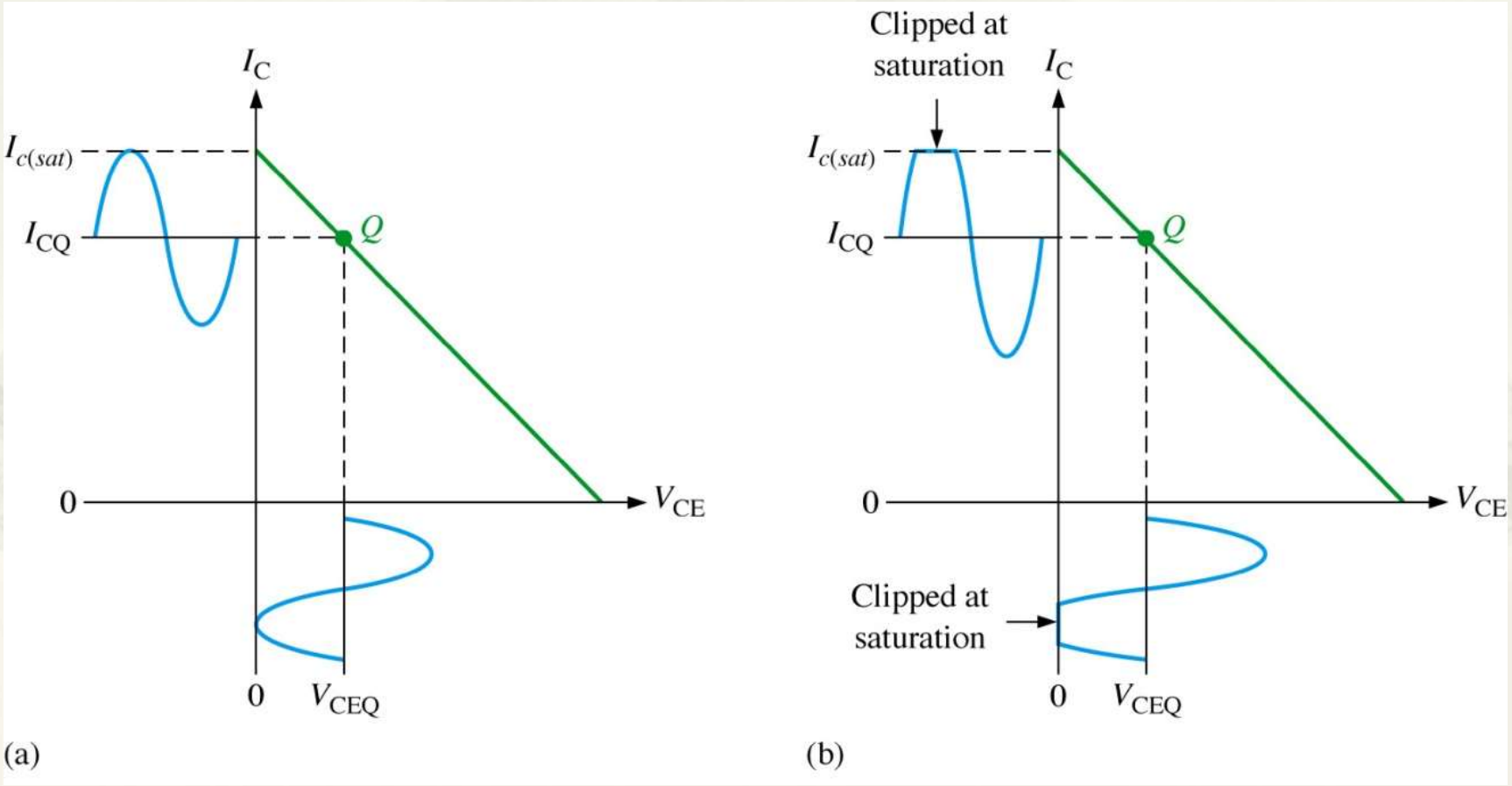
Centered Q-point

- **Centered Q-point**
  - Q-point closer to cutoff



Centered Q-point

- **Centered Q-point**
  - Q-point closer to saturation



Power Gain

- **Power gain**
  - The power gain of an amplifier is the ratio of the power delivered to the load to the input power.
  - $A_p = \frac{P_{out}}{P_{in}} = \frac{P_L}{P_{in}}$
  - $A_p = \frac{V_L^2}{V_{in}^2}\left(\frac{R_{in}}{R_L}\right) = A_v^2\left(\frac{R_{in}}{R_L}\right)$

$P_L = {V_L}^2/R_L$

$P_{in} = {V_{in}}^2/R_{in}$

For a CC amplifier, A_p is approximately equal to the ratio of the input resistance to the output load resistance.

DC Quiescent Power

DC Quiescent Power

- The power dissipation of a transistor with no signal input is the product of its Q-point current and voltage.

$$P_{\mathrm{DQ}} = I_{\mathrm{CQ}} V_{\mathrm{CEQ}}$$

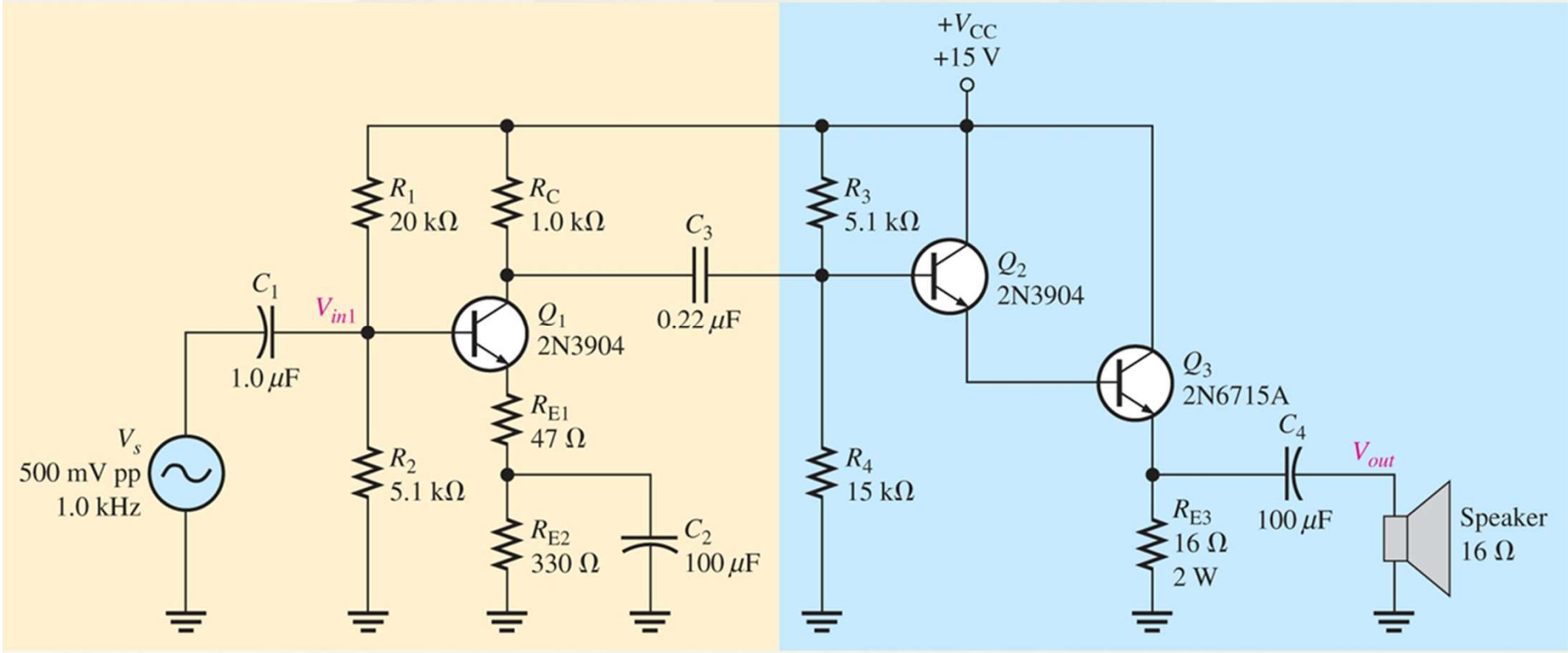
- A class A power amplifier should maintain a quiescent current as large as the peak current.
- A signal will not increase the power dissipation by the transistor but actually causes less total power to be dissipated.
- The transistor's power rating should normally exceed the quiescent power.

Output Power

- **Output power**
  - In general, the product of the rms load current and the rms load voltage.
  - The maximum unclipped ac signal occurs when the Q-point is centered on the ac load line.
  - $P_{out(max),rms} = \left(\frac{\sqrt{2}}{2} I_{c,peak}\right)\left(\frac{\sqrt{2}}{2} V_{c,peak}\right) = \frac{I_{\mathrm{CQ}} V_{\mathrm{CEQ}}}{2}$

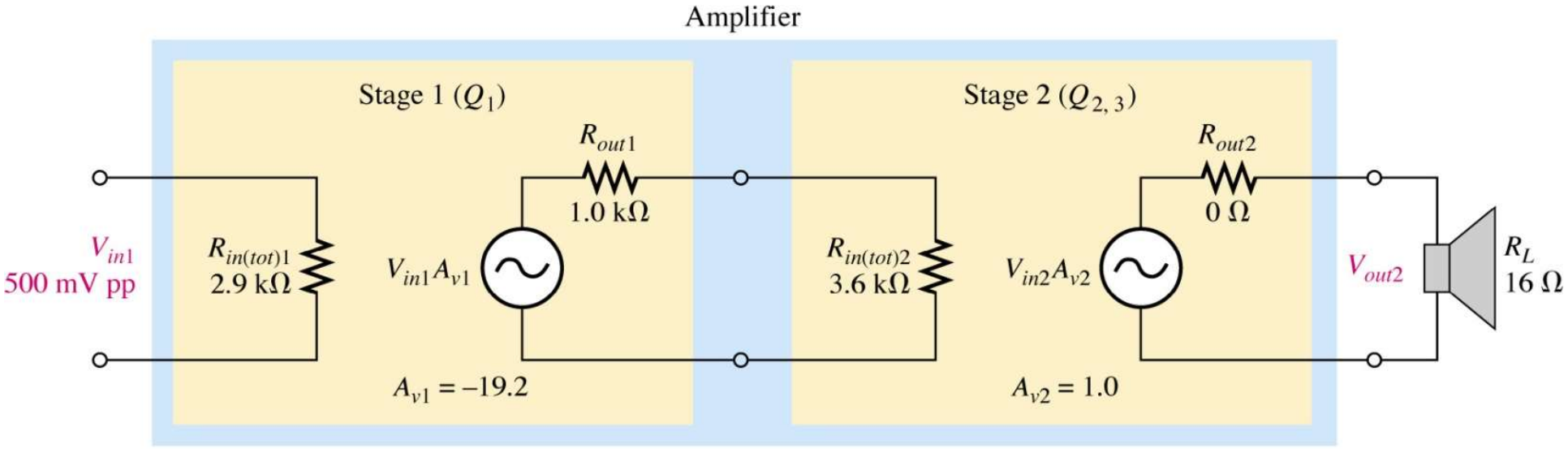
A Two-Stage Class A Amplifier

- **Darlington pair**
  - Two cascaded transistors connected so that the emitter of the first drives the base of the second.



A Two-Stage Class A Amplifier

- **Amplifier model**
  - $A_v = A_{v1} \cdot A_{v2} \cdot \dfrac{R_{in(tot)2}}{R_{out1} + R_{in(tot)2}} = -19.2 \dfrac{3.6\text{k}\Omega}{1.0\text{k}\Omega + 3.6\text{k}\Omega} = -15$
  - $A_p = A_v^2 \left(\dfrac{R_{in}}{R_L}\right) = (-15)^2 \left(\dfrac{2.9\text{k}\Omega}{16\Omega}\right) = 41000$



Efficiency

Efficiency

- is the ratio of the signal power supplied to the load to the power from the dc supply.
- The dc power: $P_{\mathrm{DC}} = I_{\mathrm{CC}}V_{\mathrm{CC}} = 2I_{\mathrm{CQ}}V_{\mathrm{CEQ}}$
- The maximum efficiency of a capacitance coupled load:

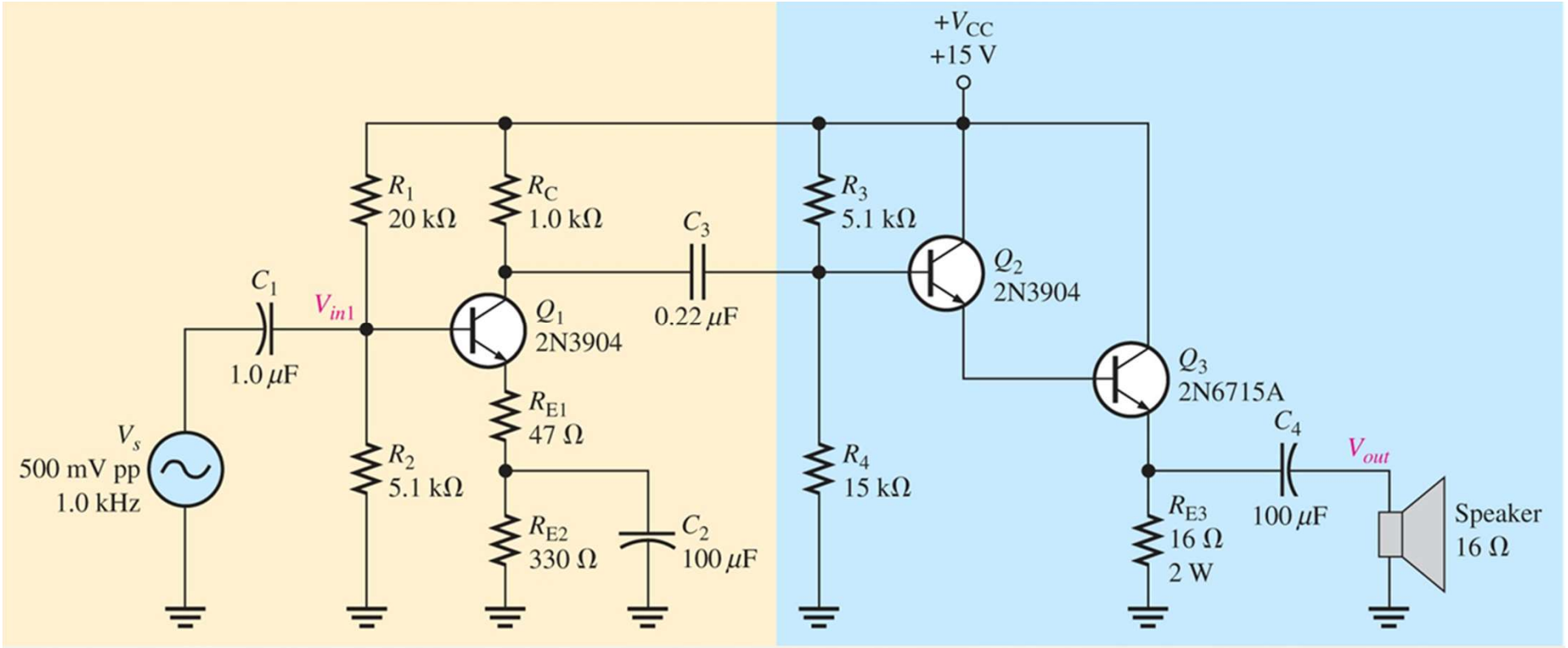
$$eff_{max} = \frac{P_{out}}{P_{\mathrm{DC}}} = \frac{0.5I_{\mathrm{CQ}}V_{\mathrm{CEQ}}}{2I_{\mathrm{CQ}}V_{\mathrm{CEQ}}} = 0.25$$

The maximum efficiency of a capacitance coupled class A amplifier cannot be higher than 0.25, or 25%, and, in practice, is usually considerably less (about 10%).

Example 5-3

Problem

Assume the dc supply for the amplifier in Figure 5–19 has a current of 0.6 A. Determine the efficiency of the power amplifier.



Example 5-3

Solution

The efficiency is the ratio of the signal power in the load to the power supplied by the dc source. The input voltage is 500 mV peak-to-peak which is 176 mV rms. The input power is, therefore,

$$P_{in} = \frac{V_{in}^2}{R_{in}} = \frac{(176\text{ mV})^2}{2.9\text{ k}\Omega} = 10.7\ \mu\text{W}$$

The output power is

$$P_{out} = P_{in}A_p = (10.7\ \mu\text{W})(41{,}000) = 0.44\text{ W}$$

The power from the power supply is

$$P_{DC} = I_{CC}V_{CC} = (0.6\text{ A})(15\text{ V}) = 9\text{ W}$$

Example 5-3

Solution

Therefore, the efficiency of the amplifier for this input is

$$eff = \frac{P_{out}}{P_{DC}} = \frac{0.44\ \text{W}}{9\ \text{W}} \cong \mathbf{0.05}$$

This represents an efficiency of 5%.

Question

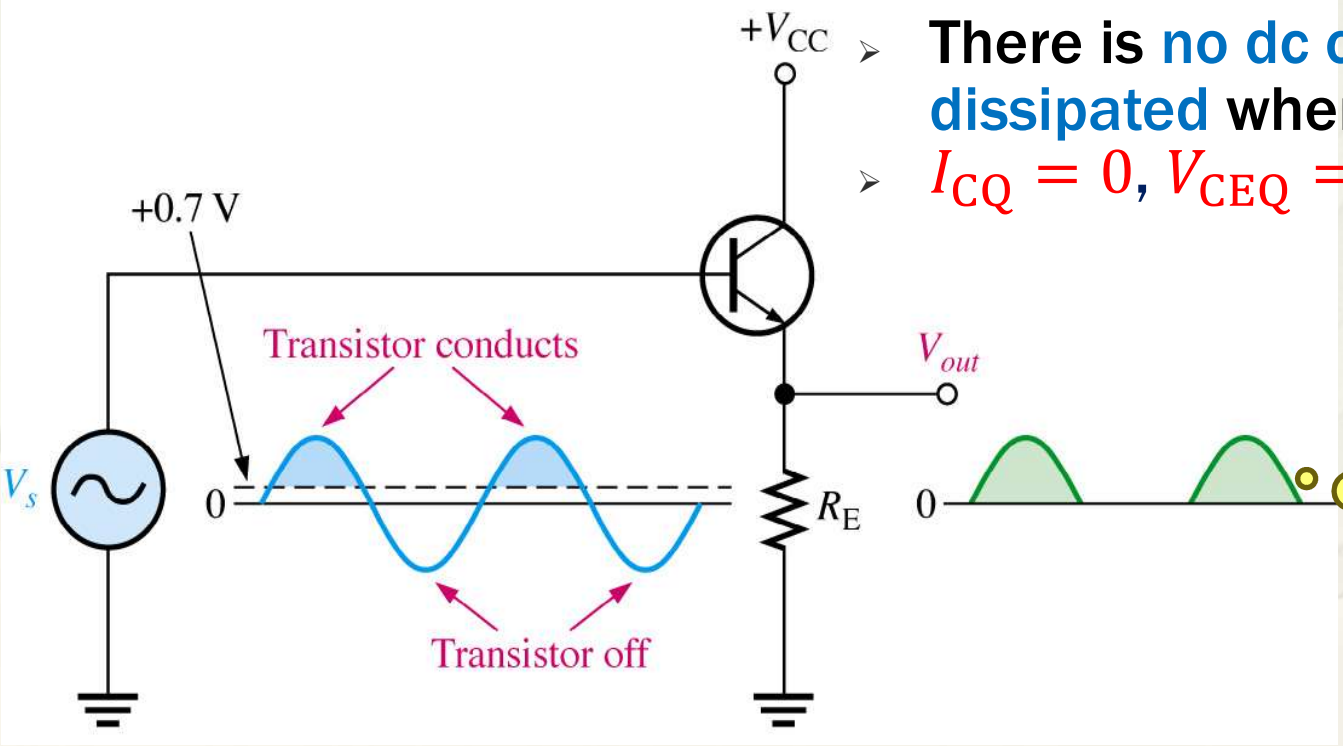
What happens to the efficiency if R_{E3} were replaced with the speaker? What disadvantage does this have?

5-2-2 Class B Power Amplifiers

- The Q-point Is at Cutoff
- Push-Pull Operation
- Crossover Distortion
- Biasing the Push-Pull Amplifier
- AC Operation
- Single-Supply Operation
- Different Types of Class AB Power Amplifiers

The Q-point Is at Cutoff

- **Bias**: operates in the linear region for half of the input cycle and is in cutoff for another half.
- **Operation**: when the Q-point is located at cutoff, causing the output current to vary only during one-half of the input cycle.

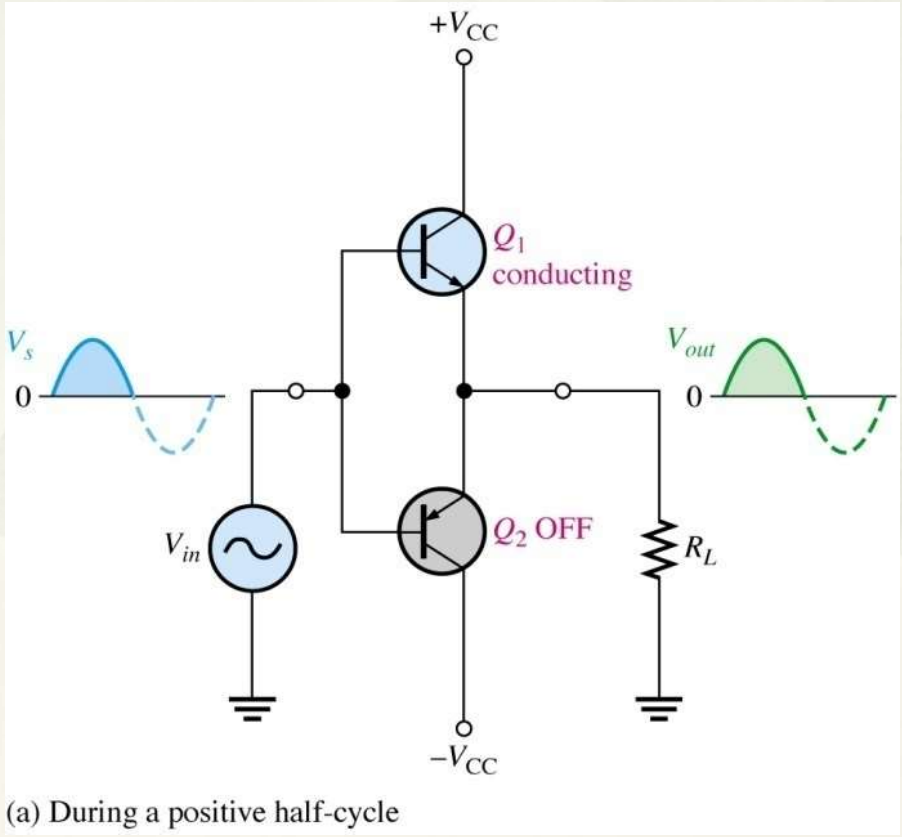


- There is no dc current or power dissipated when there is no signal.
- $I_{CQ} = 0, V_{CEQ} = V_{CE(cutoff)}$

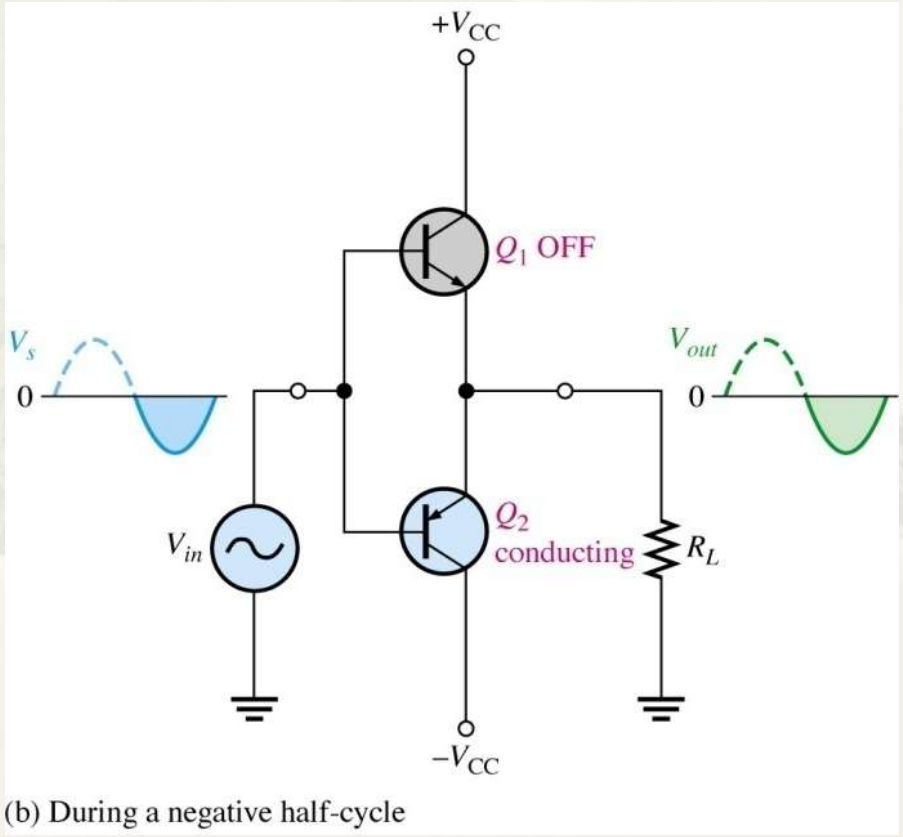
The signal is amplified for only half of the cycle.

Push-Pull Operation

- **Push-pull**
  - The combination of two class B amplifiers working together to amplifier the entire cycle.



(a) During a positive half-cycle



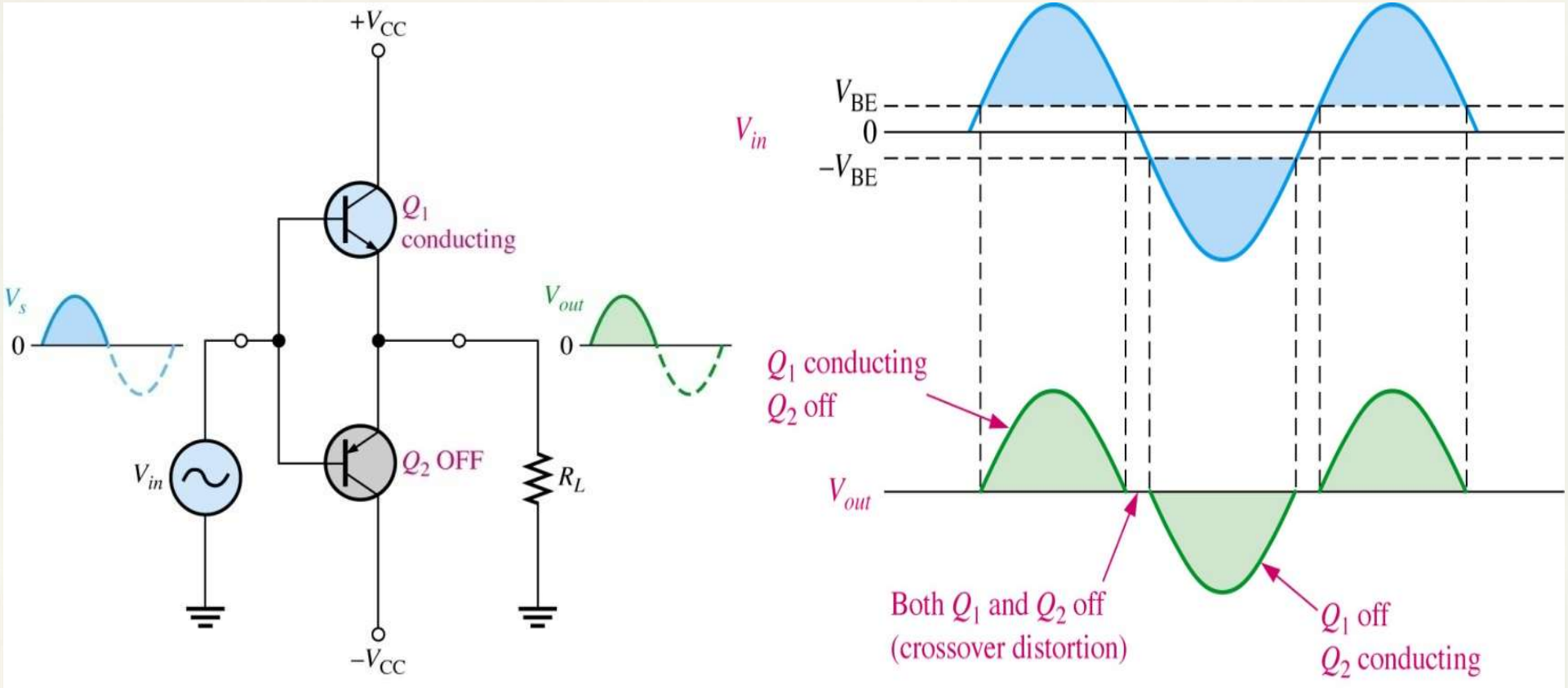
(b) During a negative half-cycle

Push-Pull Operation

- **Push-pull**
  - The combination of two class B amplifiers working together to amplifier the entire cycle.
- **Complementary symmetry transistors**
  - The most common approach
  - Matching pair: *npn/pnp* BJTs or *n-channel/p-channel* FETs

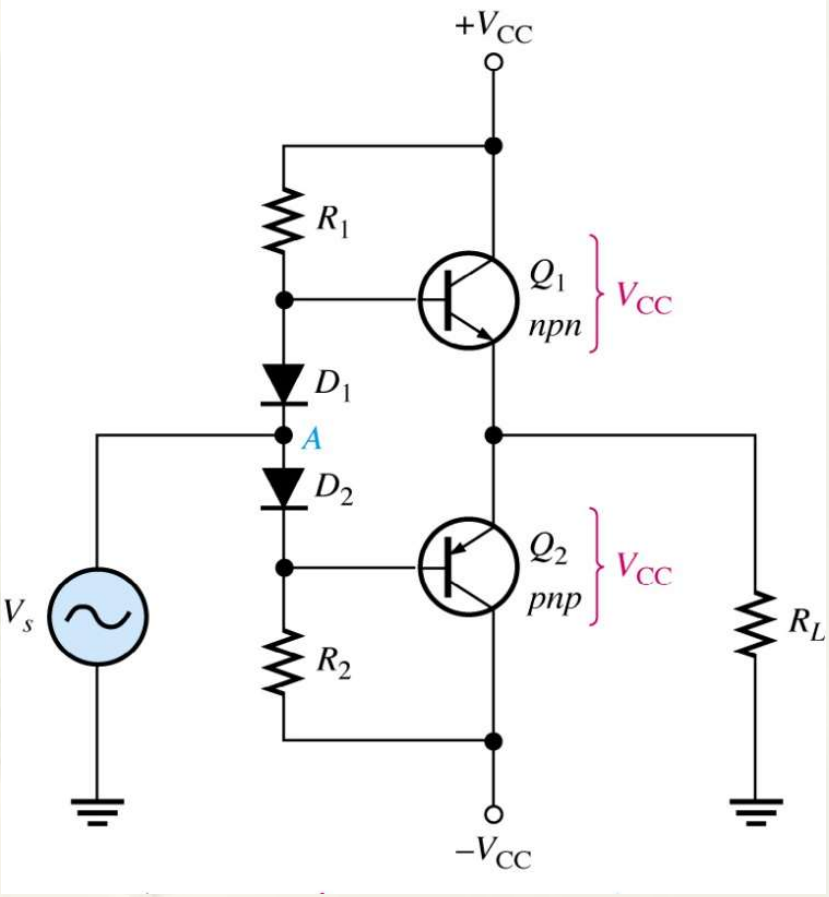
- **Efficiency:** be greatly increased
- Be widely used as power amplifiers
- Either BJTs or MOSFETs can be used as power amplifiers

Crossover Distortion



Biasing the Push-Pull Amplifier

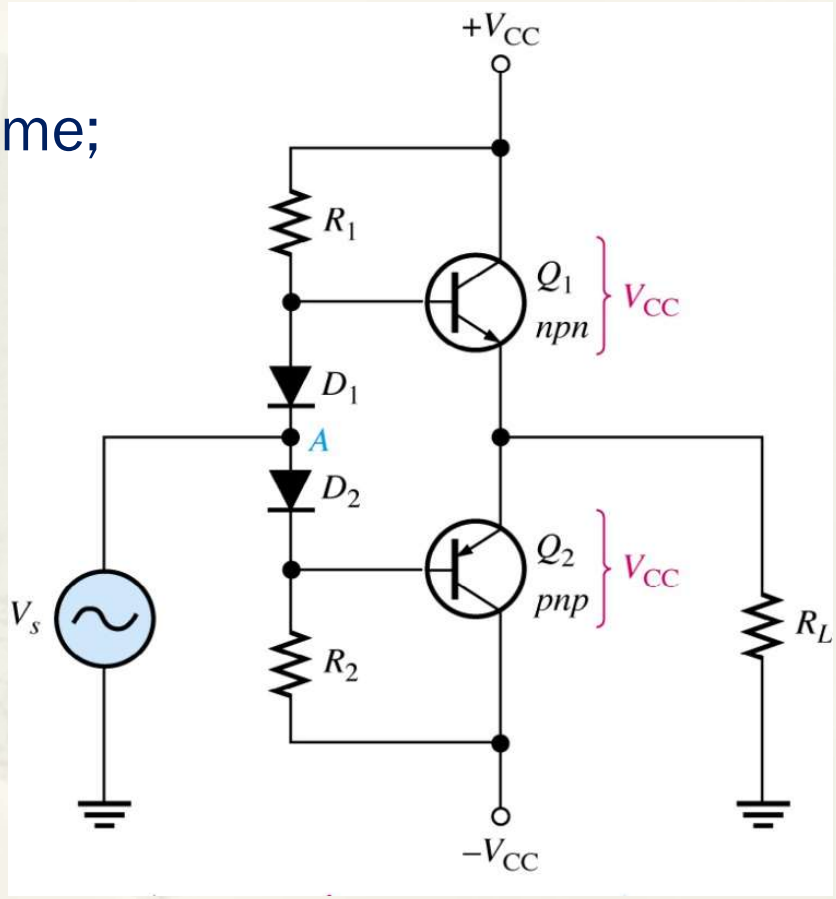
- **Class AB Power Amplifier**
  - The push-pull stages are biased into slight conduction, even when no input signal is present.
  - The diode characteristics are closely matched to the characteristics of the transistor base-emitter junctions.
  - The voltage at point A is forced to equal 0V and eliminates the need for an input coupling capacitor.



Biasing the Push-Pull Amplifier

- **Class AB Power Amplifier**
  - the current in the diodes and the current in the transistors is the same; this is called a current mirror.

$$I_{CQ} = I_{D1} = \frac{V_{CC} - 0.7\text{V}}{R_1}$$

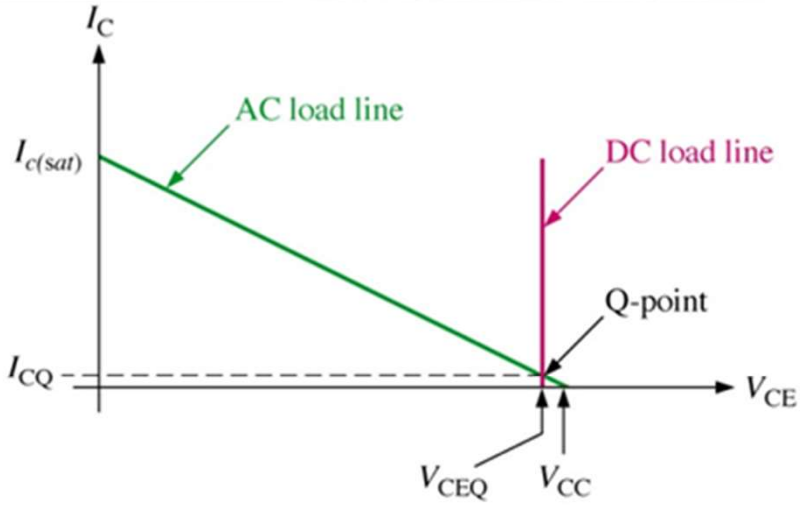


  - There is the potential for thermal instability if the transistor's V_{BE} drops are not matched to the diode drops or if the diodes are not in thermal equilibrium with the transistors.
  - Stringently, a small resistor in the emitter of each transistor can alleviate thermal runaway.

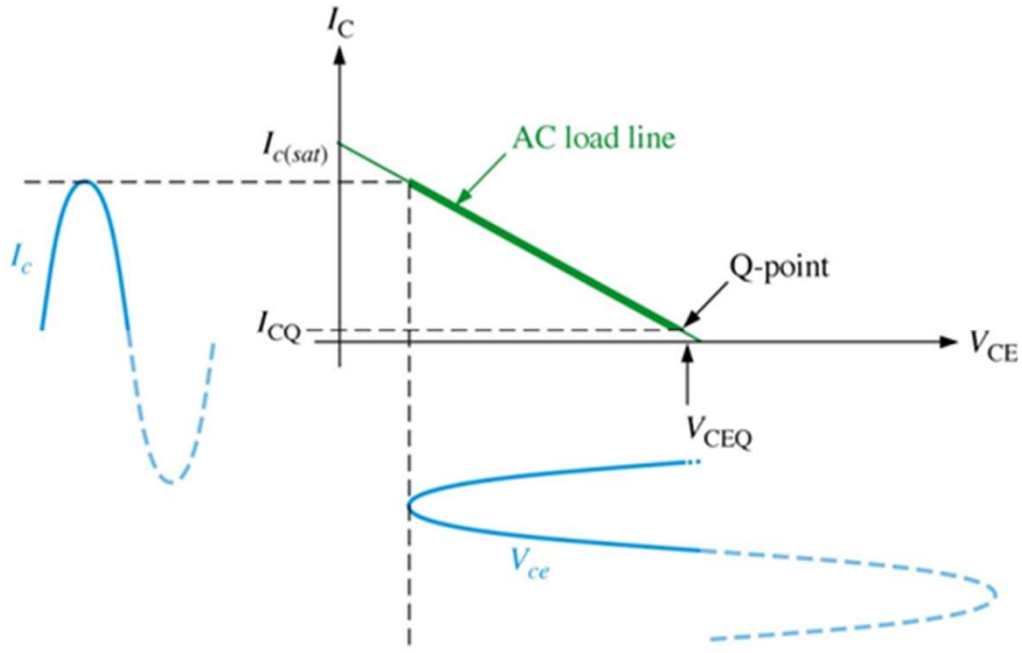
AC Operation

- **The ac saturation current for a two-supply operation with a push-pull amplifier**

$$I_{c(sat)} = \frac{V_{CC}}{R_L}$$



(a) DC and ac load lines for Q_1.



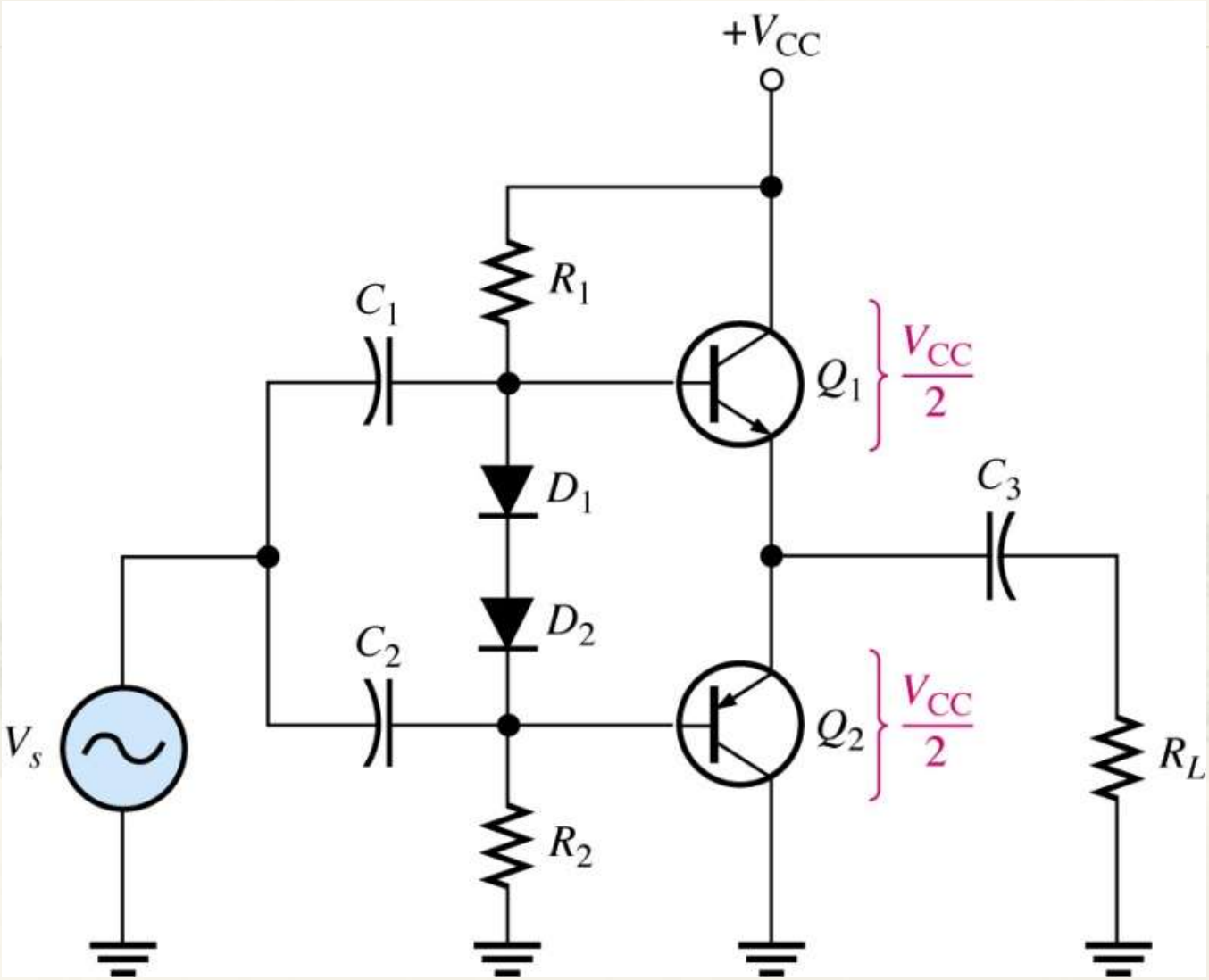
(b) AC load line operation for Q_1

AC Operation

- **The maximum output power of a class B amplifier**
  - $P_{om} = \frac{V^2_{om}}{R_L} = \frac{(V_{CC} - V_{CE(sat)})^2}{2R_L} \cong \frac{V_{CC}{}^2}{2R_L}$ (assume $V_{CE(sat)} = 0\text{V}$)
- **The power of dc power supply**
  - $P_{dc} = \frac{1}{\pi}\int_0^{\pi} \frac{V_{CC} - V_{CE(sat)}}{R_L} \sin\omega t \cdot V_{CC} \mathrm{d}\omega t$

    $= \frac{2}{\pi} \cdot \frac{V_{CC}(V_{CC} - V_{CE(sat)})}{R_L} \cong \frac{2}{\pi} \cdot \frac{V_{CC}{}^2}{R_L}$
- **The maximum theoretical efficiency of a class B amplifier**
  - $eff = \frac{P_{om}}{P_{dc}} = \frac{\pi}{4} \cong 78.5\%$

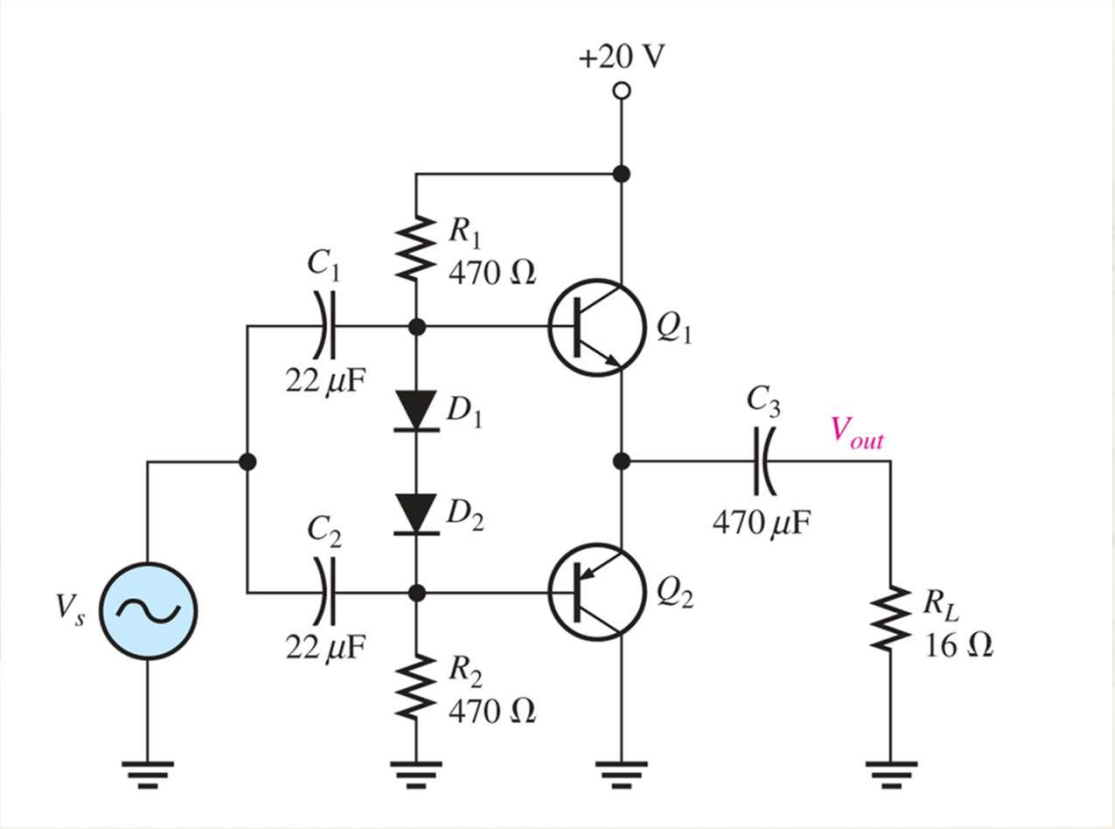
Single-Supply Operation



Example 5-4

Problem

Determine the ideal maximum peak output voltage and current for the circuit shown in Figure 5–27.



Example 5-4

Solution

The ideal maximum peak output voltage is

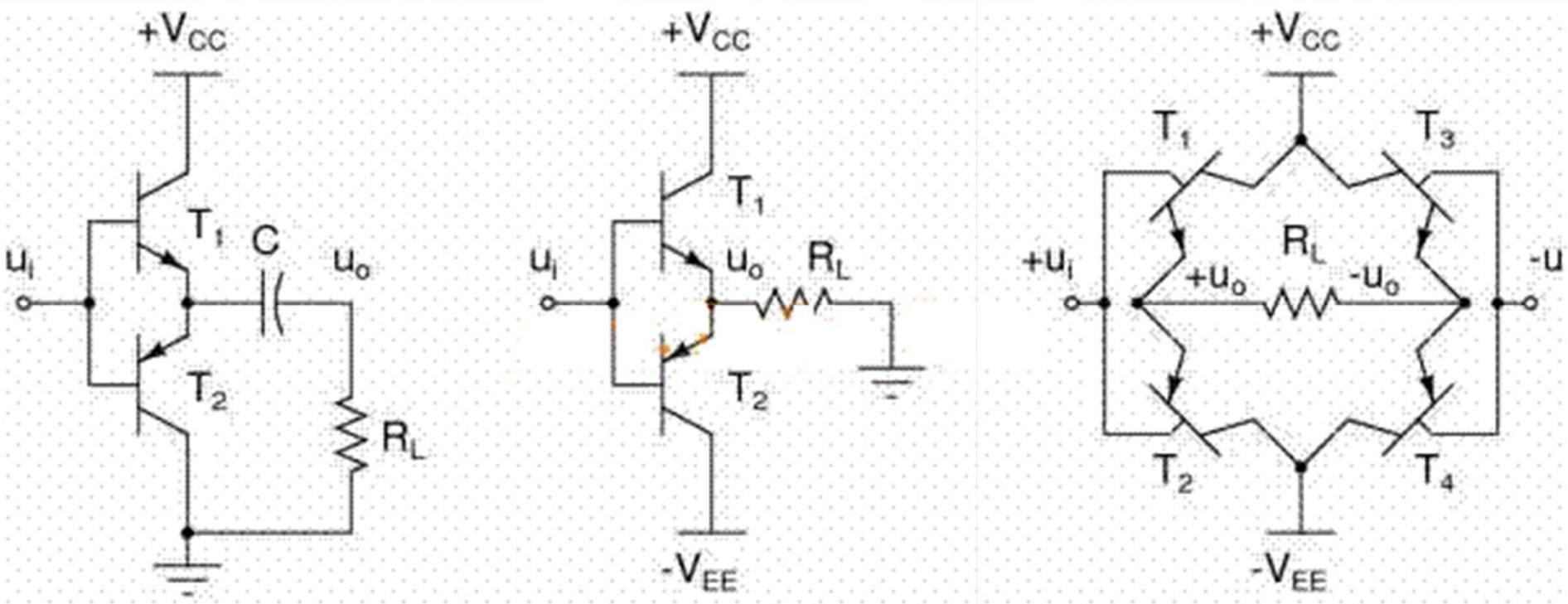
$$V_{p(out)} \cong V_{CEQ} \cong \frac{V_{CC}}{2} = \frac{20\text{ V}}{2} = \mathbf{10\text{ V}}$$

The ideal maximum peak current is

$$I_{p(out)} \cong I_{c(sat)} \cong \frac{V_{CEQ}}{R_L} = \frac{10\text{ V}}{16\ \Omega} = \mathbf{0.63\text{ A}}$$

Different Types of Class AB Power Amplifiers

- Output transformerless power amplifier (OTL)
- Output capacitorless power amplifier (OCL)
- Bridge-tied load power amplifier (BTL)



5-3 Differential Amplifiers

- Configuration
- Modes of Signal Input and Output
- Basic Operation
- Common-Mode Rejection Ratio
- Examples

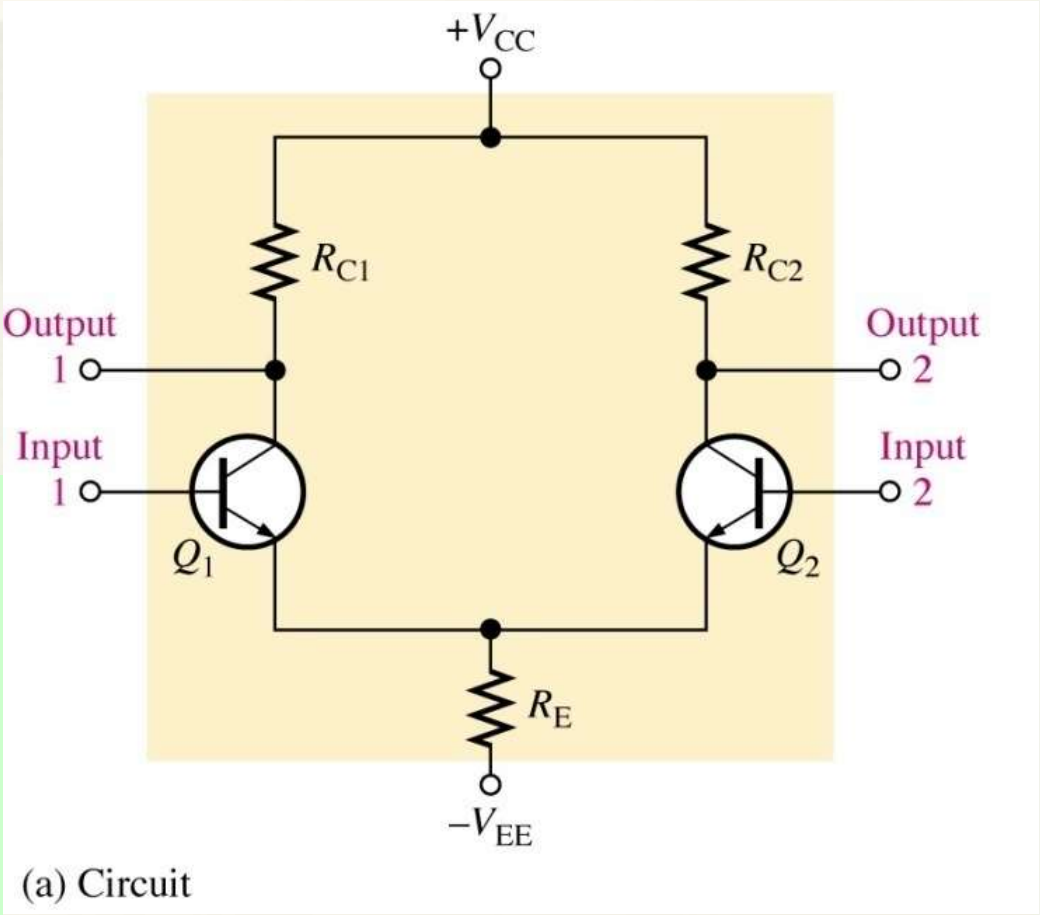
Configuration

Basic differential amplifier (diff-amp)

Consists of two common-emitter amplifiers in which components are carefully matched to have identical characteristics and parameter values

match ⇒

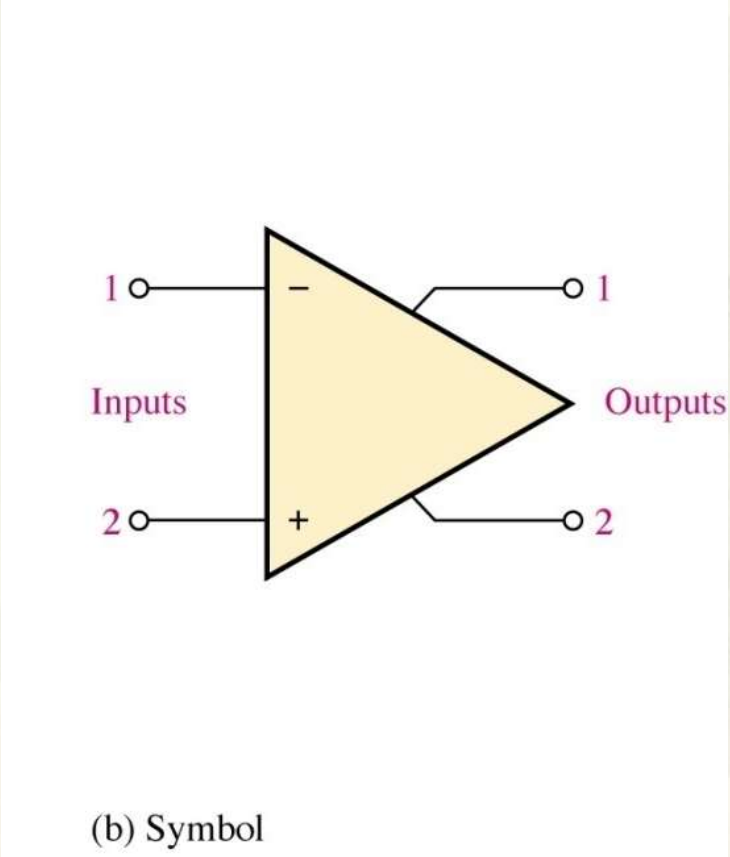
$$h_{fe1}=h_{fe2}=h_{fe}=\beta$$
$$U_{BE1}=U_{BE2}=U_{BE}$$
$$h_{ie1}=h_{ie2}=h_{ie}=r_{be}$$
$$I_{CBO1}=I_{CBO2}=I_{CBO}$$
$$R_{C1}=R_{C2}=R_C$$
$$R_{b1}=R_{b2}=R_b$$

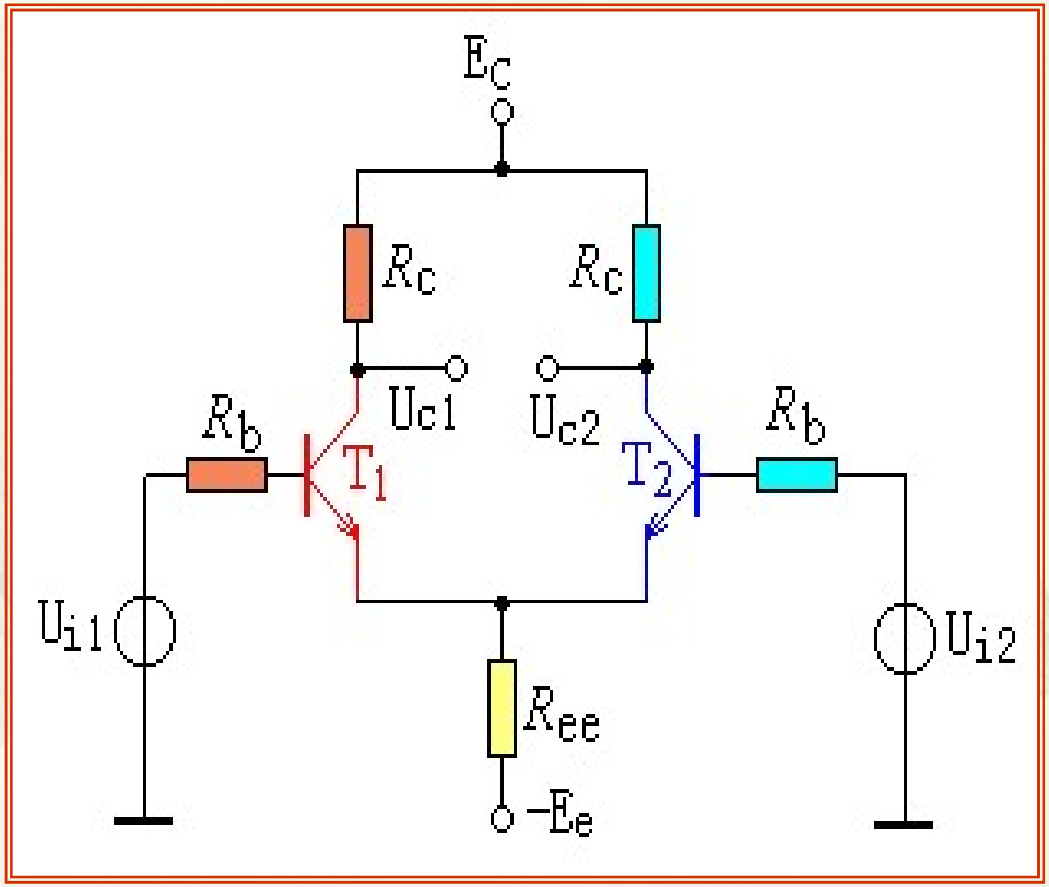


(a) Circuit

Configuration

- **Basic differential amplifier (diff-amp)**

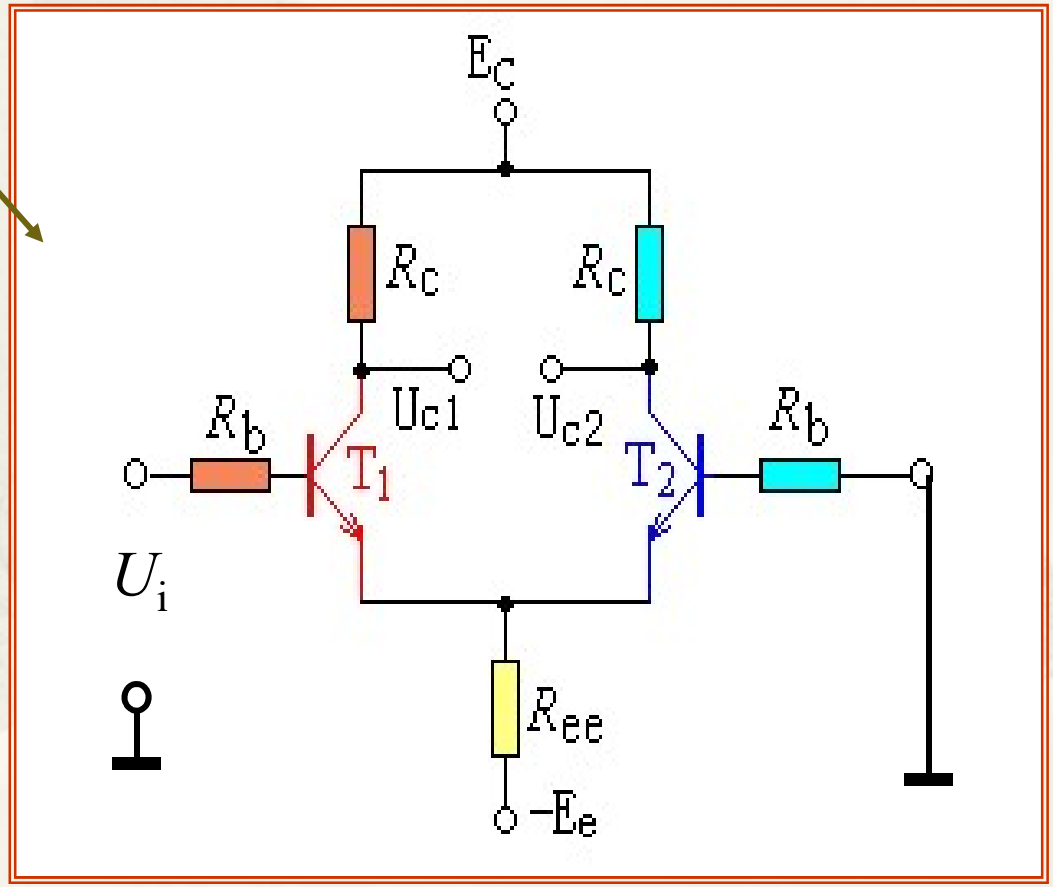




Modes of Signal Input and Output

- **Input Modes**

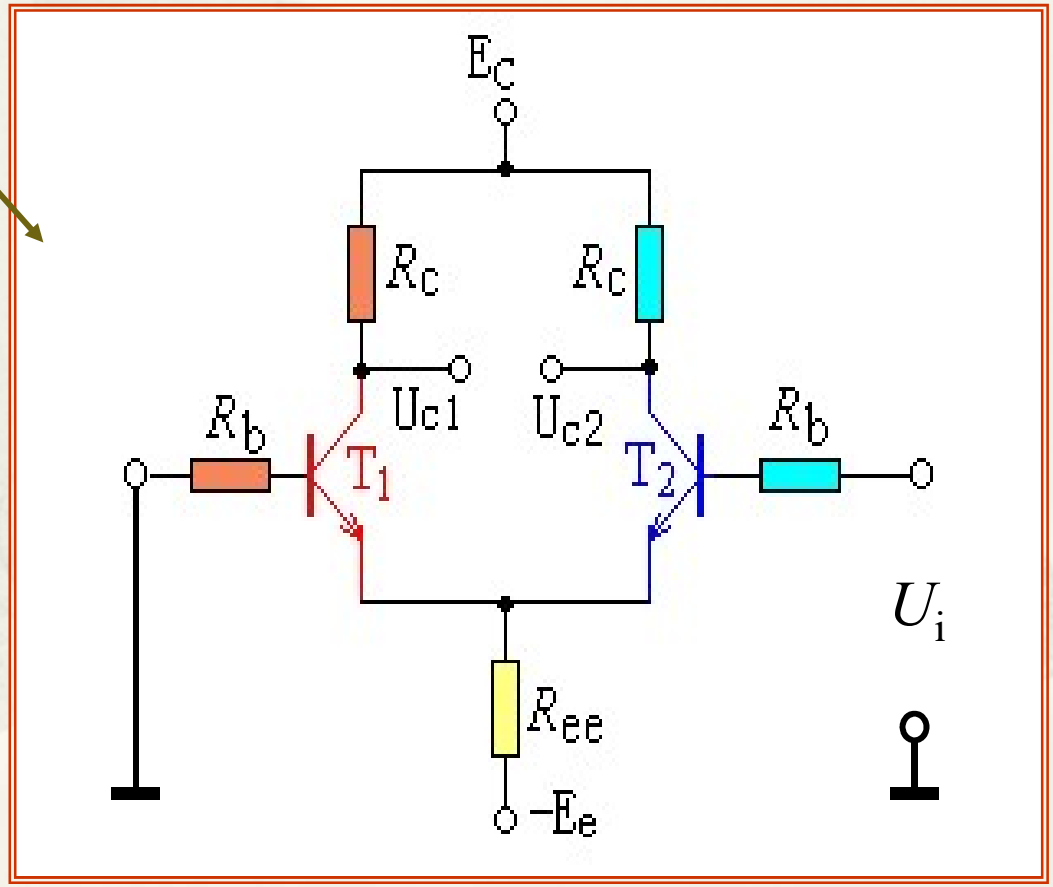
{ single-ended input



Modes of Signal Input and Output

- **Input Modes**

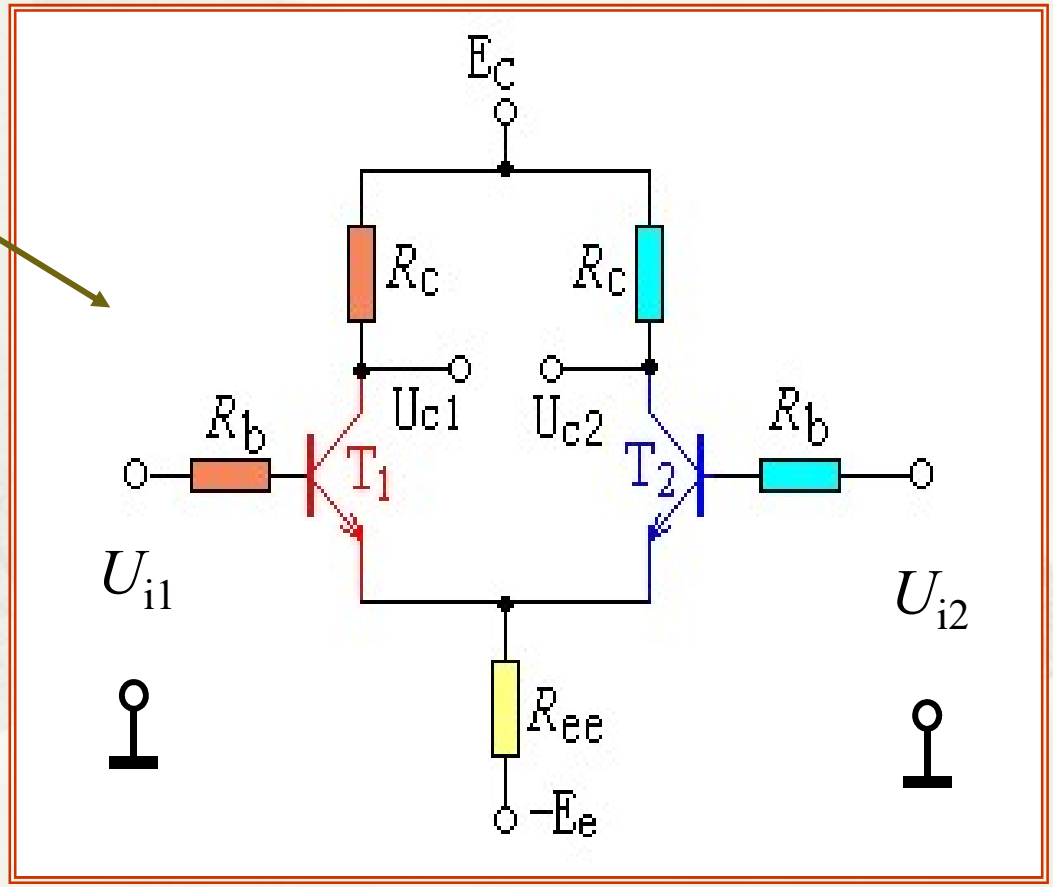
single-ended input



Modes of Signal Input and Output

- **Input Modes**

  - single-ended input
  - double-ended input



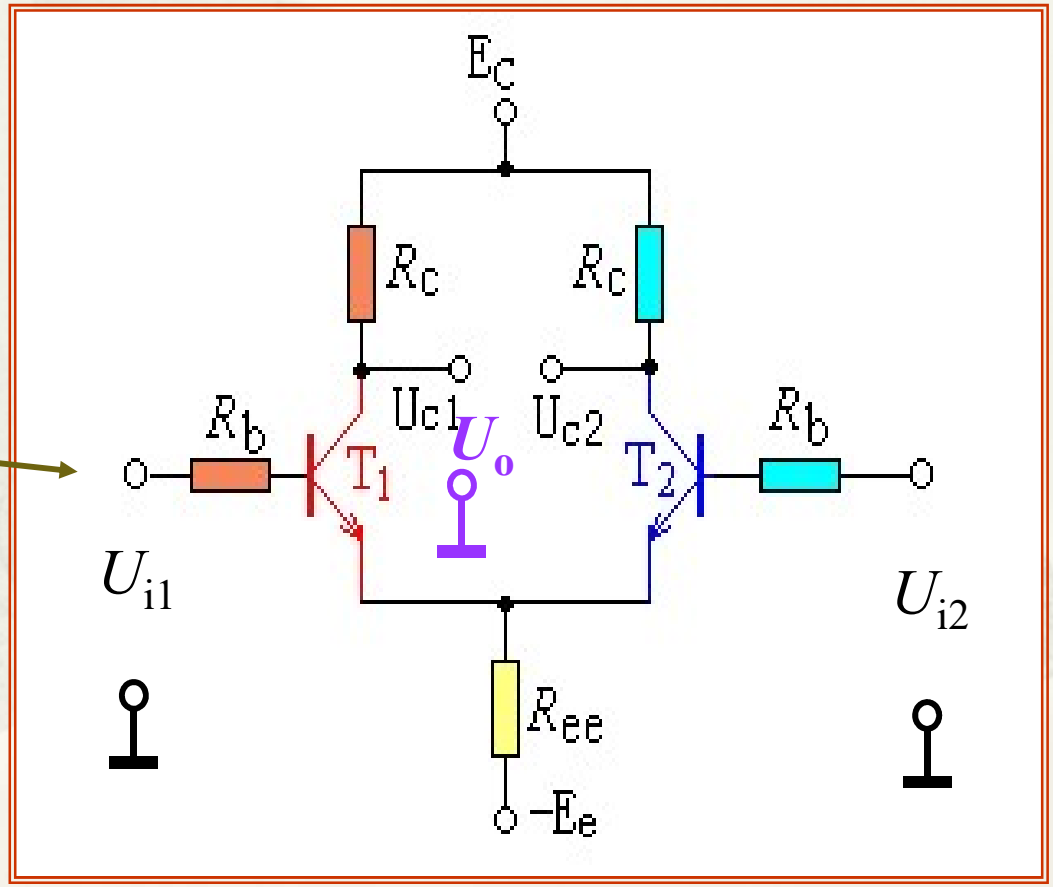
Modes of Signal Input and Output

- **Input Modes**

  - single-ended input
  - double-ended input

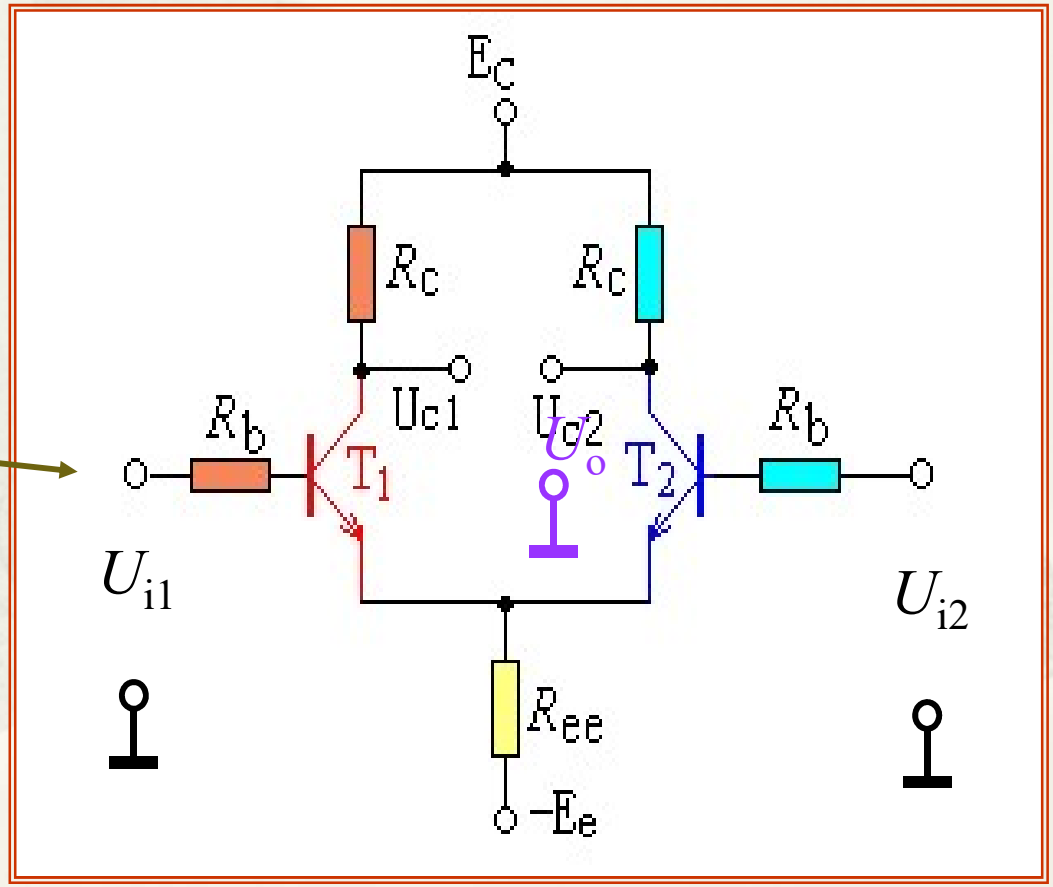
- **Output Modes**

  - single-ended output



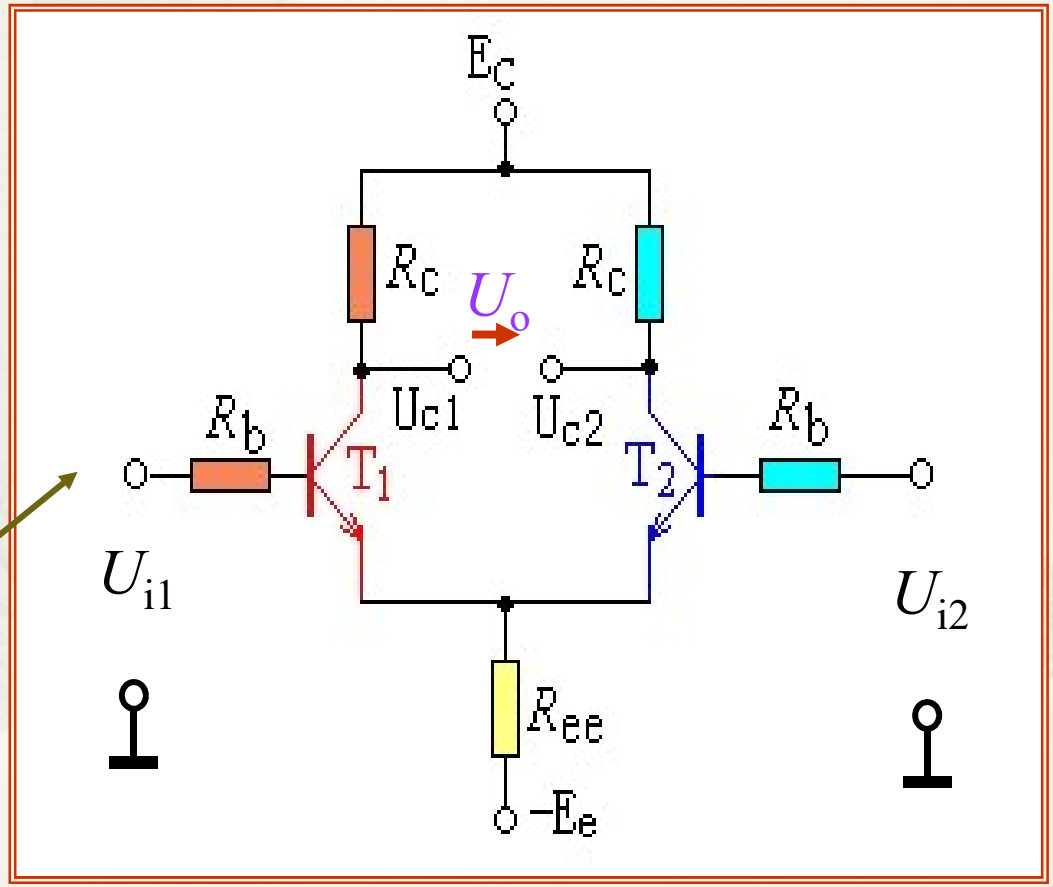
Modes of Signal Input and Output

- **Input Modes**
  - single-ended input
  - double-ended input
- **Output Modes**
  - single-ended output



Modes of Signal Input and Output

- **Input Modes**
  - single-ended input
  - double-ended input
- **Output Modes**
  - single-ended output
  - double-ended output



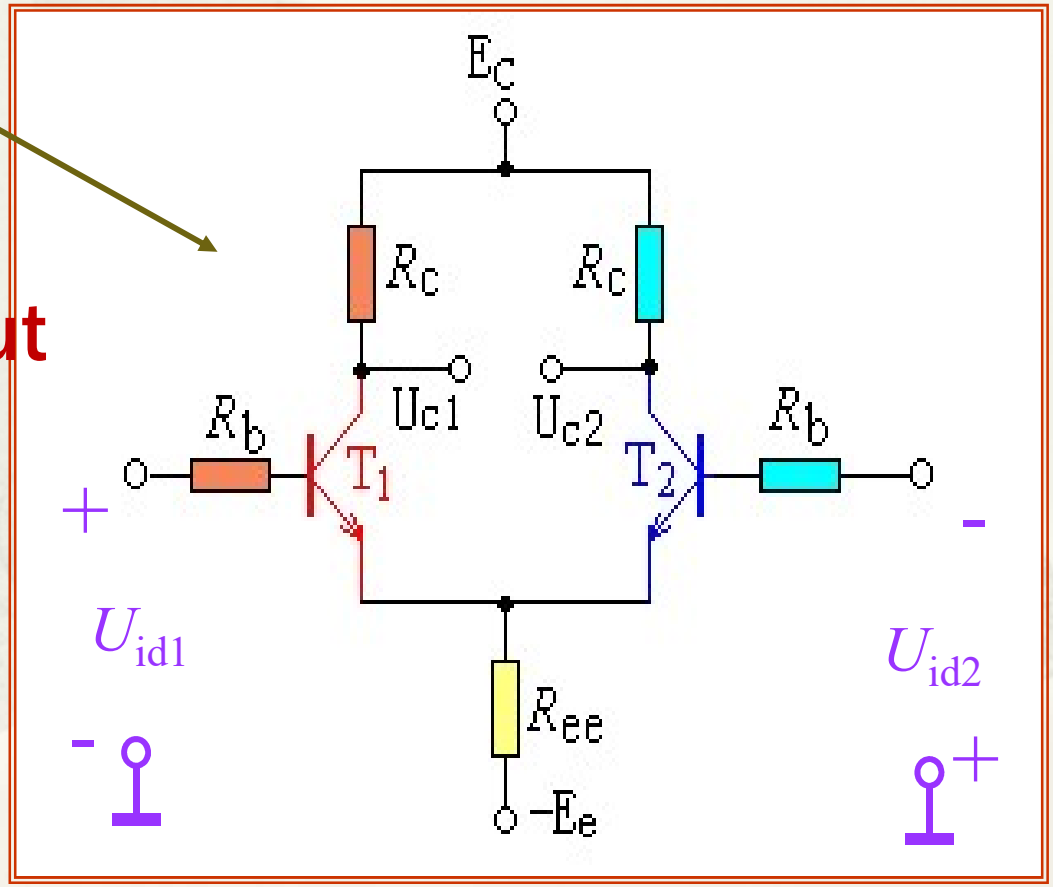
Modes of Signal Input and Output

- **Differential Input**

  two opposite-polarity (out of phase) signals are applied to the two inputs

  $$U_{id1} = - U_{id2}$$

- **Common-mode Input**



Modes of Signal Input and Output

- **Differential Input**

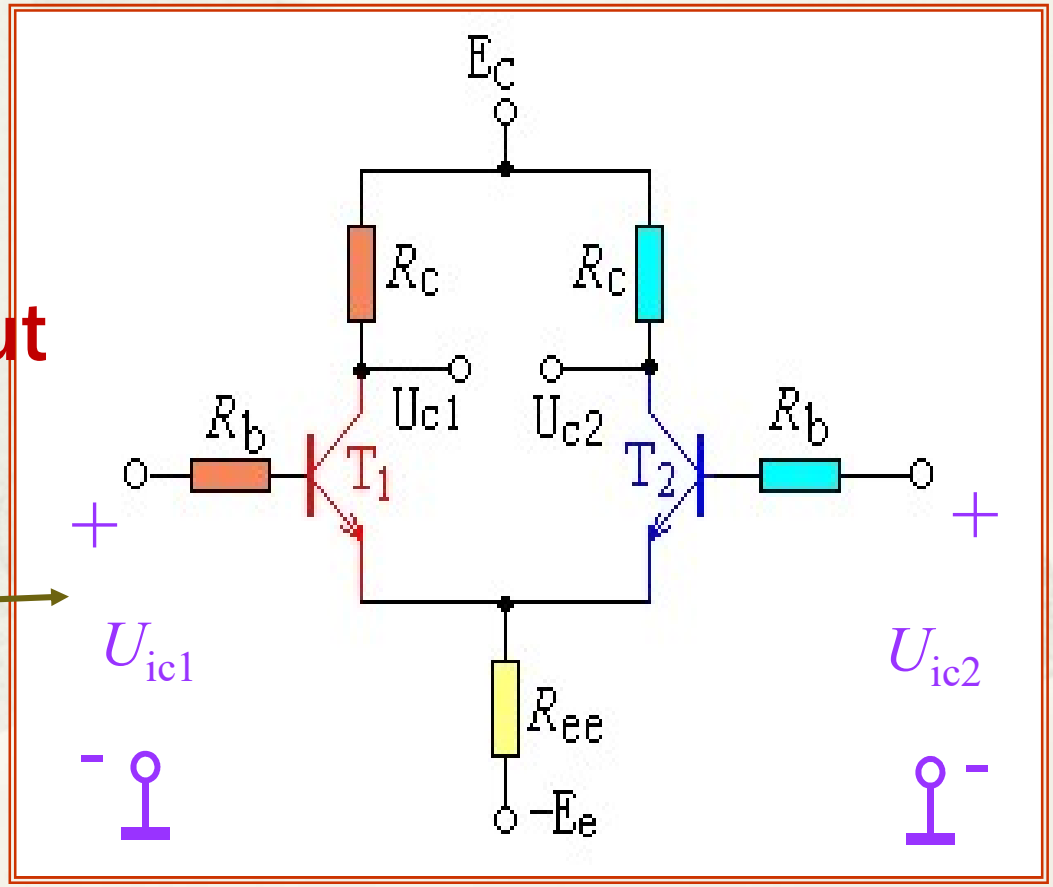
  two opposite-polarity (out of phase) signals are applied to the two inputs

  $$U_{id1} = - U_{id2}$$

- **Common-mode Input**

  two identical signals are applied to the two inputs

  $$U_{ic1} = U_{ic2}$$



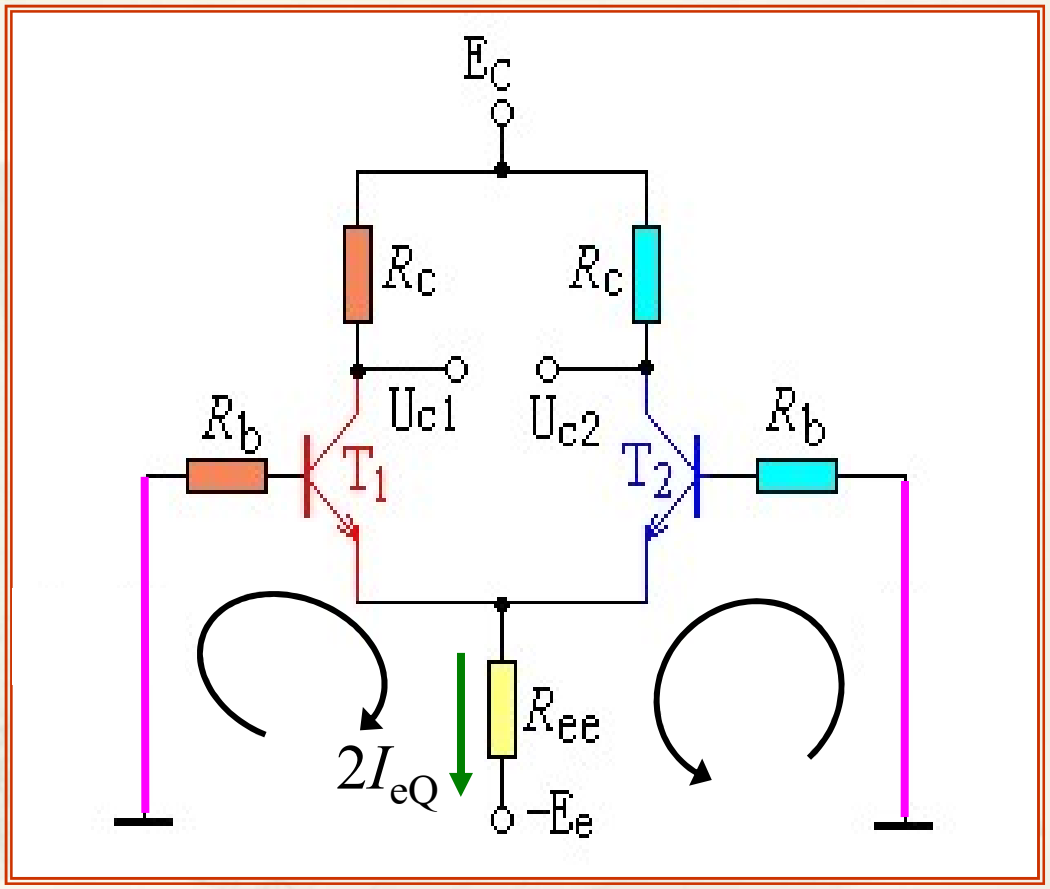
Basic Operation

DC Operation

– two inputs grounded

match →

$$I_{BQ1}=I_{BQ2}=I_{BQ}$$
$$I_{CQ1}=I_{CQ2}=I_{CQ}$$
$$U_{C1}=U_{C2}$$



$$I_{BQ}R_b + U_{BE} + 2I_{EQ}R_{ee} = E_e$$

$$I_{CQ} = \beta I_{BQ}$$

$$I_{BQ} = \frac{E_e - U_{BE}}{R_b + 2(1+\beta)R_{ee}}$$

$$U_{CEQ} = E_C - I_{CQ}R_c + \boxed{E_e - 2I_{EQ}R_{ee}}$$

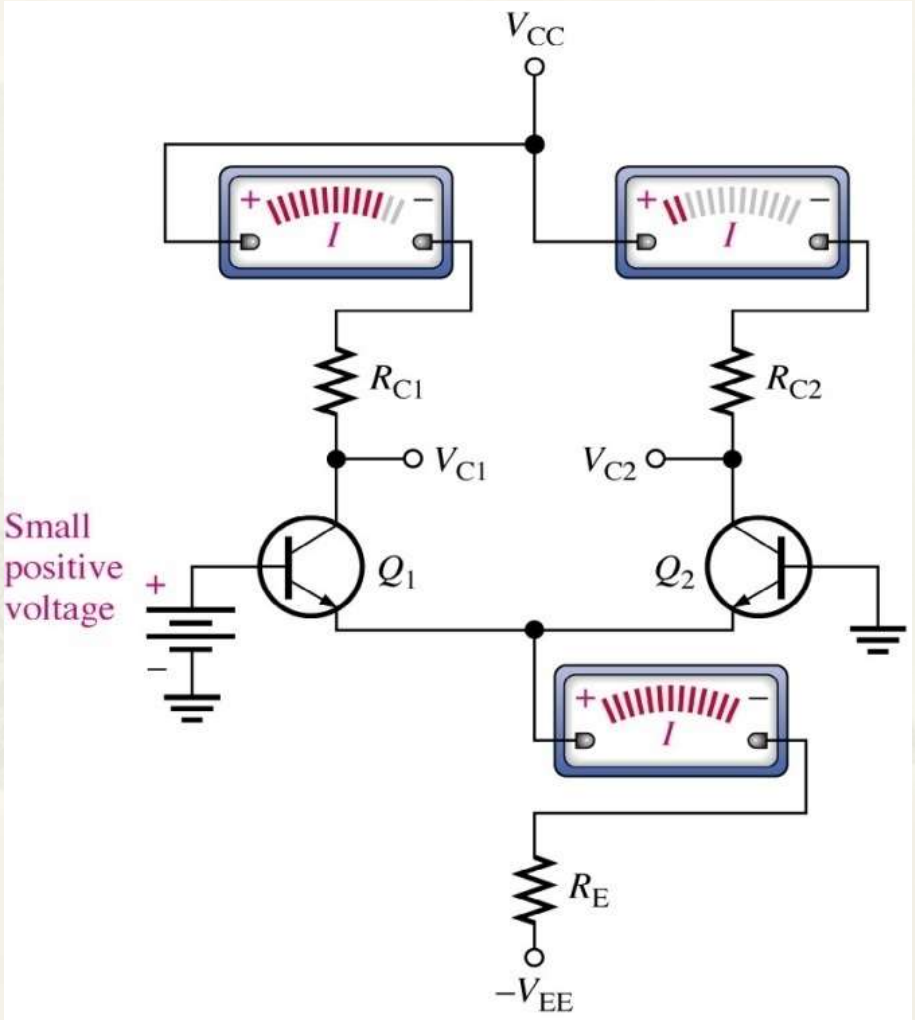
≈ 0

Basic Operation

- **DC Operation**
  - **Single-ended input**

    One input is grounded and the signal voltage is applied only to the other input.

- ✓ **Showing the effect on current when a small voltage is connected to one of the bases.**



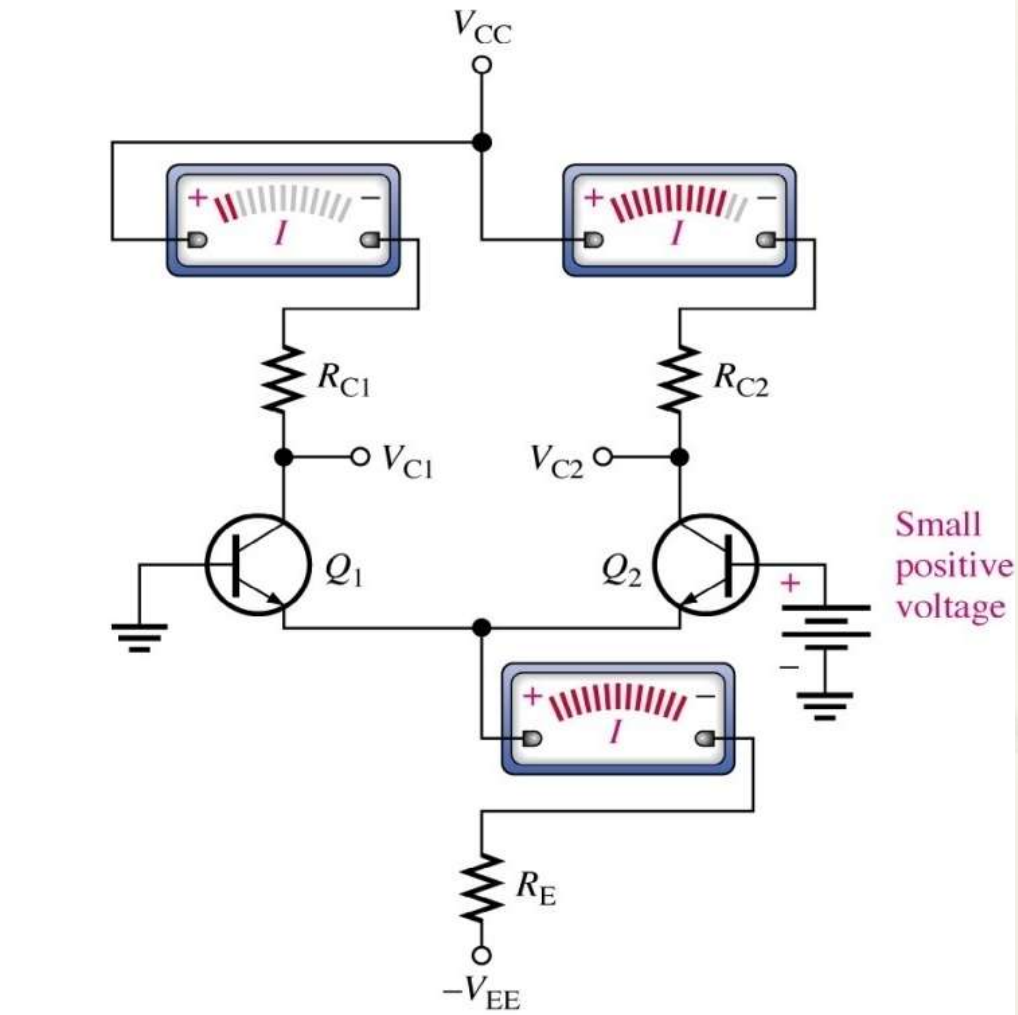
(a) A small positive voltage on Q_{B1} when Q_{B2} is grounded

Basic Operation

- **DC Operation**
  - **Single-ended input**

    One input is grounded and the signal voltage is applied only to the other input.

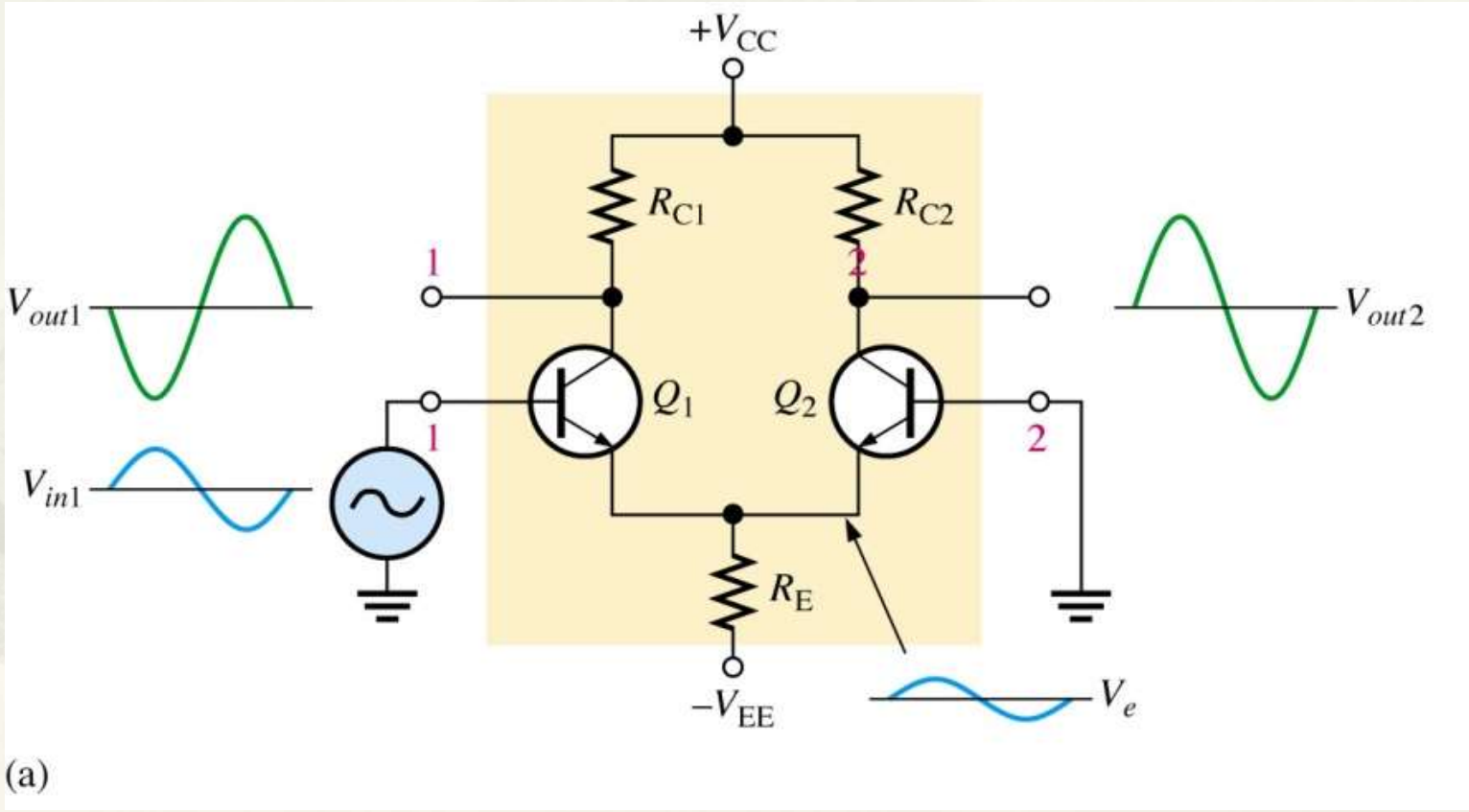
- ✓ **Showing the effect on current when a small voltage is connected to one of the bases.**



(b) A small positive voltage on Q_{B2} when Q_{B1} is grounded

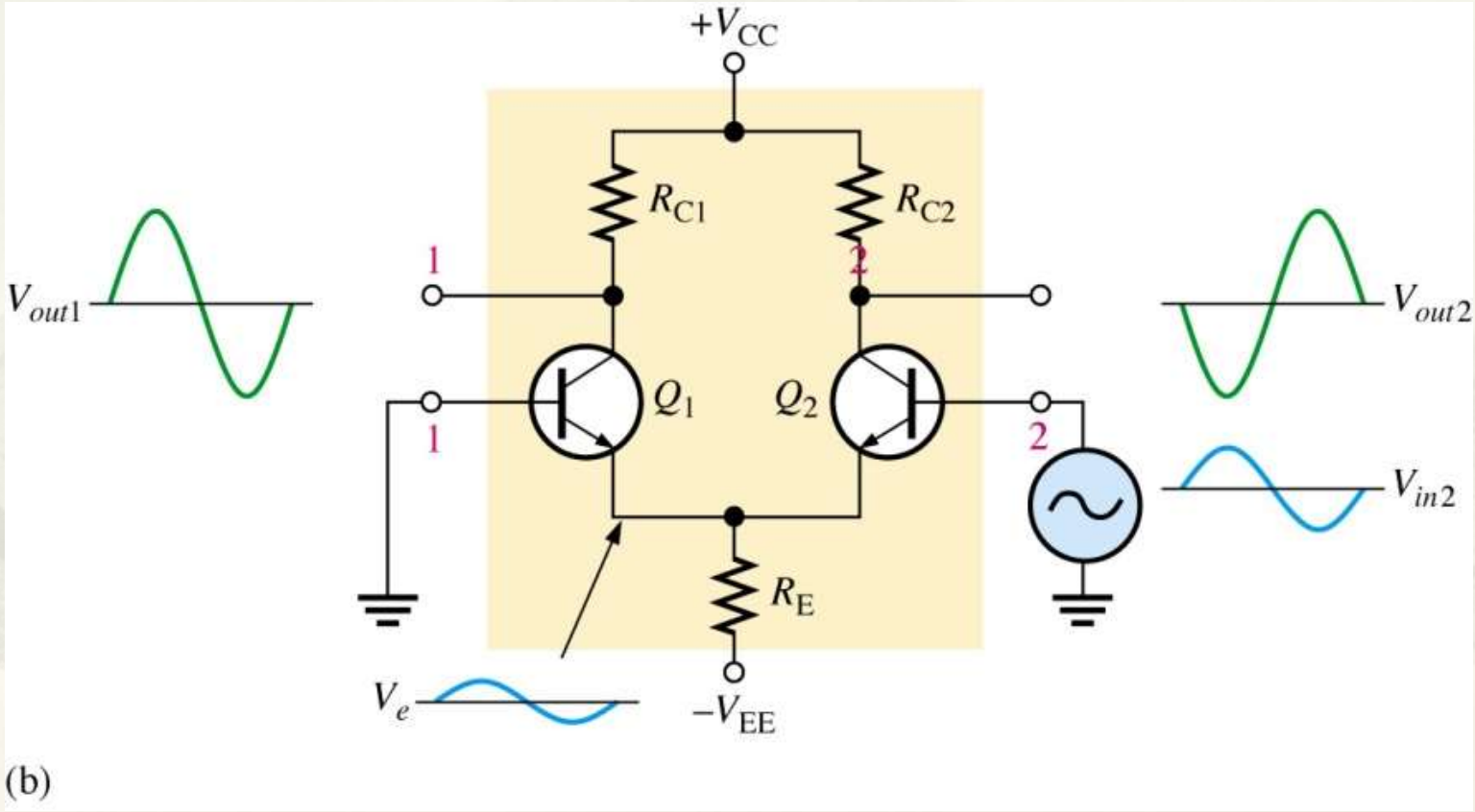
Basic Operation

- **AC Operation** – Single-Ended Input



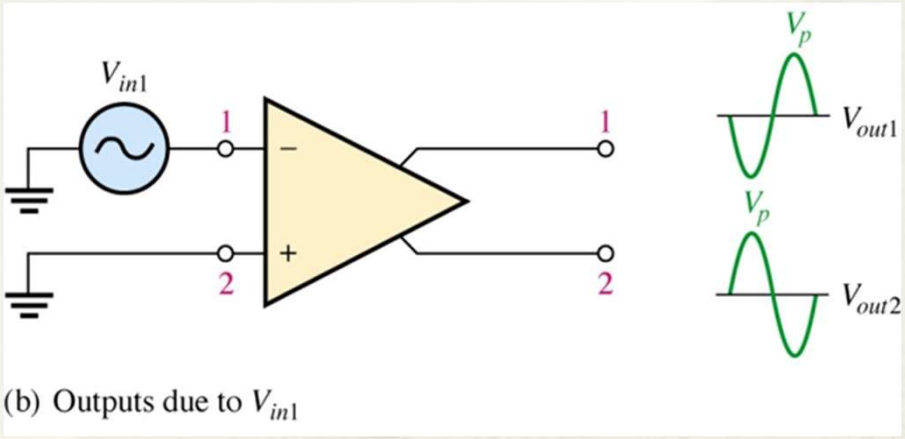
Basic Operation

- **AC Operation** – Single-Ended Input

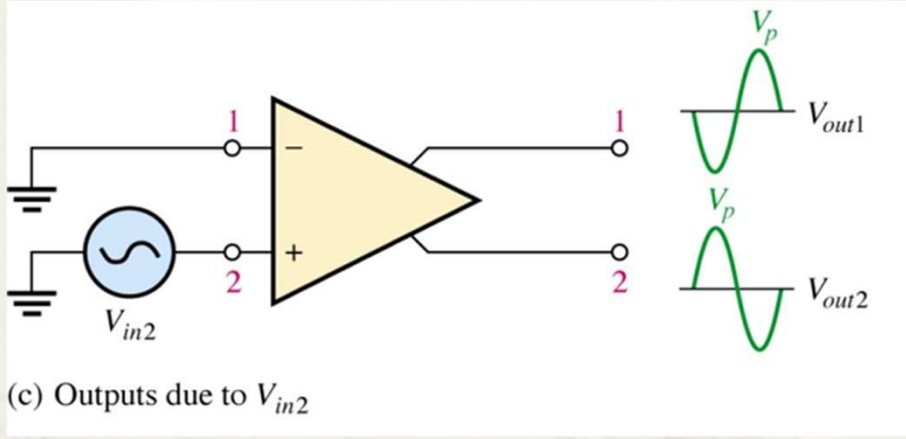


Basic Operation

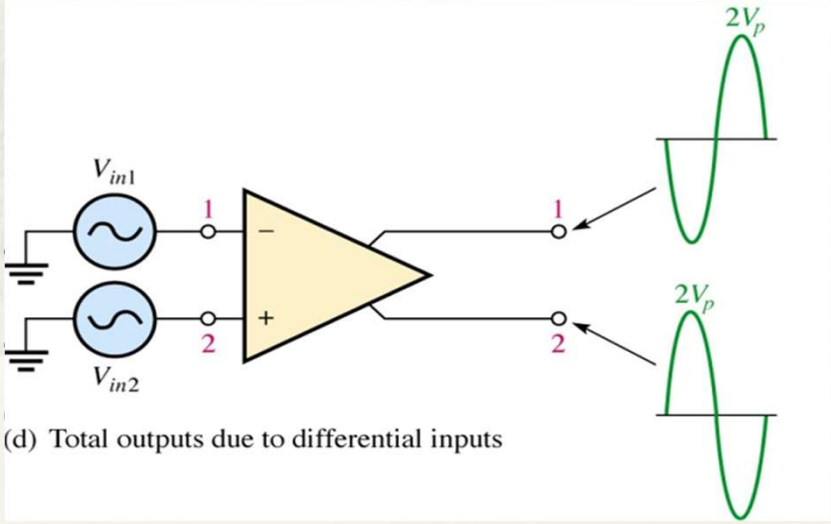
- **AC Operation – Differential Input**



(b) Outputs due to V_{in1}



(c) Outputs due to V_{in2}



(d) Total outputs due to differential inputs

Basic Operation

➢ AC Operation – Differential Input

Analysis quantitively

Step 1: AC equivalent circuit

$U_{id1} = -U_{id2}$ → $I_{b1} = -I_{b2}$

↓

$I_{e1} = -I_{e2}$ → $U_{c1} = -U_{c2}$

↓

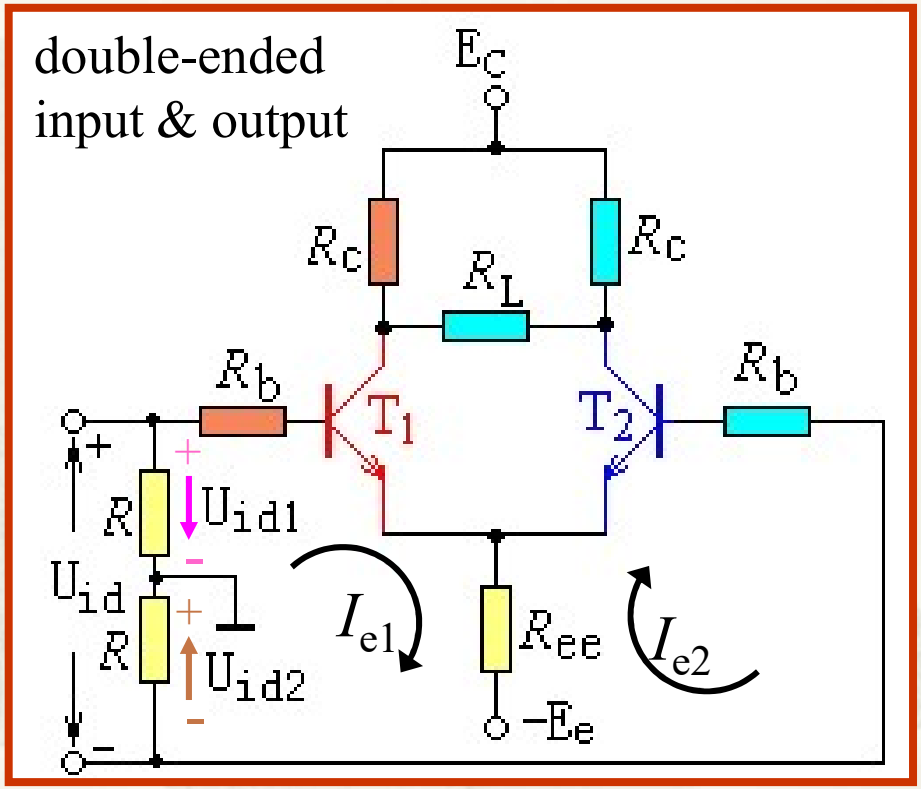
$I_e = I_{e1} + I_{e2} = 0$

↓

The AC voltage across R_{ee} reduces to **0**

↓

R_{ee} can be regarded as **short** and the two emitters are **grounded**.



positive and negative potentials are in R_L ends

↓

midpoint of R_L is **0** → load of each transistor referenced to ground is $R_L/2$

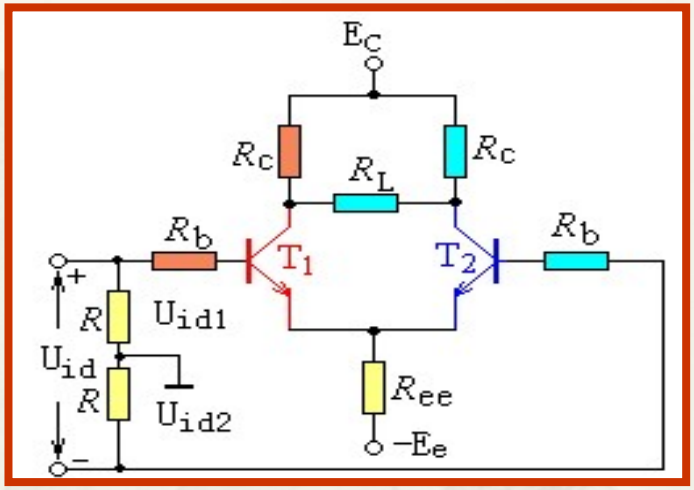
Basic Operation

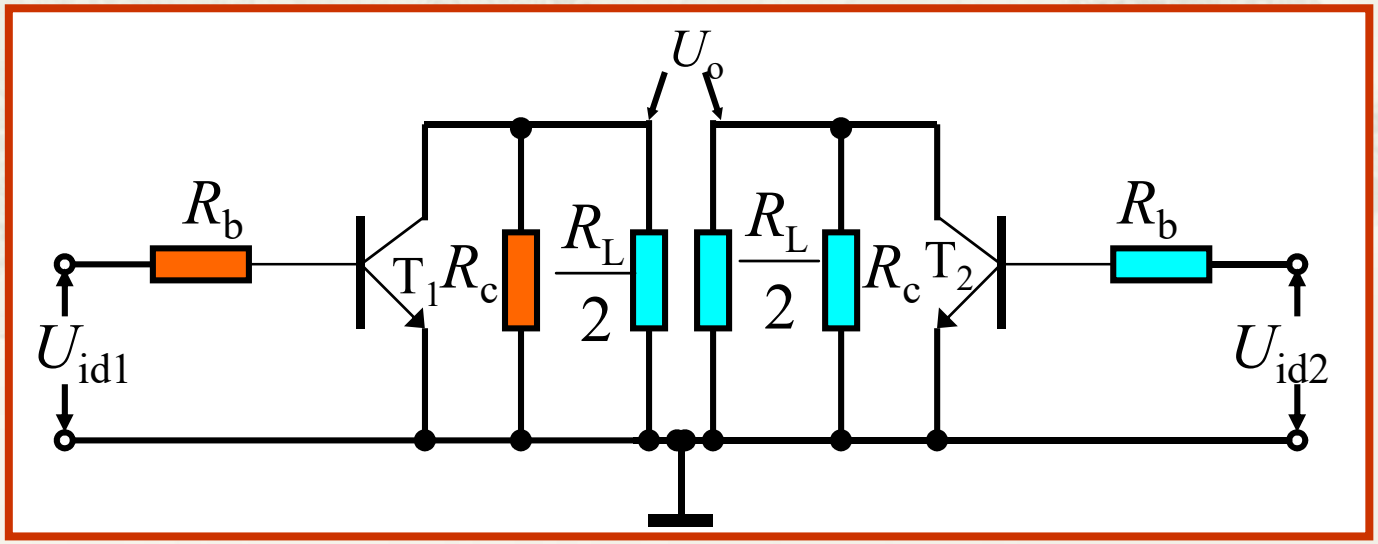
➢ AC Operation – Differential Input

Analysis quantitively

Step 1: AC equivalent circuit

- ◆ two emitters are **grounded**
- ◆ load of each transistor referenced to ground is $R_L/2$



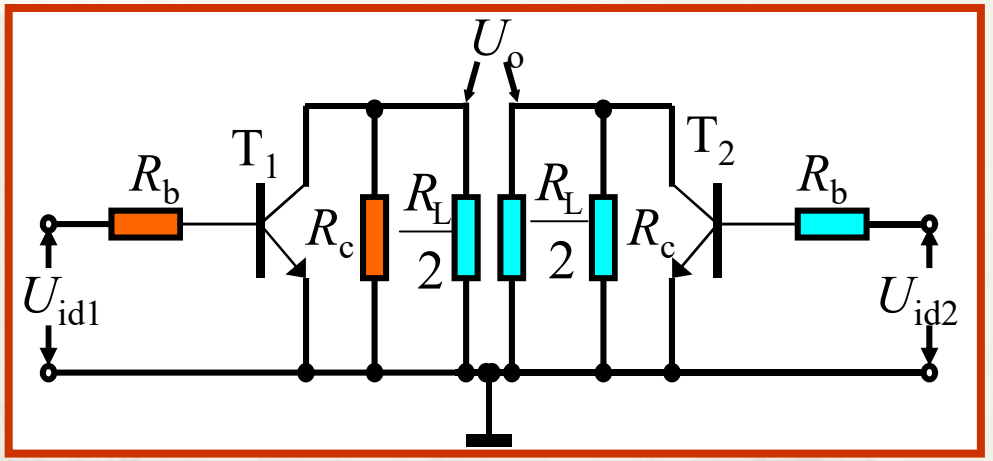


Basic Operation

➢ AC Operation – Differential Input

Analysis quantitively

Step 2: Calculating A_{Ud}, r_{id} and r_{od}



$$U_{cd1} = \frac{-h_{fe}R'_L}{R_b + h_{ie}}U_{id1}$$

$$U_{cd2} = \frac{-h_{fe}R'_L}{R_b + h_{ie}}U_{id2}$$

$R'_L = R_c // (R_L/2)$

$$A_{Ud} = \frac{U_o}{U_{id}} = \frac{U_{cd1} - U_{cd2}}{2U_{id1}} \rightarrow A_{Ud} = \frac{-h_{fe}R'_L}{R_b + h_{ie}}$$

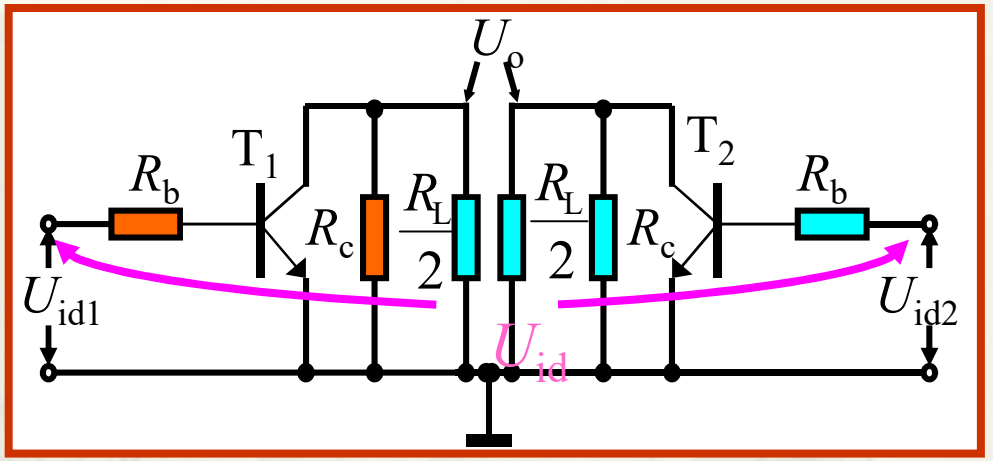
same with single transistor gain

Basic Operation

➢ AC Operation – Differential Input

Analysis quantitively

Step 2: Calculating A_{Ud}, r_{id} and r_{od}



two inputs are in series

$$r_{id}=2(R_b+h_{ie})$$

two outputs are in series

$$r_{od}=2R_c//(1/h_{oe}) \longrightarrow r_{od}\approx 2R_c$$

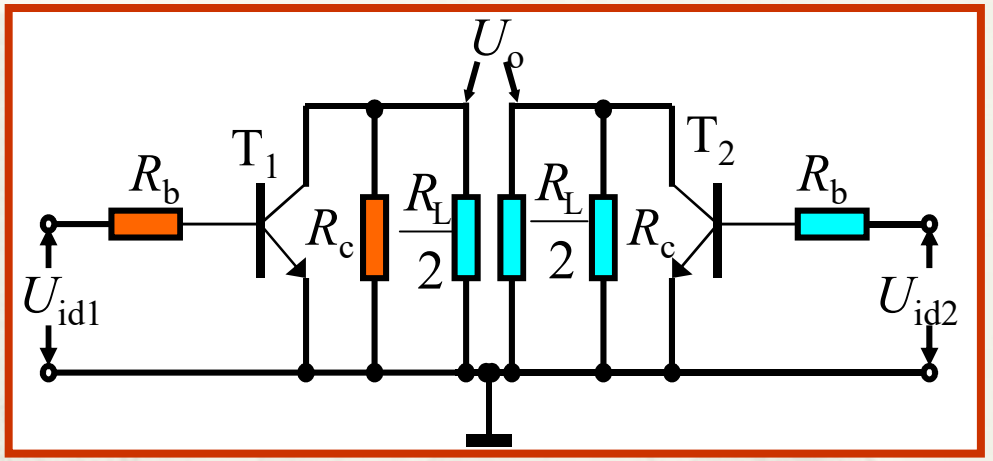
$1/h_{oe}=r_{ce}\rightarrow\infty$

Basic Operation

➢ AC Operation – Differential Input

Analysis quantitively

Step 2: Calculating A_{Ud}, r_{id} and r_{od}



double-ended output

$$A_{Ud} = \frac{-h_{fe}R'_L}{R_b + h_{ie}}$$

$r_{id}=2(R_b+h_{ie})$ $\quad r_{od} \approx 2R_c$

single-ended output

$$A_{Ud} = \frac{U_o}{U_{id}} = \frac{U_{cd1}}{2U_{id1}} = -\frac{1}{2}\frac{h_{fe}R'_L}{R_b + h_{ie}}$$

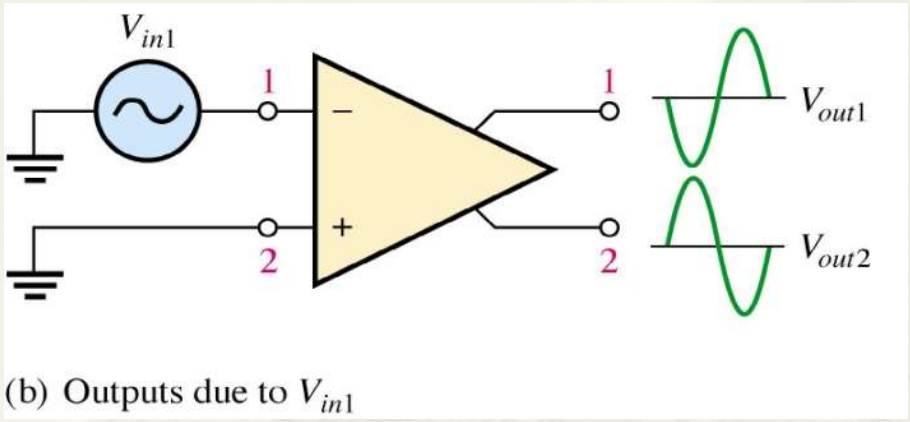
half of single transistor gain

$r_{id}=2(R_b+h_{ie})$

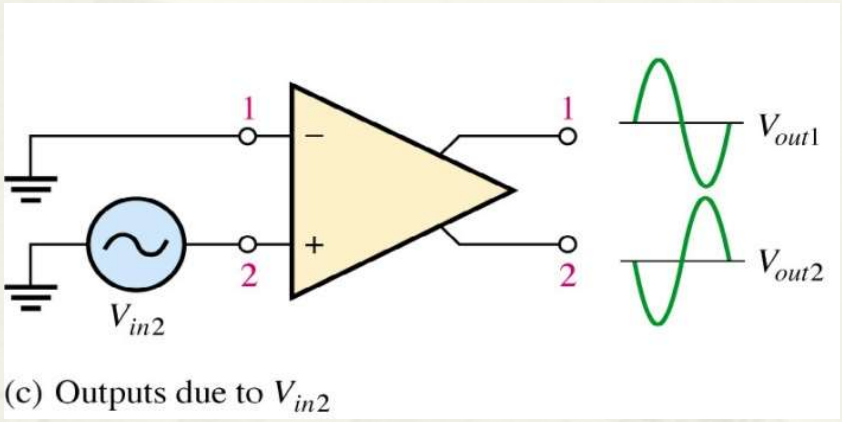
$r_{od} \approx R_c$

Basic Operation

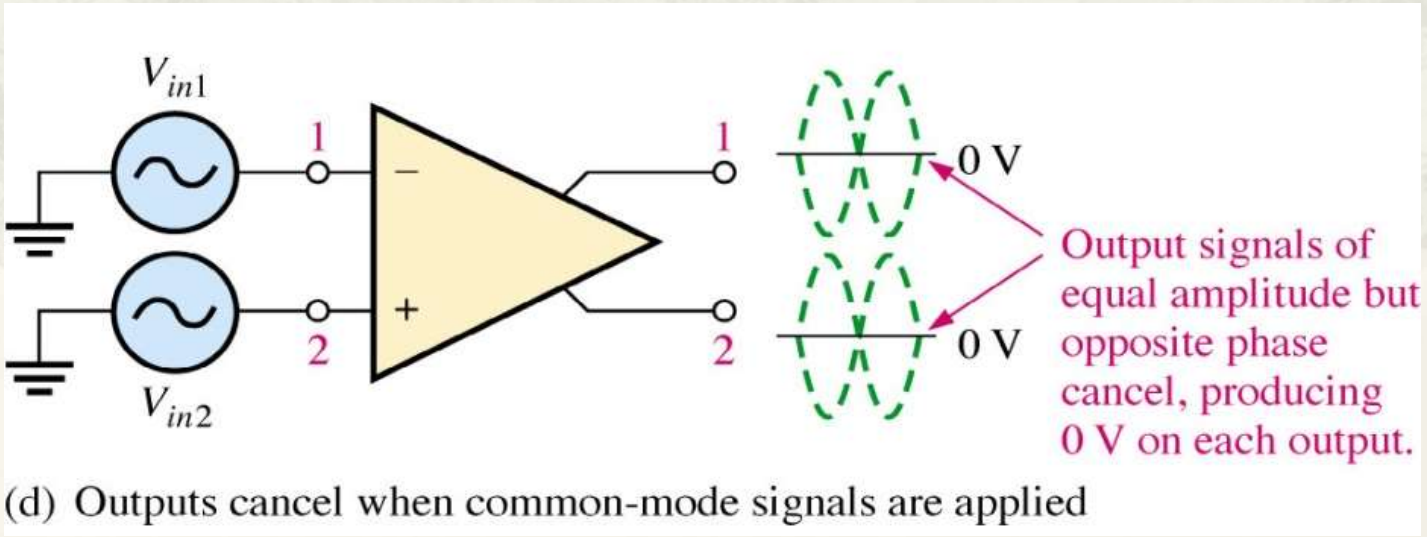
- **AC Operation** – Common-mode Input



(b) Outputs due to V_{in1}



(c) Outputs due to V_{in2}



(d) Outputs cancel when common-mode signals are applied

Basic Operation

➢ AC Operation – Common-mode Input

Analysis quantitively

Step 1: AC equivalent circuit

$$U_{ic1}=U_{ic2}=U_{ic}$$

$$\Downarrow$$

$$I_{bc1}=I_{bc2} \quad I_{ec1}=I_{ec2}$$

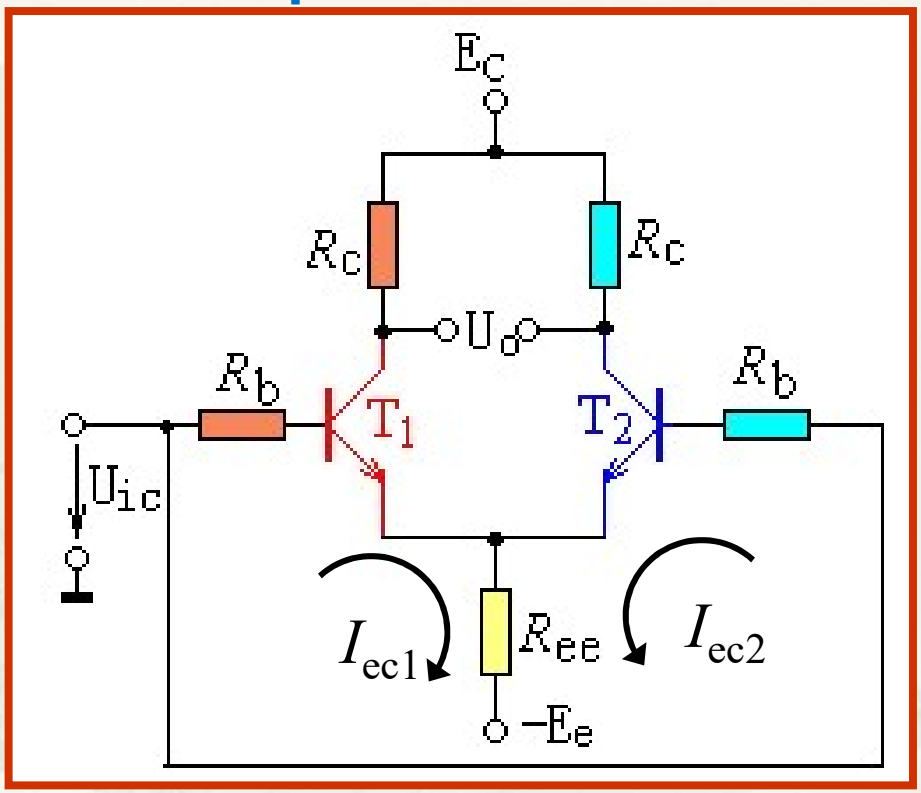
$$\Downarrow$$

$$I_{ec}=I_{ec1}+I_{ec2}=2\,I_{ec1}$$

$$\Downarrow$$

$$U_{Ree}=I_{ec1}\cdot 2R_{ee}$$

$$\Downarrow$$



The emitter resistance of each transistor can be regarded as $2R_{ee}$

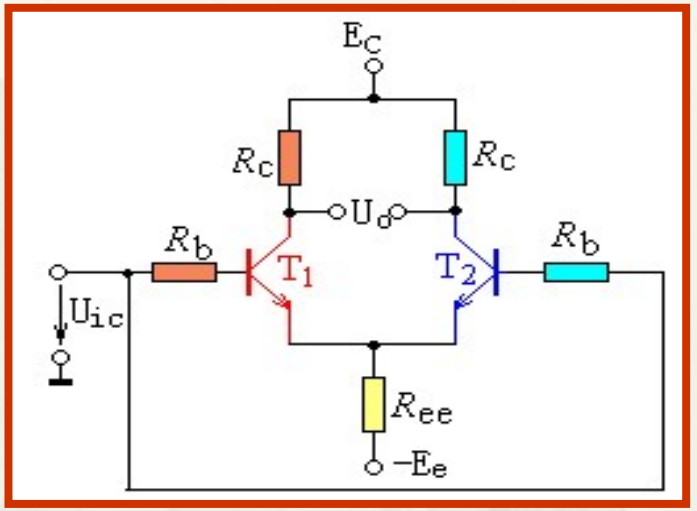
Basic Operation

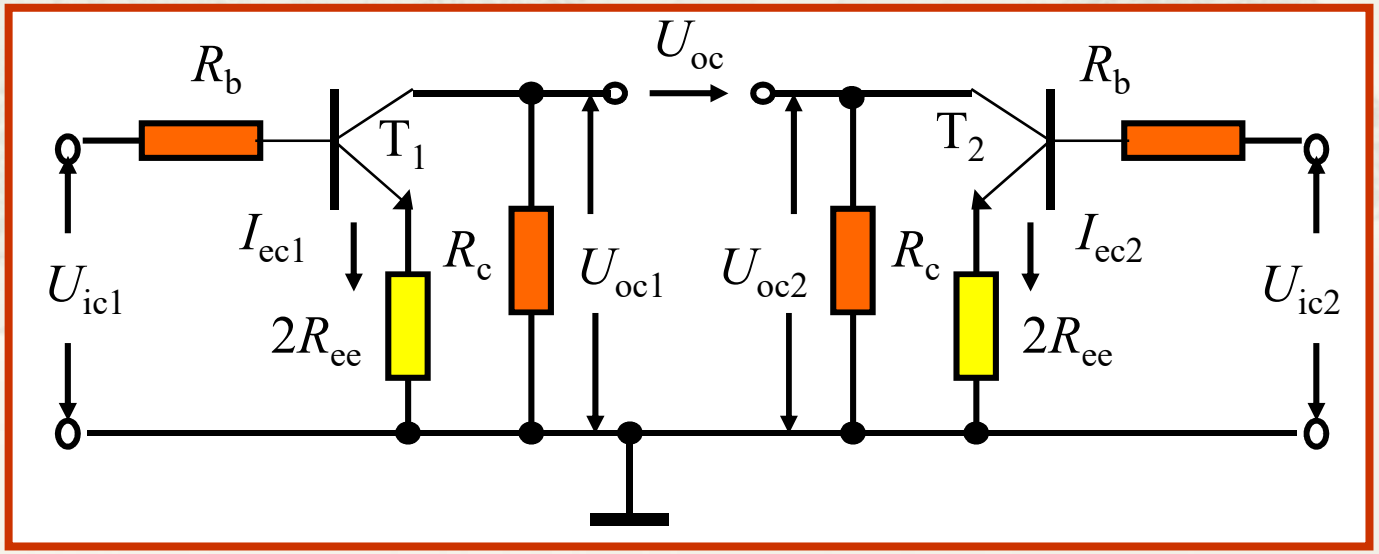
➢ AC Operation – Common-mode Input

Analysis quantitively

Step 1: AC equivalent circuit

The emitter resistance of each transistor can be regarded as $2R_{ee}$



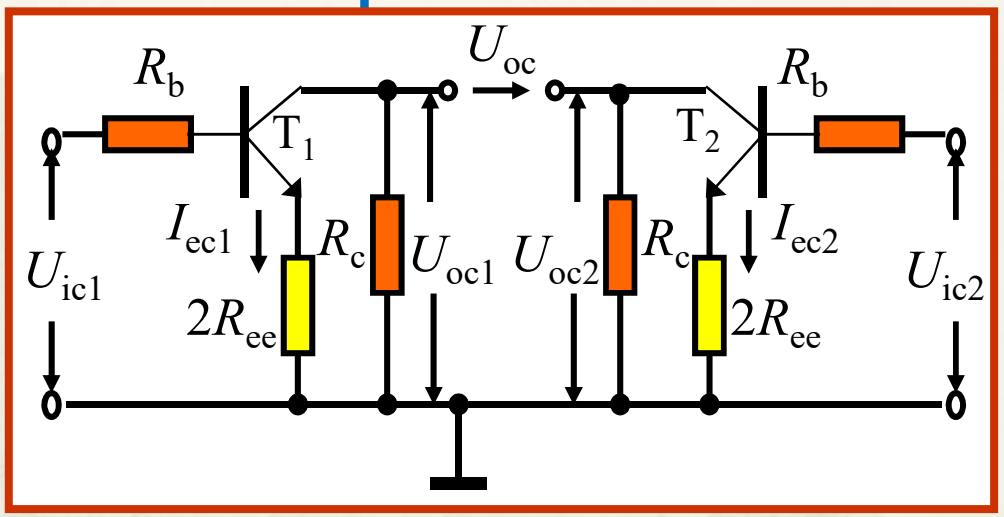


Basic Operation

➢ AC Operation – Common-mode Input

Analysis quantitively

Step 2: Calculating A_{Uc}, r_{ic} and r_{oc}



single-ended output

$$A_{Uc} = \frac{-h_{fe} R_c}{R_b + h_{ie} + (1+h_{fe}) \cdot 2R_{ee}}$$

$$= -\frac{R_c}{2R_{ee}}$$

$(1+h_{fe}) \cdot 2R_{ee} >> R_b + h_{ie}$

double-ended output

$$A_{Uc} = \frac{U_{oc}}{U_{ic}} = \frac{U_{oc1} - U_{oc2}}{U_{ic}} = 0$$

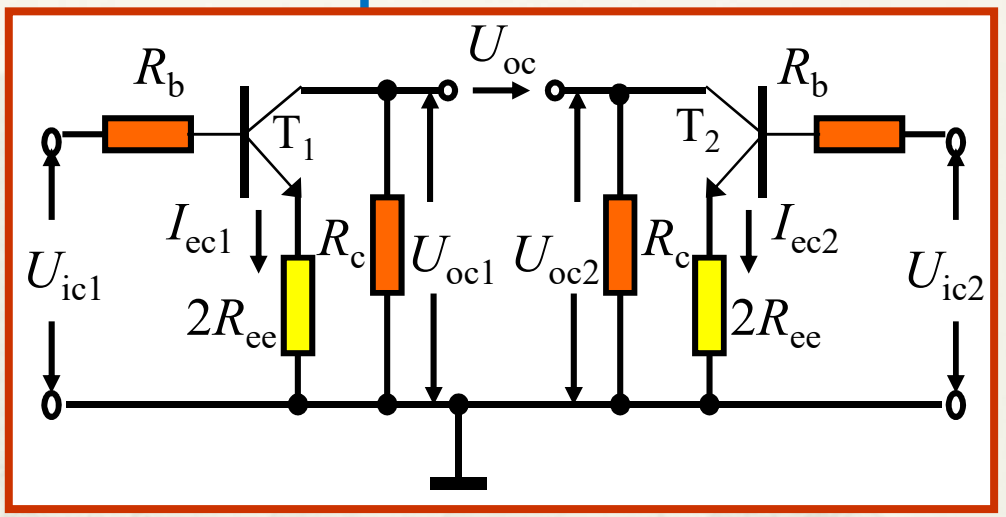
- strong rejection capability to common-mode signals.
- the higher R_{ee} is, the stronger the rejection capability is
- $A_{uc} << 1$
- drifts in same direction caused by environment are rejected, such as temperature.

Basic Operation

➢ AC Operation – Common-mode Input

Analysis quantitively

Step 2: Calculating A_{Uc}, r_{ic} and r_{oc}



single-ended input

$$r_{ic} = R_b + h_{ie} + (1 + h_{fe}) \cdot 2R_{ee}$$

double-ended input

two inputs are in parallel

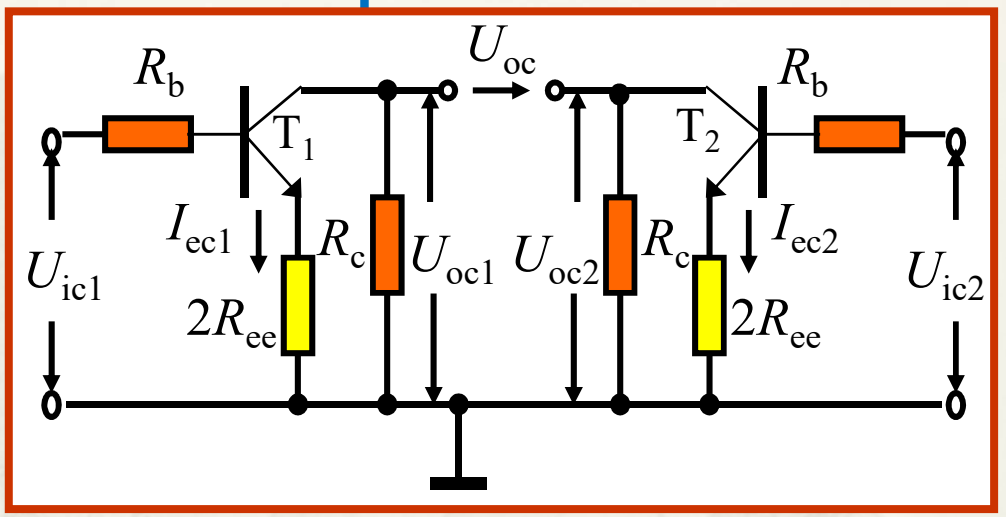
$$r_{ic} = \frac{1}{2}[R_b + h_{ie} + (1 + h_{fe}) \cdot 2R_{ee}]$$

Basic Operation

AC Operation – Common-mode Input

Analysis quantitively

Step 2: Calculating A_{Uc}, r_{ic} and r_{oc}



single-ended output

$$r_{oc} = R_c//(1/h_{oe}) \approx R_c$$

double-ended output

two outputs are in series

$$r_{oc} = 2R_c//(1/h_{oe}) \approx 2R_c$$

Common-Mode Rejection Ratio

CMRR

- The measure of an amplifier's ability to reject common-mode signals
- an important parameter of diff-amp
- $\mathrm{CMRR} = \left|\frac{A_{v(d)}}{A_{cm}}\right|$ or $\mathrm{CMRR} = 20\lg\left|\frac{A_{v(d)}}{A_{cm}}\right|$ (dB)

Where,

$A_{v(d)}$ -- differential gain

A_{cm} -- common-mode gain

double-ended output

$A_{\mathrm{Uc}} = 0 \longrightarrow \mathrm{CMRR} \to \infty$

single-ended output

$$\mathrm{CMRR(single)} = \left|\frac{A_{\mathrm{Ud}}(\mathrm{single})}{A_{\mathrm{Uc}}(\mathrm{single})}\right|$$

Example 5-5

Problem

A certain diff-amp has a differential voltage gain of 2000 and a common-mode gain of 0.2. Determine the CMRR.

Solution

$A_{v(d)} = 2000$, and $A_{cm} = 0.2$. Therefore,

$$\text{CMRR} = \frac{A_{v(d)}}{A_{cm}} = \frac{2000}{0.2} = \mathbf{10{,}000}$$

Examples

AC analysis for arbitrary input signal

- two inputs can be divided two pairs:
  - a pair of common-mode signals
  - a pair of differential signals
- calculate their gains respectively

$$U_{i1} = U_{ic1} + U_{id1}$$
$$U_{i2} = U_{ic2} + U_{id2}$$

$$\boldsymbol{U_{ic1} = U_{ic2} = (U_{i1} + U_{i2})/2}$$
$$\boldsymbol{U_{id1} = -U_{id2} = (U_{i1} - U_{i2})/2}$$

例题

U_{i1}=10mV U_{i2}=8mV

solution：

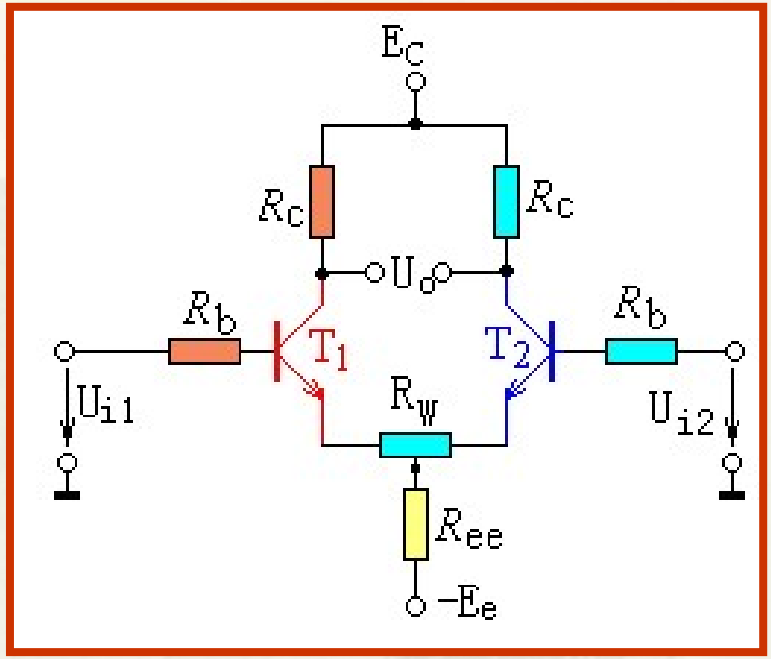
$$U_{ic1} = U_{ic2} = (U_{i1} + U_{i2})/2 = 9\text{mV}$$
$$U_{id1} = -U_{id2} = (U_{i1} - U_{i2})/2 = 1\text{mV}$$

$$U_{i1} = 9 + 1 = U_{ic1} + U_{id1}$$
$$U_{i2} = 9 - 1 = U_{ic2} + U_{id2}$$

Examples

Example

E_c=12V, E_e=6V, β=h_{fe}=60, $r_{bb'}$=100Ω, R_c=R_b=R_{ee}=10KΩ, R_W=100Ω, U_{i1}=5mV, U_{i2}=-3mV. Determine U_{id1}, U_{id2}, U_{ic}, ΔU_{od} and ΔU_{oc}



solution:

- determine h_{ie}

$$I_{BQ}R_b + U_{BE} + (1+\beta)I_{BQ}\cdot R_W/2 + (1+\beta)I_{BQ}\cdot 2R_{ee} - E_e = 0$$

$$I_{BQ} = \frac{E_e - U_{BE}}{R_b + (1+\beta)\cdot(R_W/2 + 2R_{ee})} = 4.37\mu A$$

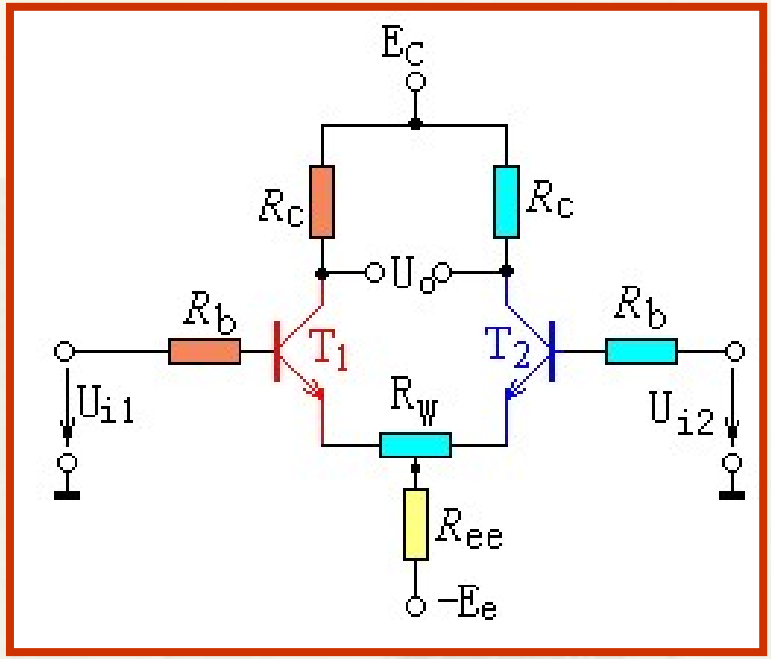
$$I_{CQ} = \beta I_{BQ} = 0.26mA \approx I_{EQ}$$

$$r_{be} = hie = r_{bb'} + (1+h_{fe})\frac{26mV}{I_{EQ}(mA)} = 6.2K\Omega$$

Examples

Example

E_c=12V, E_e=6V, β=h_{fe}=60, $r_{bb'}$=100Ω, R_c=R_b=R_{ee}=10KΩ, R_W=100Ω, U_{i1}=5mV, U_{i2}=-3mV. Determine U_{id1}, U_{id2}, U_{ic}, ΔU_{od} and ΔU_{oc}



solution:

- determine U_{id1}, U_{id2}, U_{ic}

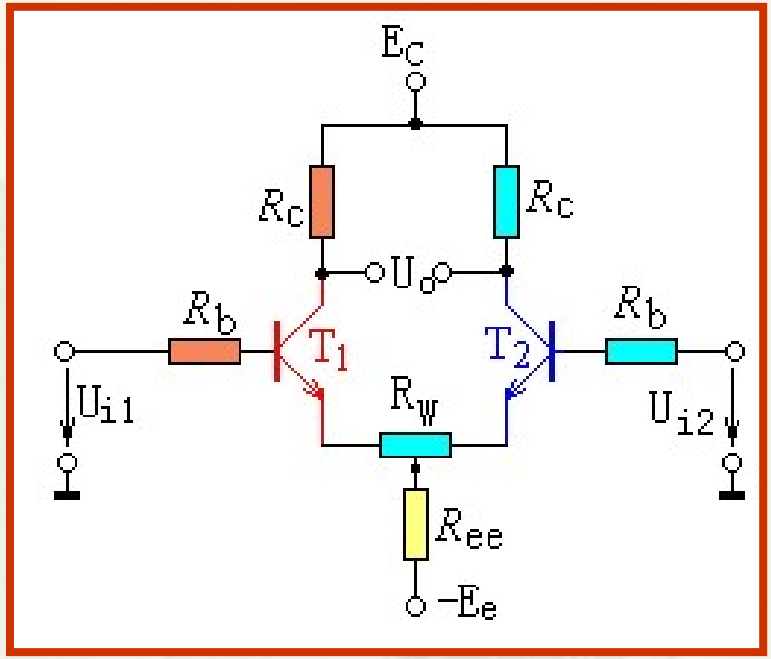
$U_{ic1} = U_{ic2} = (U_{i1} + U_{i2})/2 = (5-3)/2 = 1$ mV

$U_{id1} = -U_{id2} = (U_{i1} - U_{i2})/2 = (5+3)/2 = 4$ mV

Examples

Example

E_c=12V, E_e=6V, β=h_{fe}=60, $r_{bb'}$=100Ω, R_c=R_b=R_{ee}=10KΩ, R_W=100Ω, U_{i1}=5mV, U_{i2}=-3mV. Determine U_{id1}, U_{id2}, U_{ic}, ΔU_{od} and ΔU_{oc}



solution:

- determine $A_{Ud} \rightarrow \Delta U_{od}$

$$A_{Ud} = \frac{-h_{fe} R_c}{R_b + h_{ie} + (1+h_{fe}) \cdot R_W/2} = -31$$

$$\Delta U_{od} = A_{Ud} \cdot \Delta U_{id} = -31 \times 8 = -248 \text{ mV}$$

- determine $A_{Uc} \rightarrow \Delta U_{oc}$

double-ended output: $A_{Uc} = 0 \rightarrow \Delta U_{oc} = 0$

Examples

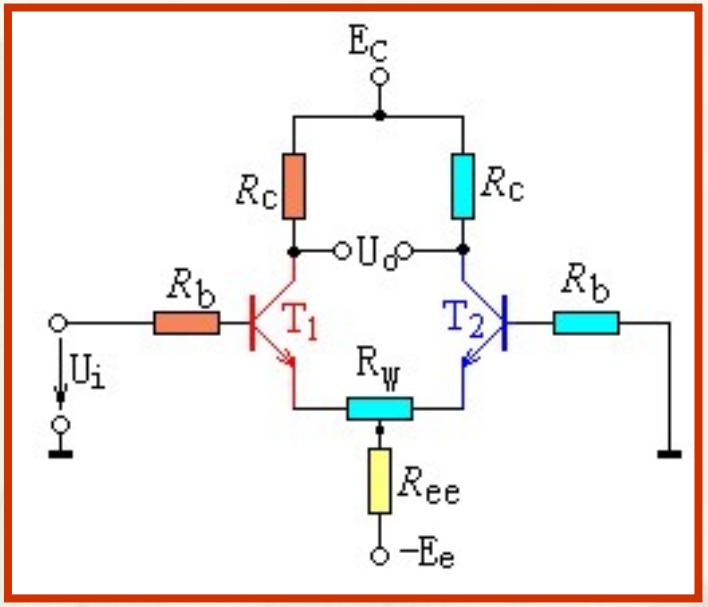
- **AC analysis for single-ended input signal**

$U_{i1}=U_i$, $U_{i2}=0$

be regarded as: arbitrary signals

$$U_{ic1} = U_{ic2} = (U_i + 0)/2 = U_i/2$$

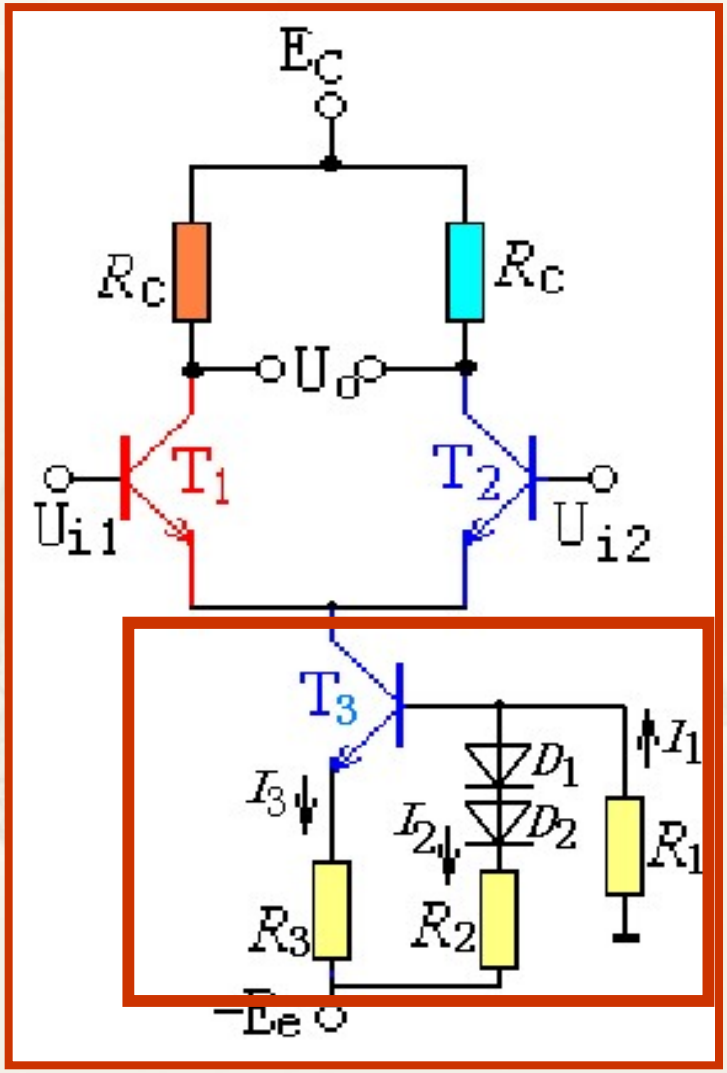
$$U_{id1} = -U_{id2} = (U_i - 0)/2 = U_i/2$$



Examples

Diff-amp based on constant current source

- increase R_{ee} to improve common-mode rejection ratio. However
  - ✓ increase negative power supply to keep Q-point. uneconomic
  - ✓ difficult to be manufactured
- replaced by constant current source
  - ✓ high ac resistance
  - ✓ stable bias
  - ✓ low CE drop



Examples

Diff-amp based on constant current source

Constant current

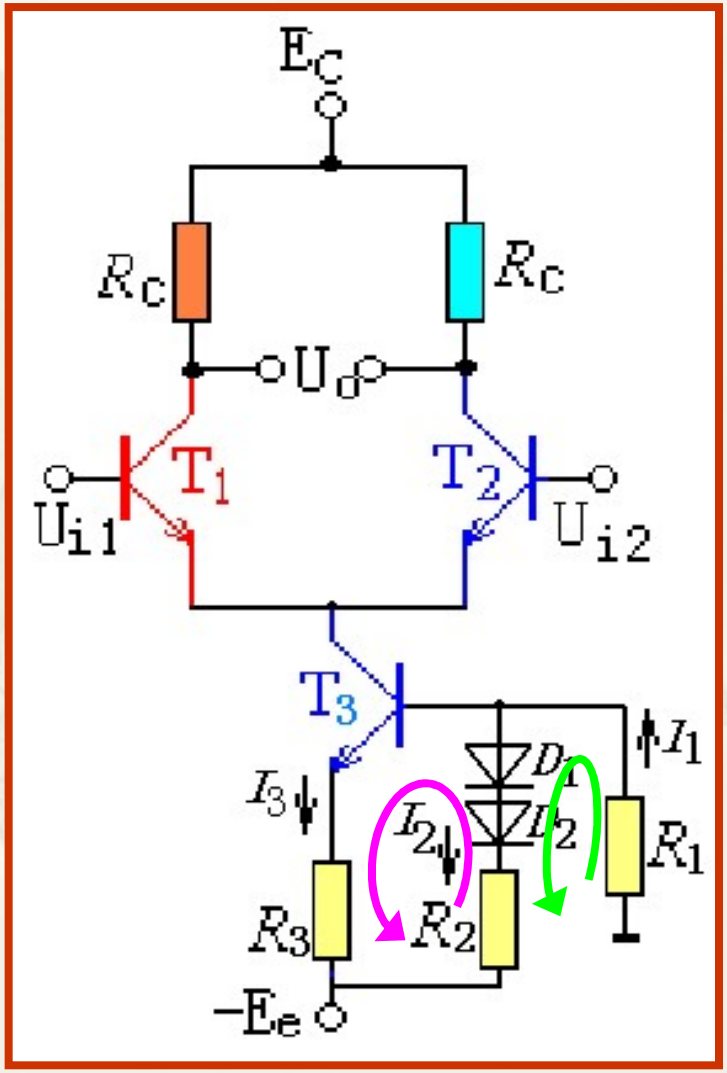
$$U_{be} + I_3 R_3 = 2U_D + I_2 R_2$$

I_{B3} is small $\rightarrow$ $I_1 \approx I_2$

$$I_1 \approx I_2 = \frac{E_e - 2U_D}{R_1 + R_2}$$

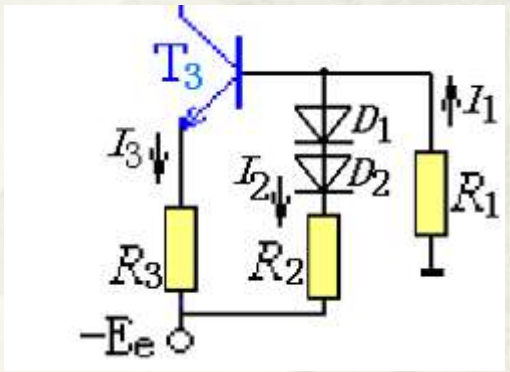
assume $R_1 = R_2$, $U_{be} = U_D$

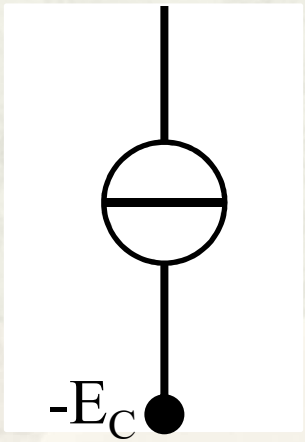
$$\rightarrow \quad I_{c3} \approx I_3 = E_e / 2R_3$$

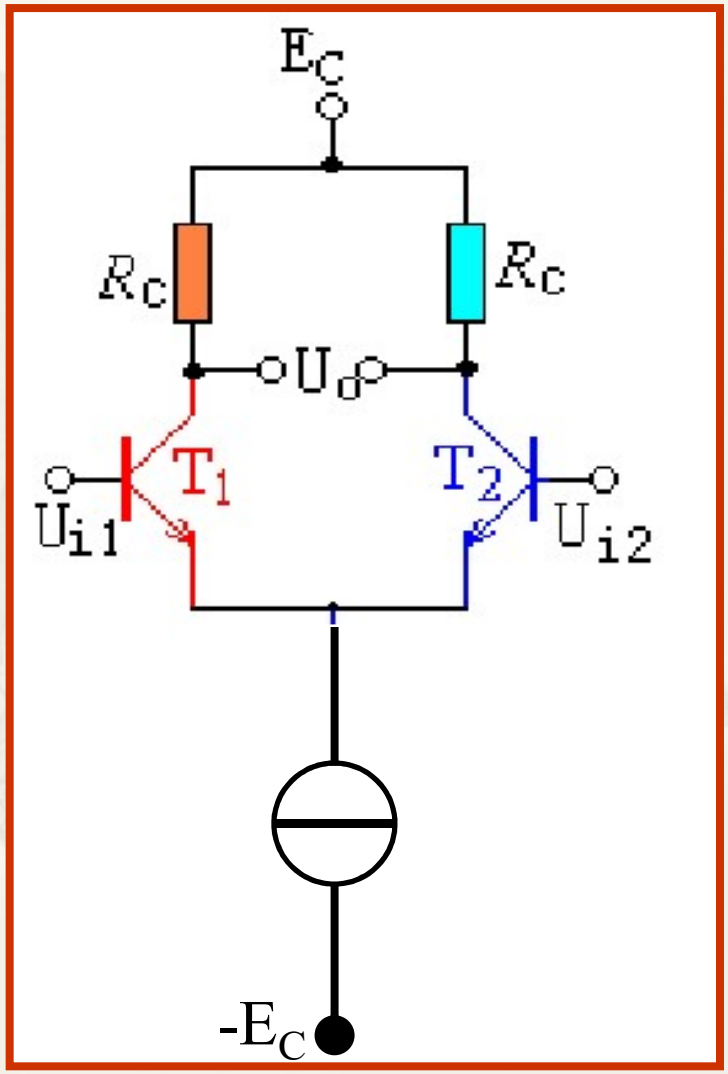


Examples

- **Diff-amp based on constant current source**
  - Constant current







Examples

Diff-amp based on constant current source

active load -- high AC resistance

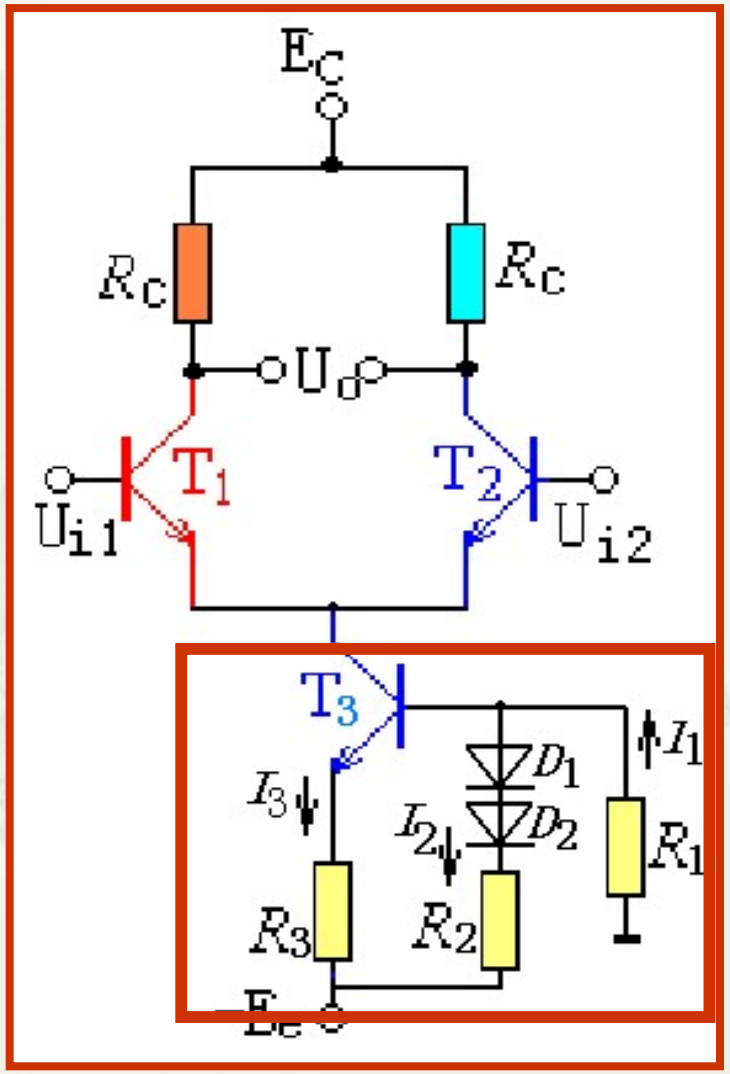
a solid resistor can be replaced by the equivalent resistor which is the output of a constant current source, named active load.

$$r_{o3} \approx \boxed{\frac{1}{h_{oe3}}}\left(1+\frac{h_{fe3}R_3}{R_{b3}+h_{ie3}+R_3}\right)$$

r_{ce}, high

$$R_{b3}=R_1//R_2$$

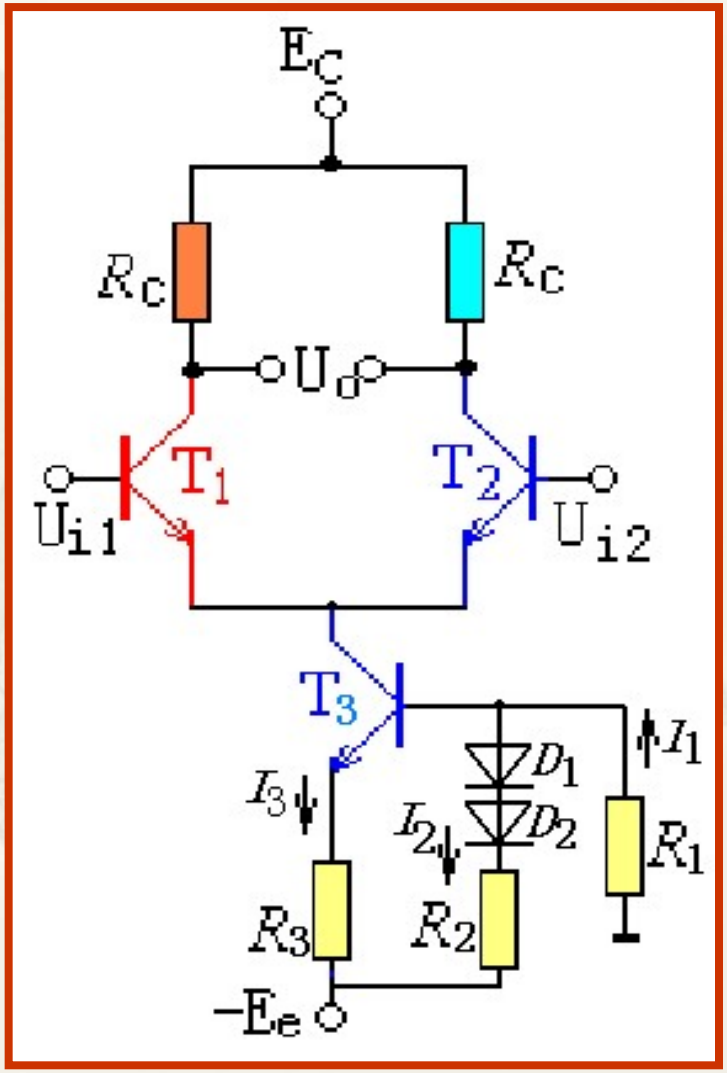
(assume $h_{fe3}*1/h_{oe3} >> h_{ie3}+R_{b3}$)



Examples

- **Diff-amp based on constant current source**
  - AC analysis

?



Examples

Combinational Diff-amp

CE-CB diff-amp

- ✓ improve the speed of input stage
- ✓ high voltage gain

a CE diff-amp consists of T_1/T_2

a CB diff-amp consists of T_3/T_4

a current mirror as a active load consisting of T_5/T_6

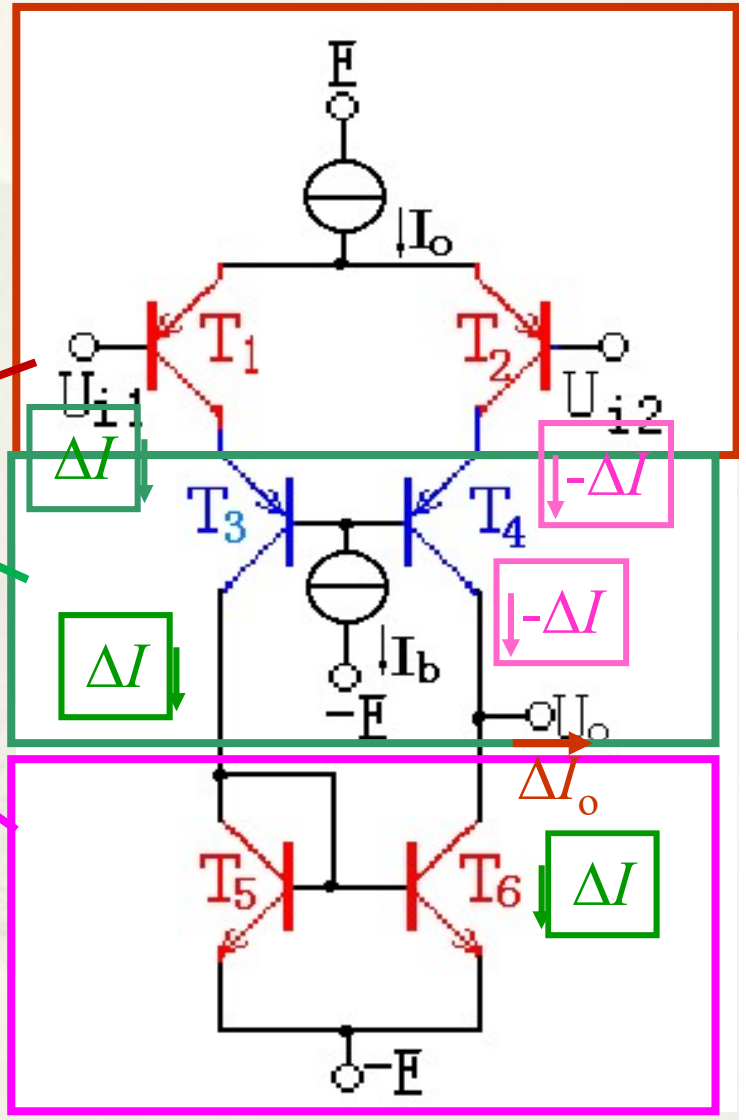
- are matched, and
- have identical base/emitter potentials

⇓

identical currents through T_5/T_6

⇓

$$\Delta I_o = -\Delta I - \Delta I = -2\Delta I$$



Examples

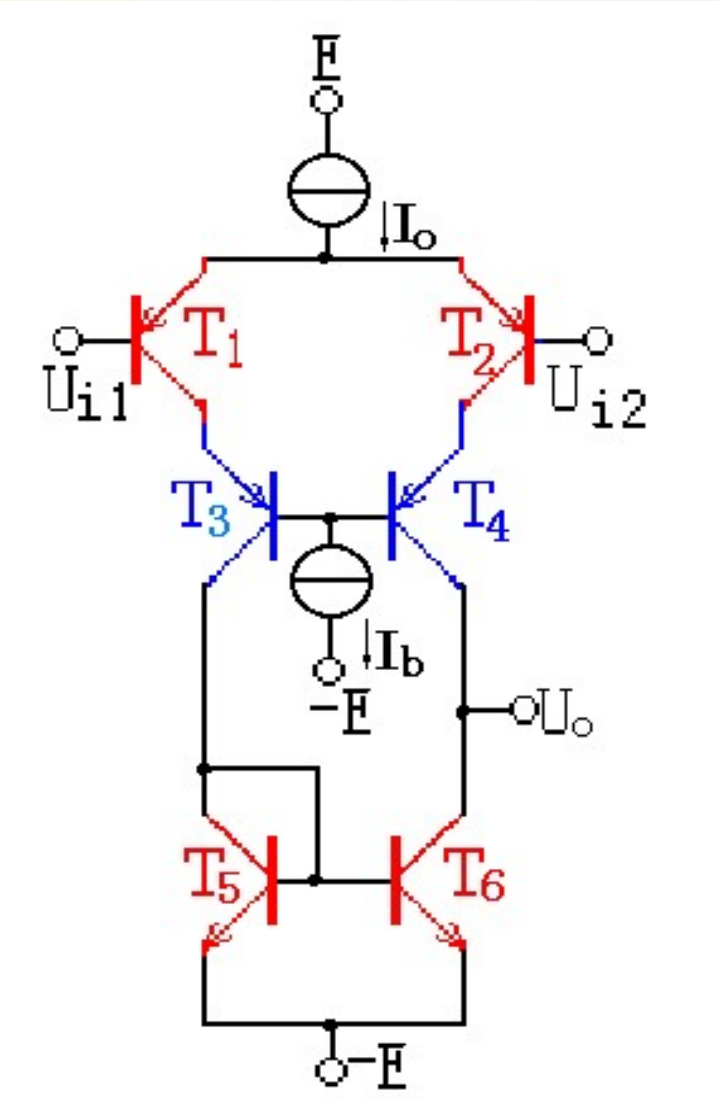
Combinational Diff-amp

CE-CB diff-amp

two stages are double-ended input/output

$$A_{U1} = -\frac{h_{fe1}h_{ib3}}{h_{ie1}} \qquad A_{U3} = \frac{h_{fb3}r_{o5}}{h_{ib3}}$$

$$A_{Ud} = A_{U1} \cdot A_{U3} \qquad r_{id} = 2h_{ie}$$



Examples

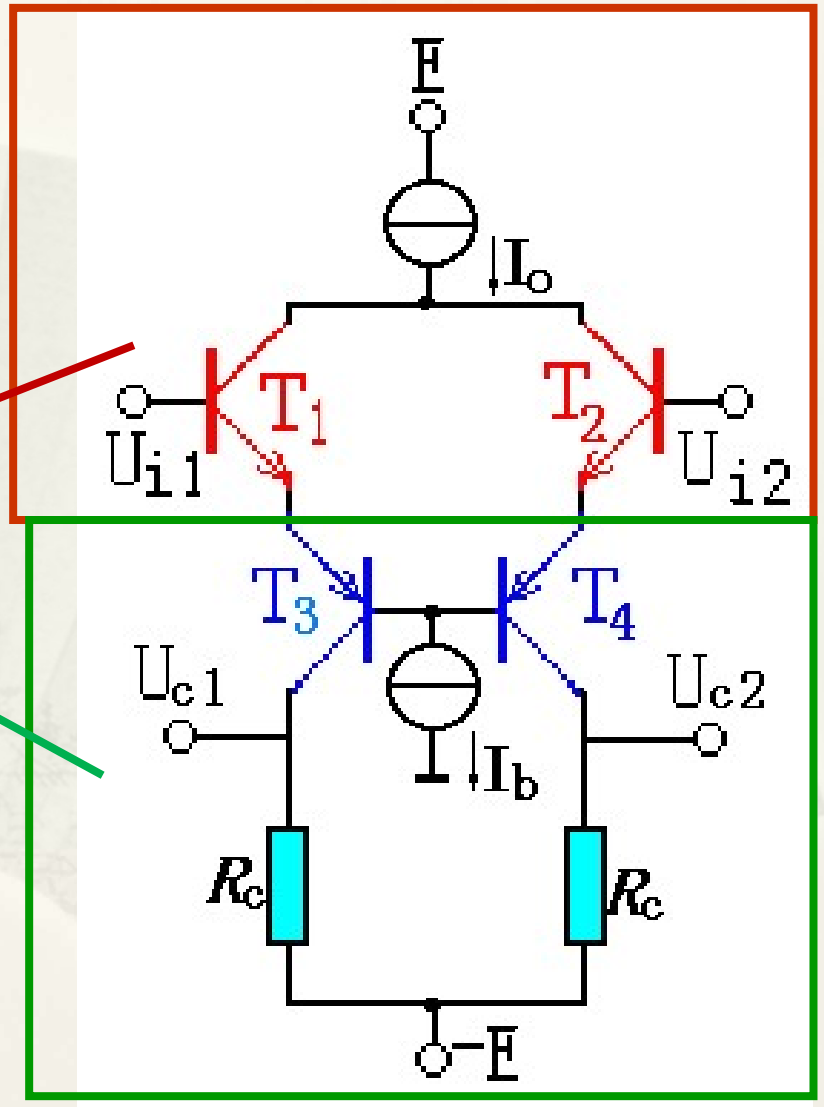
Combinational Diff-amp

CC-CB diff-amp

- ✓ high input resistance
- ✓ high voltage gain
- ✓ high current gain

a CC diff-amp consists of T_1/T_2

a CB diff-amp consists of T_3/T_4



KEY OBJECTIVES

1. Determine the ac parameters for a capacitively coupled multistage amplifier.
2. Describe the characteristics of transformer-coupled amplifiers, tuned amplifiers, and mixers.
3. Determine dc and ac parameters for direct-coupled amplifiers.
4. Compute dc and ac parameters for a class-A power amplifier and describe operation along the ac load line.
5. Computer dc and ac parameters for a class-B and class AB power amplifier and describe the load lines.
6. Explain the operation of a differential amplifier including how it can significantly reduce common-mode noise.

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 3, 4, 7, 8, 13, 14, 19, 20
- **Basic-plus Problems**
  - 24, 25, 26, 28
- **Complementary Problems**
  - Analyze the voltage gain, the input resistance, the output resistance of all the four types of BJT diff-amp circuits.